YouTube Video Analysis of Casey Neistat's Channel - A Cross-Sectional Study and Predictive Modelling

ARAVIND KALISSERY VIJAYAKUMAR

CANDIDATE NUMBER: 516318

SUPERVISED BY: CATHERINE SLATER

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Declaration

I declare that I have personally prepared this report and that it has not in whole or in part been submitted for any other degree or qualification. Nor has it appeared in whole or in part in any textbook, journal or any other document previously published or produced for any purpose. The work described here is my own, carried out personally unless otherwise stated. All sources of information, including quotations, are acknowledged by means of reference, both in the final reference section and at the point where they occur in the text.

Table of Contents

Acknowledgement 2

Declaration 2

Table of Figures 6

List of Tables 8

Executive Summary 9

1.0 Introduction 11

1.1 Understanding YouTube: The World of Online Video-Sharing 11

1.2 YouTube Content Creation as a Profession 13

1.3 Reason for choosing Casey Neistat youtube channel 13

2.0 Research Objective 14

3.0 Literature Review 15

3.1 Chapter introduction 15

3.2 Themes 16

3.2.1 The relationship between Likes and Engagement ratio in YouTube videos 16

3.2.2 The relationship between Comments and Engagement ratio in YouTube videos 18

3.2.3 The relationship between views and Engagement ratio in YouTube videos 19

3.2.4 The relationship between duration and Engagement ratio in YouTube videos 23

3.2.5 The relationship between the Time of uploading and Engagement ratio in YouTube videos 26

3.2.6 The relationship between Category/Genre and Engagement ratio in YouTube videos 29

3.2.7 The relationship between Tags and Engagement ratio in YouTube videos 32

3.2.8 Critical Comparison and contrast of the Hypothesis to be used for each iv to dv. Via heat map for assigning the weight 34

3.2.9 The connection between the machine learning regression model and YouTube video analysis 37

3.3 Literature Gap 39

3.4 Chapter Summary 40

4.0 Significance of the Study 40

5.0 Methodology 41

5.1 The Process of Data Collection in Analysing YouTube Videos 41

5.1.1 The YouTube Video API: An Access Point to Data 42

5.1.2 Points of Interest in the Data: Collected Information 42

5.1.3 Ethical Considerations 42

5.2 Data Cleaning: 43

5.3 Feature Engineering 43

5.4 Visualization Techniques and Exploratory Data Analysis 46

5.5 Outlier Treatment 47

5.6 Data Preparation and Preprocessing for Engagement Score Predictive Modelling 48

5.6.1 Feature Selection: 48

5.6.2 One-Hot Encoding: 48

5.6.3 Data Splitting: 48

5.6.4 Standardisation: 48

5.7 Regression Modelling 49

5.7.1 Multiple Linear Regression 49

5.7.2 Decision Tree Regression 50

5.7.3 K-Nearest Neighbours (KNN) Regression 50

5.7.4 Random Forest Regression 51

5.7.5 Gradient Boosting Regression 51

5.7.6 Neural Network Regression 52

5.7.7 Extreme Gradient Boosting Regressor (XGBRegressor) 52

5.8 Model performance evaluation techniques 53

1) Key Evaluation Metrics for YouTube Video Analysis Regression Models 53

2) Scatter Plots 54

3) Residual plots 55

4) Learning Curve 55

6.0 Data Analysis 55

6.1 Descriptive Statistics 55

6.1.1 Numerical Data 55

6.1.2 Categorical Data 57

6.2 Exploratory Data Analysis 58

6.2.1 Univariate Graphs 58

6.2.2 Multivariate graphs 59

6.3 Outlier Treatment 68

6.4 Baseline Model Evaluation for Engagement Score Prediction 69

6.5 Regression Models - Results 69

6.5.1 Multiple Linear Regression 69

6.4.2 Decision Tree Regression 72

6.5.3 K-Nearest Neighbours (KNN) Regression 76

6.5.4 Random Forest Regression 79

6.5.5 Gradient Boosting Regression 83

6.5.6 Neural Network Regression 86

6.5.7 Extreme Gradient Boosting Regressor (XGBRegressor) 89

6.6 Comparison of Models 92

7.0 Conclusion 94

7.1 Limitations 95

7.2 Recommendations 95

7.3 Future Research Opportunities 95

8.0 References 97

9.0 Appendices 104

Table of Figures

Figure 1: The ratio of Engagement rate and likes in YouTube videos) (Source: Valentina Dencheva, 2023) 16
Figure 2: Engagement ratio calculation) (Source: Keyhole, 2023) 18
Figure 3: Understanding YouTube Engagement Metrics and How They Are Counted (Source: Kapoor, 2023) 20
Figure 4: Distribution of total YouTube video views worldwide as of December 2018 by category (Source: Statista, 2019) 21
Figure 5: Views and YouTube videos Source: (Statista, 2019) 22
Figure 6: Average YouTube video length as of December 2018 by category (Source: Ceci, 2021) 24
Figure 7: Weekly time spent on online videos Source: (Statista, 2023) 25
Figure 8: Dominance of Bollywood among YouTube channels Source: (Statista, 2023) 27
Figure 9: Live Stream popularity of YouTube Source: (Uscreen, 2022) 28
Figure 10: Popularity of different YouTube channels Source: (Statista, 2023) 30
Figure 11: Content choice of YouTube users Source: (Uscreen, 2022) 31
Figure 12: Popular hashtags used by democrats in US Congress Source: (Statista, 2022) 33
Figure 13: Viral nature of YouTube Source: (Uscreen, 2022) 34
Figure 14: Heat Map Source: (self-designed) 35
Figure 15: Different types of Machine learning regression model) (Source: Gong, 2022) 38
Figure 16: Distribution of Variables 58
Figure 17: Distribution of variables over time 59
Figure 18: Views Vs Category 61
Figure 19: Likes Vs Category 62
Figure 20: Comments Vs Category 63
Figure 21: Engagement Scores Vs Category 64
Figure 22: Average Views by the Time of the Day 65
Figure 23: Average Views by Having and Not Having Tags 66
Figure 24: Engagement Score by Having and Not Having Tags 67
Figure 25: Before Outlier Treatment 68
Figure 26: After Outlier Treatment 68
Figure 27: Multiple Linear Regression model Actual and Predicted Values 71
Figure 28: Multiple Linear Regression Residual Plot 71

Figure 29: Multiple Linear Regression Learning Curve 72
Figure 30: Decision Tree Regression Actual and Predicted Scatter Plot 74
Figure 31: Decision Tree Regression Residual Plot 75
Figure 32: Decision Tree Regression Residual Plot 75
Figure 33: K-Nearest Neighbours (KNN) Regression Actual and Predicted Values Scatter Plot 78
Figure 34: K-Nearest Neighbours (KNN) Regression Residual Plot 78
Figure 35: K-Nearest Neighbours (KNN) Regression Learning Curve 79
Figure 36: Random Forest Regression Actual and Predicted Values Scatter Plot 81
Figure 37: Random Forest Regression Residual Plot 82
Figure 38: Random Forest Regression Learning Curve 82
Figure 39: Gradient Boosting Regression Actual and Predicted Values Scatter Plot 84
Figure 40: Gradient Boosting Regression Residual Plot 85
Figure 41: Gradient Boosting Regression Learning Curve 85
Figure 42: Neural Network Regression Actual and Predicted Values Scatter Plot 87
Figure 43: Neural Network Regression Residual Plot 88
Figure 44: Neural Network Regression Learning Curve 88
Figure 45: Extreme Gradient Boosting Regressor (XGBRegressor) Actual and Predicted Values Scatter Plot 90
Figure 46: Extreme Gradient Boosting Regressor (XGBRegressor) Residual Plot 91
Figure 47: Extreme Gradient Boosting Regressor (XGBRegressor) Learning Curve 91
Figure 48: Performance Evaluation of the models 92
Figure 49: Appendix 1 104
Figure 50:Appendix 2 104
Figure 51: Appendix 3 105
Figure 52: Appendix 4 105
Figure 53: Appendix 5 106
Figure 54: Appendix 6 106

List of Tables

Table 1: The Classification of Publication Timing 44
Table 2: Weight Assignment 45
Table 3 Descriptive Statisitcs of Numerical Variables 55
Table 4 Descriptive Statistics of Categorical Variables 57

Executive Summary

The primary objective of this research endeavour was to conduct a thorough analysis of YouTube videos, focusing specifically on the channel belonging to Casey Neistat. Several primary goals were established in order to attain a broader understanding of the variables that impact viewer engagement. The study encompassed a comprehensive examination, graphical representation of the interconnections between variables, development of predictive models, and generation of valuable insights.

The data utilised in this analysis was obtained through the YouTube API, which yielded significant insights into various variables including views, likes, comments, video duration, publishing time, video category, and the presence of tags. The main objective of the study was to construct regression models that could effectively forecast engagement scores for Neistat's videos.

The evaluation of the performance of different regression models involved the utilisation of metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Prominent models encompassed in the analysis consisted of Multiple Linear Regression, Decision Tree Regression, Random Forest Regression, K-Nearest Neighbours (KNN) Regression, Gradient Boosting Regression, Neural Network Regression, and Extreme Gradient Boosting (XGB) Regressor.

Furthermore, the visualisation of variable relationships yielded perceptive patterns. Research findings indicate that there is a positive correlation between the timing of video uploads, specifically during the morning hours (between 7 am and 11:59 am), and the level of engagement scores observed. Moreover, the presence of tags linked to videos had a positive influence on their extent of promotion. Furthermore, it was observed that videos classified under the "Education" category exhibited higher levels of engagement, as indicated by the number of likes, views, and comments, in comparison to other genres explored by Casey Neistat.

In summary, this study successfully accomplished its stated goals and additionally yielded practical insights that can be utilised by individuals involved in content creation, marketing, and those with a keen interest in YouTube. The thorough examination of Casey Neistat's YouTube channel and the formulation of predictive models hold promise in informing strategies aimed at improving user engagement on the platform.

1.0 Introduction

One-third of all internet traffic comes from YouTube videos, which have a monthly audience of over two billion users. YouTube has become a powerful and revolutionary platform in the constantly changing world of digital media, revolutionising how users interact with creators and consume content. YouTube has had a significant impact on contemporary media culture as the second-largest search engine in the world and a centre for user-generated content. Beyond simple viewership figures, the success of YouTube as a platform depends on the intensity and scope of viewer engagement. Likes, comments, shares, and subscriber interactions are just a few of the many components that make up engagement metrics, which are crucial measures of a video's effect on viewers. Understanding the dynamics of engagement is essential for researchers, advertisers, and content producers alike in order to identify the driving forces behind the development of the digital content landscape.

Casey Neistat's YouTube channel is proof of how dynamic and constantly changing content creation on the site is. Neistat's channel is a multifaceted digital playground that offers viewers an extensive variety of experiences, insights, and emotions. He is known for his engrossing storytelling, modern cinematography, and unfiltered authenticity. With millions of subscribers and billions of views, Casey Neistat has built a diverse content library that defies categorization and reflects his limitless imagination and relentless search for new experiences.

With a focus on Casey Neistat's channel, this research project sets out on an intriguing journey into the world of YouTube content analysis. In order to predict the engagement ratio of his video uploads, our main goal is to use data-driven predictive modelling. We aim to reveal the latent patterns, trends, and insights that may be extrapolated to the more extensive field of YouTube content creation by closely examining the numerous elements that go into Casey Neistat's videos' success. Our project brings analytics and creativity together, reflecting the platform's dynamic nature as a place where content producers can use data-driven insights to guide their creative choices.

1.1 Understanding YouTube: The World of Online Video-Sharing

Founded in 2005, YouTube has significantly transformed the landscape of online video sharing. With a monthly logged-in user base exceeding 2 billion, this platform has emerged as a dominant force on

a worldwide scale, fundamentally reshaping the way in which we engage with and produce video content. The platform has facilitated the emergence of a novel cohort of individuals engaged in content creation, presenting a wide array of films encompassing educational, entertaining, and other genres. The recommendation method, which utilises machine learning, effectively maintains viewer engagement while also stimulating discussions around content discovery and filter bubbles. The impact of YouTube on digital culture is indisputable, as it persists in shaping the trajectory of online video and redefining the parameters of artistic manifestation.

According to estimates made in 2021, YouTube played a significant role in the generation of employment opportunities, with around 425,000 jobs being facilitated or supported in the United States and 122,000 jobs in Brazil. This contribution to the labour market has had a notable impact on the respective areas' Gross Domestic Product (GDP), amounting to 25 billion U.S. dollars in the United States and six billion Brazilian reals in Brazil. Based on a poll conducted among professional YouTube video creators in the United States, it was found that a significant majority of 81 percent of participants acknowledged the platform's efficacy in facilitating the dissemination of their material to global audiences that would otherwise be inaccessible. Within the United Kingdom, a significant majority of creative businesses, specifically eight out of ten, have expressed that YouTube provides them with unique prospects for creativity and financial gain that are not readily available through conventional media channels.

Nevertheless, it is important to note that a substantial quantity of views does not necessarily result in financial gains for content creators. The matter of ad revenue redistribution has gained more significance since the implementation of the YouTube Shorts monetization programme in February 2023, which superseded the YouTube Shorts Fund. The newly developed programme use an algorithm to compute income shares, which are determined by the advertisements displayed during video playback. This allocation of funds serves the purpose of covering music expenses and providing compensation to content providers. Although creators may have some reservations regarding the monetization potential of Shorts, they ultimately see the value of this format in enhancing the discoverability and visibility of their material.(L. Ceci, Aug 31, 2023)

1.2 YouTube Content Creation as a Profession

Since its founding, YouTube has even gained great recognition and continues to be a very relevant profession in the modern era. The following are a few justifications as to why youtubing is still a viable and current career:

1. Massive Audience: YouTube is one of the most popular social media sites with over 2 billion monthly active users worldwide. It offers accessibility to a sizable and varied global audience.

2. Content Consumption: Online video content consumption is still growing. The fact that people of all ages are spending more time watching videos on YouTube gives content producers the chance to reach a large audience.

3. Monetization: YouTube provides content creators with several monetization options, including affiliate marketing, channel memberships, Super Chats, merchandise shelf integration, and ad revenue. Successful YouTubers can generate considerable income.

4. Diverse Niches: YouTube supports a wide range of content categories, including gaming, education, beauty, health, and technology. Because of the diversity, people can produce content in the subjects they are most interested in.

5. Influence and Brand Collaborations: Popular YouTubers frequently rise to prominence in their respective fields as influential figures. They can work with brands, obtain sponsorships, and promote goods to increase their income.

6. Global Reach: By removing geographical barriers, YouTube enables creators to interact with a global audience. The unique opportunities and cultural exchanges are made possible as a result of this global reach.

7. Business Opportunities: Many YouTubers have established businesses based on their content, such as clothing lines, books, podcasts, or live events, to take their brand beyond the platform.

1.3 Reason for choosing Casey Neistat youtube channel

Casey Neistat's YouTube channel is an intriguing focal point for our video analysis project because of its extraordinary diversity of content, a feature that is both indicative of his creativity and indicative of YouTube's enormous potential for content creators. This diversity goes far beyond simple topical

variation; it embodies a wide range of storytelling techniques, content formats, and genres that call for in-depth analysis. The following are the main justifications for our analytical investigation's choice of Casey Neistat's channel:

1. An exemplary example of YouTube's versatility:

The versatility of YouTube itself is reflected in Casey Neistat's channel. He has shown that YouTube is open to all genres and formats as one of the most well-known creators on the site. This adaptability reflects the user-driven, organic nature of YouTube as a platform where content producers have the freedom to express themselves through a variety of content types.

2. Analysing Engagement Across Genres:

We have the chance to look at how engagement dynamics vary across genres by choosing a channel that produces content in a variety of genres. This information is crucial for researchers, advertisers, and content producers who are trying to figure out what influences viewer interaction across different categories of content.

3. Analysis of Audience Reaction:

Given the variety of Casey's content choices, we can examine how his audience reacts to various types of videos. For content creators looking to effectively connect with their target audience, understanding which content resonates most strongly with viewers offers useful insights.

4. Statistical Significance:

Casey Neistat's channel has a large viewership, ensuring that our analysis is statistically significant. We are able to make more reliable and extrapolable conclusions about engagement trends due to the size of the sample of viewers.

2.0 Research Objective

Through this research, we endeavor to achieve several key objectives:

In-Depth Analysis: To gain a profound understanding of Casey Neistat's YouTube channel and the factors influencing viewer engagement.

Visualising Variable Relationships: Visualising variable relationships is crucial to our research to identify patterns, trends, and correlations between views, likes, comments, video duration, publishing time, and tags.

Predictive Modelling: To develop regression models capable of accurately predicting engagement scores for Neistat's videos.

Insight Generation: To unearth actionable insights that can benefit content creators, marketers, and those interested in the YouTube platform.

3.0 Literature Review

3.1 Chapter introduction

In the following chapter, the researcher critically evaluates the numerical and non-numerical aspects of YouTube videos and their critical relationship with the engagement ratio of those videos. The researcher will present in this chapter the aspects such as Likes, views, comments, duration, time of uploading category/genre, and tags impact on YouTube video engagement ratio and how significant these impacts on the machine learning for forecasting reach increase. Furthermore, in this chapter, the researcher will also explore the critical comparison and contrast of hypotheses and Literature Gaps.

3.2 Themes

3.2.1 The relationship between Likes and Engagement ratio in YouTube videos

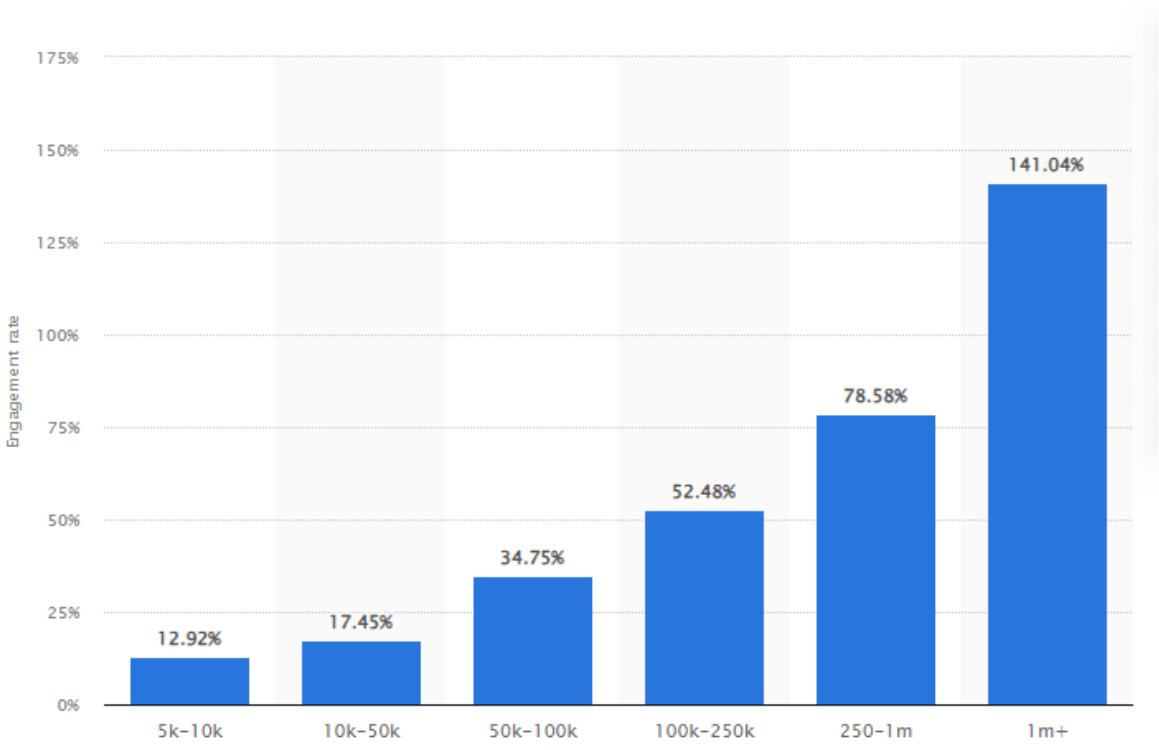


Figure 1: The ratio of Engagement rate and likes in YouTube videos) (Source: Valentina Dencheva, 2023)

The above-mentioned figure shows the ratio of engagement rate and likes on YouTube videos as per the 2023 survey of Valentina Dencheva (2023). This figure shows that between January 2023 and June 2023, in the United Kingdom, influencers who have more than 100 thousand followers reached more than 125% engagement rate and gained 10 million likes in their videos. According to the research study of Levinson, Greenfield, and Signorelli (2020), on social media platforms like YouTube, likes on videos represent the number of weavers who enjoy the content and express their appreciation to the creator by providing a thumbs-up button to the content. Similarly, the engagement ratio is a bigger cadent that calculates different types of forms of viewer interaction that have been taken into account by the video. However, the engagement ratio is mainly calculated by dividing the total view of the video by the total engagement action such as likes, comments, or shares and then dividing that measure by a hundred. In the context of the critical relationship between like and engagement ratio, my understanding of the critical aspect that day depends on, for

example, how positively the video is impacting the viewers, engagement ratio and overall engagement in content, and balance and interpretation. The study of Klug et al. (2021) followed the hypothesis of branding posts presented in reach media format to generate more likes than posts containing no rich media. However, the authors noted in the findings that the type of content that has been uploaded is very important for likes and overall engagement ratio. The researcher mentions that video content critically attracts viewers more than textual content (4.206, $p<0.001$), and similar views can be observed in the case of likes. Therefore, likes, on average, can be observed 15 times on video content (15.114, $p<0.001$). As per the view of Munaro et al. (2021), The ERR is critically linked to the number of followers. The study shows an ERR% increase with followers. For example, in different cases, YouTubers who have over 1 million followers have an average ERR of 2.66%. On the other hand, YouTubers who have more than 5K to 20K have an ERR of 32.52%. On the other hand, in the study of Su, Wu, and Wu (2022), for the spectrum of engagement, only focusing on likes on the content is not critically affected because the like shows the positive sentiment of the viewers. It has been observed that more likes in a video did not contribute to increasing the engagement ratio specifically. Furthermore, the engagement ratio shows the deeper connection between the audience and the content that critically provides a comprehensive understanding of how the viewers are actively engaging with the content created and his work. Additionally, balance and interpretation are required to gain an understanding of the relationship between likes and engagement ratio. However, a video that contains a high number of likes can have a lower engagement rate due to few comments and shares [{ERR= Likes+Comments+shares}{ERR%= (Likes+Comments+shares)/100]. For example, a video that contains high likes and comments but has fewer shares can represent a low engagement ratio [L+S<S of ERR%]. On the other hand, a video that shares meaningful discussion or gains a high amount of shares among these but does have a lower number of likes can contain a high engagement ratio. Martin (2023) stated that, as an example, if an account on YouTube or any other SNS platform can reach a 10% engagement ratio, it can be deemed high. For example, the engagement ratio is mainly identified by calculating the overall engagement divided by the viewers and then multiplying by a hundred [E/V*100]. In this particular context, any account which has 1 million views and earns, particularly 100,000 likes, comments, and shares, has a 10% ERR.

Moreover, according to Aldous, An and Jansen (2019), the analysis of different topics on social media platforms like YouTube can help study the engagement of users by studying the likes of different posts. The authors in that paper collected data from 3163373 posts on various topics within eight months. This allowed the authors to analyse the differences in the engagement of users on YouTube by focusing the responses on the likes received by different videos and posts. Therefore, the engagement levels were found to vary across different topics and posts. As a consequence, the

accuracy of predicting whether articles and posts can be liked and preferred by users can be increased to 80%. Therefore, the likes received by different posts and videos on YouTube can help commercial and news-based organisations increase their engagement rates.

3.2.2 The relationship between Comments and Engagement ratio in YouTube videos

According to the research study of Shahbaznezhad, Dolan, and Rashidirad (2021), the comments in YouTube videos play a significant role in influencing the overall engagement ratio of any content, which shows that the comments and engagement ratio are closely intertwined with each other. However, in engagement ratio calculation, the comments are vital components as the overall engagement ratio is calculated by the engagement actions, which are likes, comments, and shares [1.574E 1 11, $p < 0.001$]. Therefore, on YouTube videos, comments are a form of engagement that expresses the viewer's engagement and interaction with the uploaded content. The comment shows the action of active engagement, which critically contributes to the engagement ratio and includes the critical thoughts, suggestions, questions, and discussions of the weavers related to the content. Furthermore, it has been observed that the engagement ratio calculation involves likes, shares, and other components with the comments but comes and shows the quality and depth of engagement of the viewers. As per the study of Bentley *et al.* (2021), the comments have a more significant impact that shows a deeper connection to content and willingness to participate in the viewers, which contributes more to the engagement ratio in comparison to the other concepts of likes or shares.

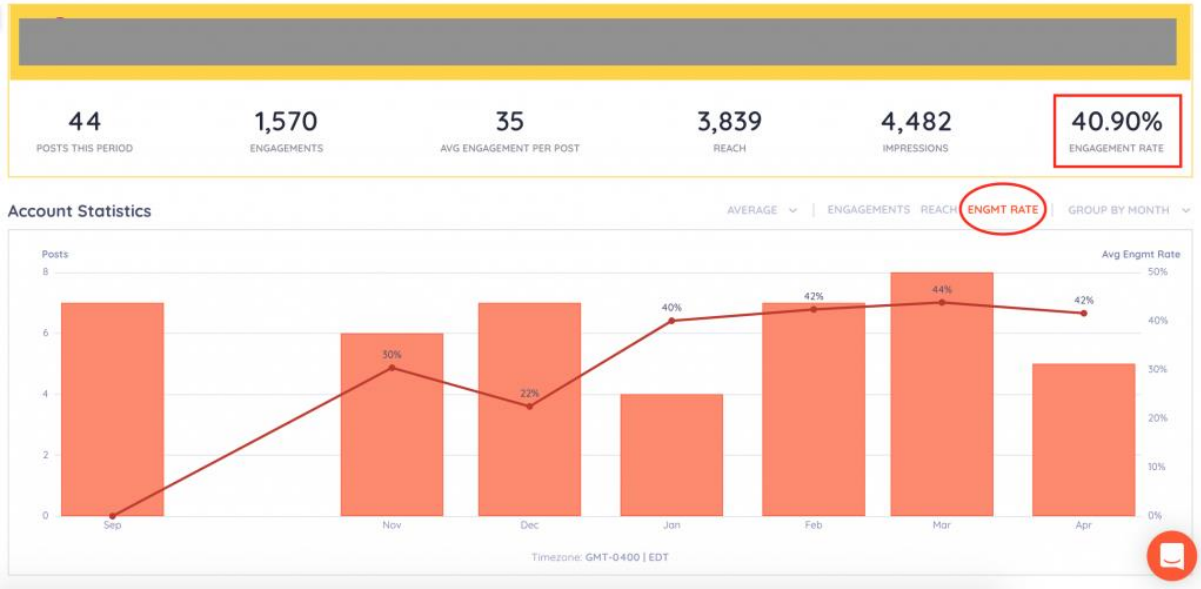


Figure 2: Engagement ratio calculation) (Source: Keyhole, 2023)

The above-shown figure presents the statistics of accounts which gain an average engagement of 35 per post, and the overall engagement rate of the account is 40. 90%. It has been observed that the total engagements that the account has gained is 1570. In the overall engagement, the comments influence the engagement ratio as a part of the numerator in the engagement ratio. Munaro *et al.* (2021) noted that while comments are one of the most important concepts of engagement ratio, they should be aligned with the other concepts, such as likes and shares, to gain an overall healthy balance of indicators that can provide a well-rounded audience engagement.

The comments on different YouTube videos and posts act as conversation starters for different topics and also help in shaping the perceptions of viewers (Risch and Krestel, 2020). In this regard, the comments on different YouTube videos are shaped according to the quality of such videos and vary as per different types of topics and content. Therefore, the comments and consequently replies and further counter comments of different YouTube videos make such videos engrossing, where user engagement rate is influenced by debates. This is because the comments made by different users on YouTube of various videos are expressions of their perceptions. Such perceptions, in turn, help users and channels on YouTube to engage further viewers, thereby making the platform viral in the spread of infotainment.

3.2.3 The relationship between views and Engagement ratio in YouTube videos

As per Munaro *et al.* (2021), a significant way to assess the popularity of YouTube content and viewer interaction is the relationship between views and engagement ratio in YouTube videos. The engagement of viewers with videos and channels is highlighted by YouTube engagement metrics such as likes, dislikes, subscriptions, and interactions.

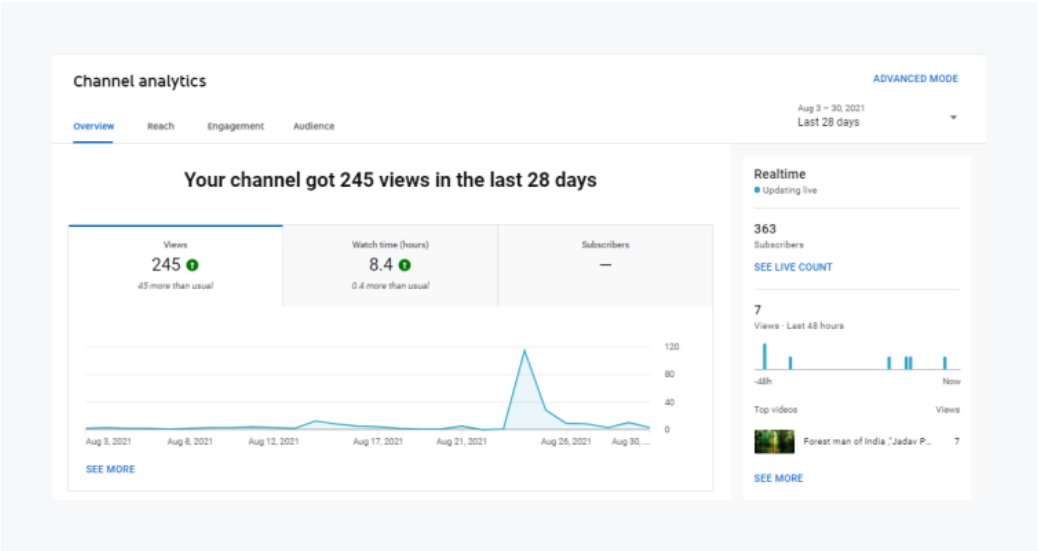


Figure 3: Understanding YouTube Engagement Metrics and How They Are Counted (Source: Kapoor, 2023)

YouTube confirms the other video engagement events on a real-time basis, other than the specific measure of views, as it can be frozen for a specific time being due to the adjustment and verification of YouTube for data accuracy. During the time of viewers' interaction with the YouTube content, the paid advertising views, including skippable in-stream and in-feed video advertisements, also add up to the video views of the original content. The authentic insights on the real-time activity and engagement ratio of the viewers to the particular content are evaluated by close monitoring of the views through YouTube analytics, which can lead content creators to measure the performance of a particular content and enhance engagement strategies.

Furthermore, as per Yang *et al.* (2022), there is a significant relationship between video characteristics, social endorsement cues, and engagement of users with YouTube videos. The research of Yang *et al.* (2022) has analysed a varied range of dependent variables to evaluate engagement aspects of YouTube videos, including view counts of the content, shares, comments, average view duration, average percentage viewed, and subscribers gained. The research found that there is an inverse correlation between the length of content and the views, with a significant coefficient of -.17 ($p < .001$). This indicated that YouTube videos with longer lengths are found to be alluring with fewer viewers. Therefore, the creators need to develop shorter content to attract viewers. Also, the research found that there is a strong positive relationship between likes displayed and views ($B = 2.42$, $p < .001$). The research highlighted strong incident rate rates with these

coefficients, indicating a ten-time increase in likes can enhance the engagement metrics of views to 1020.12%.

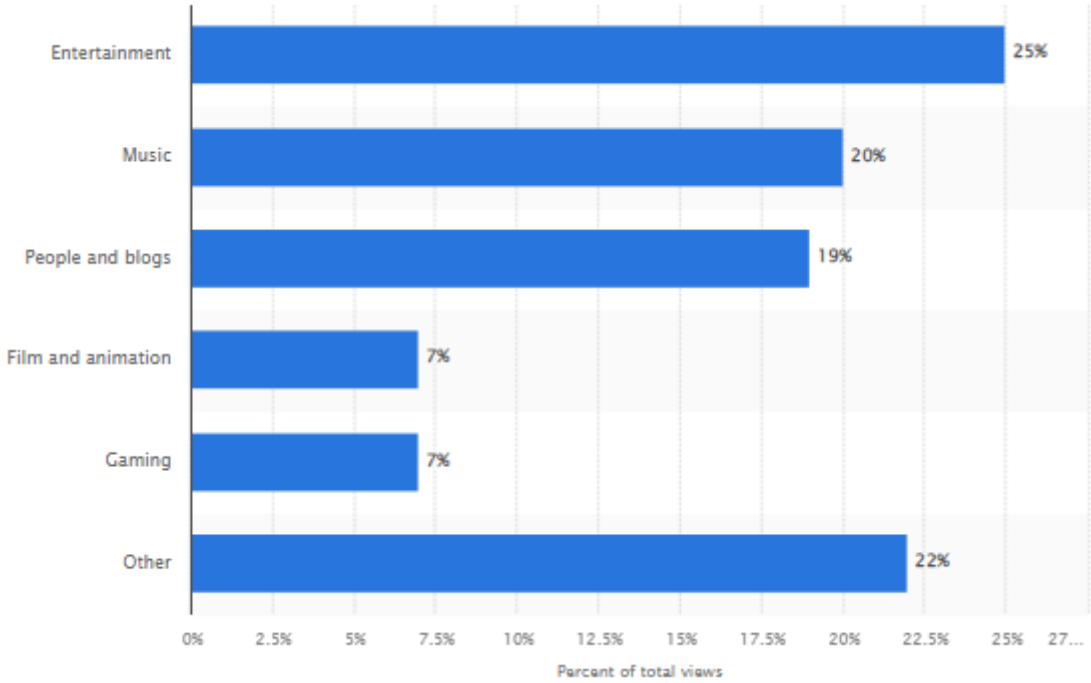


Figure 4: Distribution of total YouTube video views worldwide as of December 2018 by category (Source: Statista, 2019)

The above figure presents the Distribution of total YouTube video views worldwide as of December 2018 by category. As of December 2018, the most popular YouTube content category, entertainment, has attained 25% of the overall views. A high view count of YouTube videos indicates popularity; however, a balanced engagement ratio calculated from the likes, comments, and shares can indicate the active engagement and involvement of the viewers with the content.

Moreover, as per Munaro *et al.* (2021), there is a significant relationship between the views on a YouTube video and the engagement metrics. The research has examined the influence of linguistic attributes on the engagement and interaction of the viewer. The researchers have evaluated the impacts of linguistics styles, category of content, and length of the video on the engagement metrics involving views. The researchers have examined over 11,000 videos and identified that linguistic aspects such as function words, personal pronouns, verbs, and analytical thinking can significantly influence the perspective and reaction of a viewer. The findings of the research indicated that the interaction of the viewers with the content is significantly influenced by the linguistic style, emotional tone, and length of the video, which can significantly impact the viewership count on particular content.

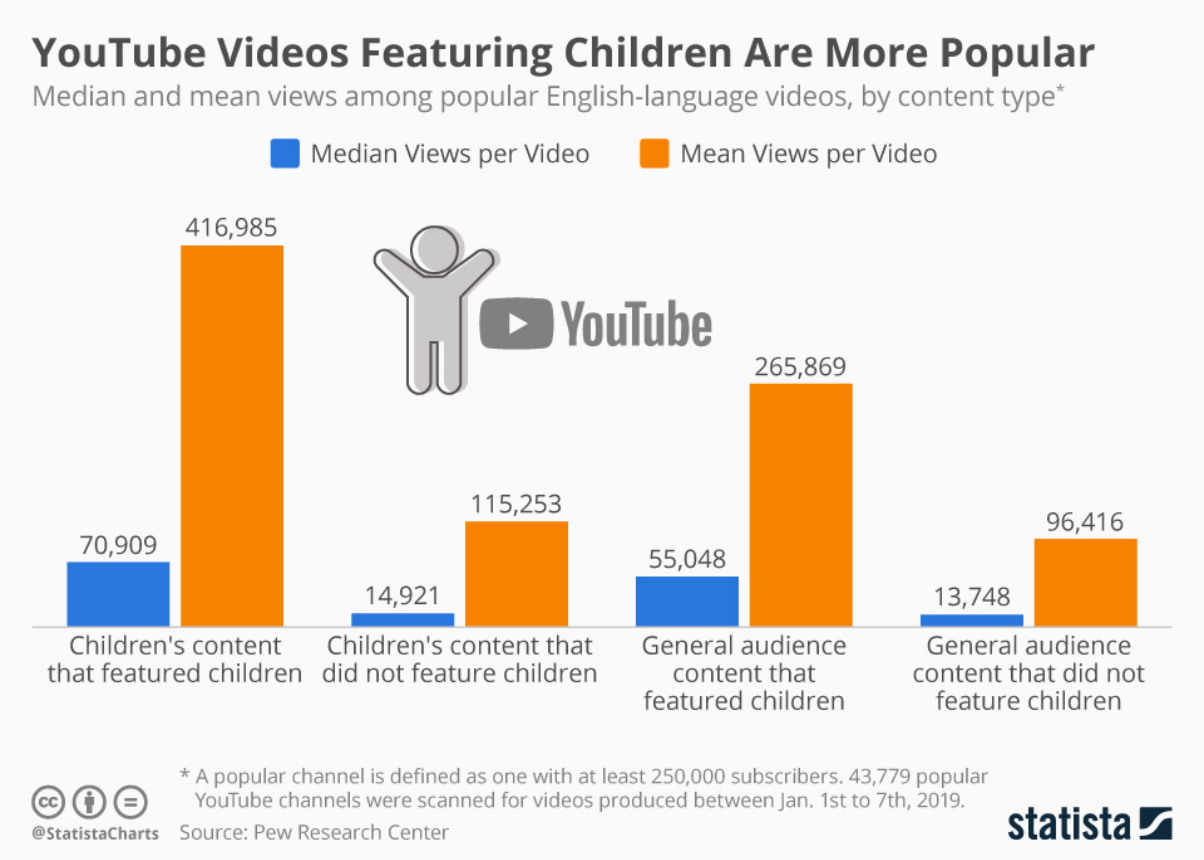


Figure 5: Views and YouTube videos Source: (Statista, 2019)

In the above figure, the popularity of videos on YouTube featuring children is expressed based on the mean and median views. Here, mean views imply the average views per video, which are published in the English language with different contents. As such, the mean views of content designed for children, which features children, has topped the list with 416985 views. Again, general audience content which featured children in their videos on YouTube received 265869 views. By contrast, children's content, which did not feature children, received considerably fewer 115253 views, whereas general audience content not featuring any children in the videos on YouTube performed poorly and received 96416 views. Therefore, there is a strong positive correlation between the number of views and the engagement rate of users, especially children of different videos on YouTube. As a result, the engagement of children and teenagers on YouTube can be effectively engaged with the featuring of individuals from their age group in such videos.

3.2.4 The relationship between duration and Engagement ratio in YouTube videos

As per the view of Szmuda *et al.* (2020), there is a significant relationship between the duration of a YouTube video and the engagement ratios, depending on user behaviour and preferences. The researchers Szmuda et al. (2020) have examined two distinct datasets of random YouTube videos and other datasets involving the videos watched by respondents through Chrome extensions to identify the factors impacting viewer engagement. The findings of the study indicated that there is a strong and positive association between the video view duration and the engagement metrics, including view counts, likes per view, and negative words in the comments section of the content. A significant predictive capacity is also facilitated by these metrics over the duration of engagement for the viewers.

Furthermore, as per Saurabh and Gautam (2019), there is a significant correlation between video duration and the engagement of viewers with YouTube content. The research has identified with a Scatter-Pot analysis that most popular videos on YouTube typically range from 10 to 15 minutes. The research has highlighted that shorter videos can lack the explanation of a particular subject, and many viewers can lose interest in watching a video if the video is lengthy, and only 30-50% of a long YouTube video is majorly watched by the viewers. Therefore, the research recommended that content creators on YouTube create more medium-length content, and they should make engaging videos with the potential to hold viewers' interest from start to finish. These factors aid in comprehending the significant role of video duration in improving viewer engagement and user retention on the YouTube platform.

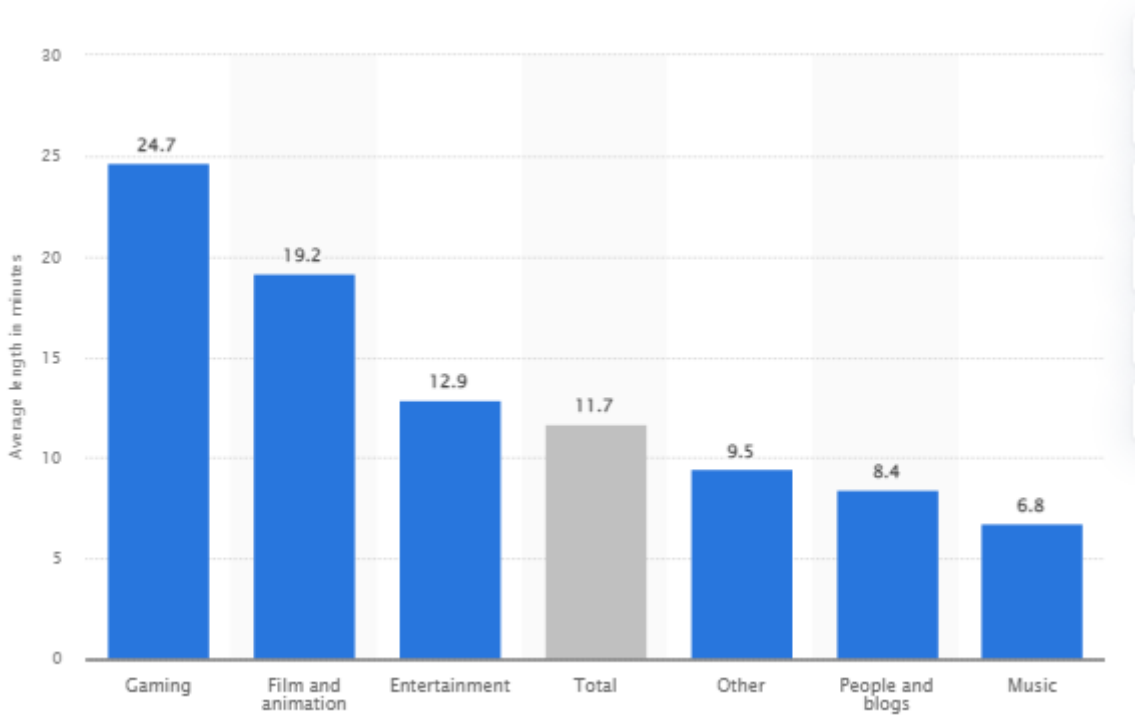


Figure 6: Average YouTube video length as of December 2018 by category (Source: Ceci, 2021)

The above figure depicts the average length of YouTube videos based on category as of December 2018. Gaming videos can have the majority of the length among other types of content. An average length of gaming video can reach up to 24.7 minutes. Furthermore, as per Jothi (2019), a significant factor in comprehending user preferences and video popularity on YouTube is assessing the relationship between the duration of YouTube videos and the engagement ratios of the viewers. The proportion of likes, comments, and shares in relation to the total number of views on a particular content is referred to as the engagement ratio in the research.

The research demonstrated that the engagement of viewers is significantly impacted by the duration of YouTube content. Shorter videos are more engine to the viewers, as seen from statistics. Videos less than 2 minutes in duration often have a higher engagement ratio of more than 10%, as compared to lengthy videos more than 20 minutes long, which exhibit only 2% of the engagement ratio. However, the preferences of viewers to consume concise and focused content to capture their engagement can impact this trend.

Moreover, different purposes can be served by videos of different durations, such as shorter videos, which can be made for quick entertainment, tutorials, or information related to specific things. At the same time, lengthy videos can be made to facilitate in-depth discussions, blogs, and educational content.

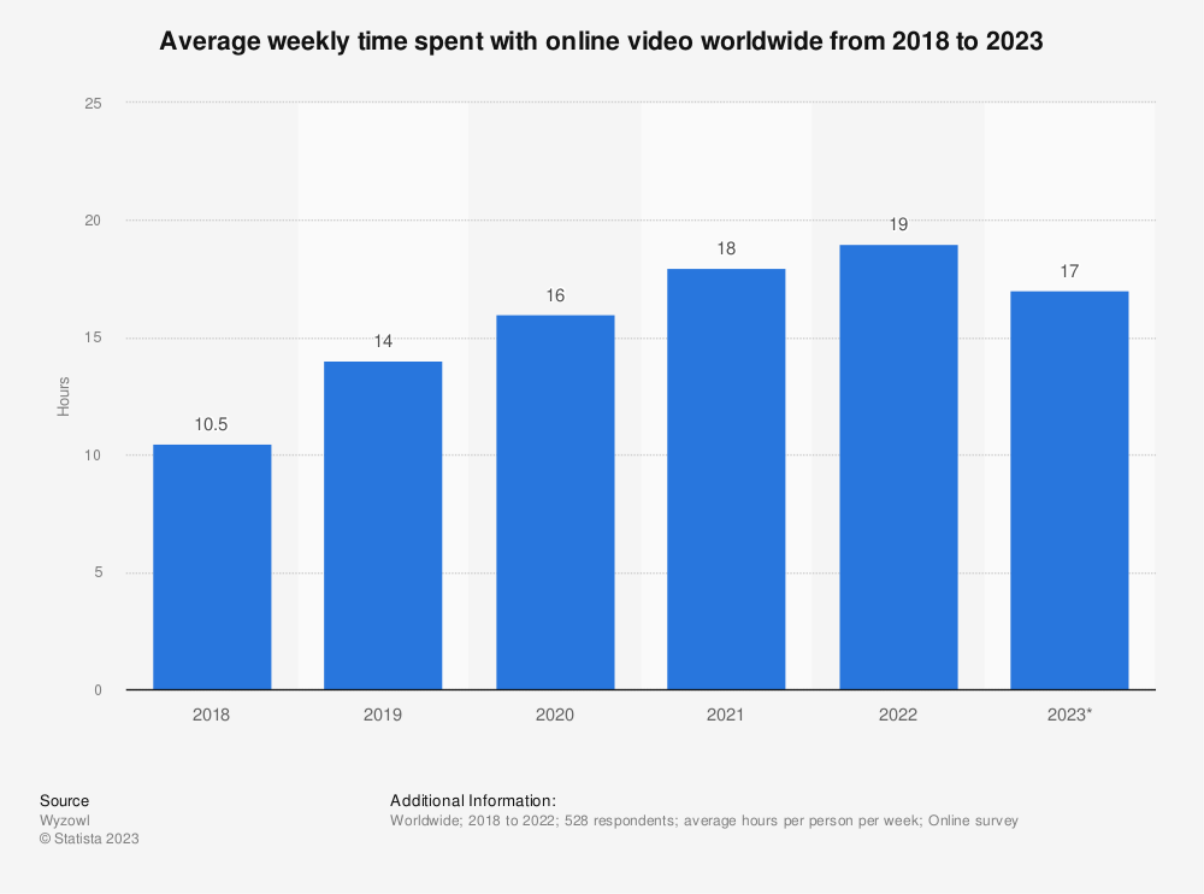


Figure 7: Weekly time spent on online videos Source: (Statista, 2023)

In the above figure, the average time spent per week watching online videos worldwide for all users is given. This average time of viewing online videos on major platforms like YouTube has been increasing steadily from 2018 onwards, as evident from the above infographic. In 2019, users watched an average of 14 hours of online videos per week, which increased to 16 and 18 hours per week on average in 2020 and 2021, respectively. However, the highest was 19 hours spent on average per week viewing online videos in 2022, which declined to 17 hours per week in 2023. Therefore, the general trend in the increase in the number of mean hours per week of viewing online videos on YouTube shows the preference of users to watch lengthy videos. As a result, increasing the duration of YouTube videos with greater depth and details can help channels in engaging a large number of users. This is because of the preference for viewing long-duration videos by YouTube users.

3.2.5 The relationship between the Time of uploading and Engagement ratio in YouTube videos

According to Munger and Phillips (2022), the perspectives of demand and supply can help in understanding the dynamics of consumers and producers of right-wing videos and the associated engagement among them. In this regard, these authors have discussed several technological affordances which can help in explaining the demand, supply, collection of videos on YouTube and associated engagement. Among these, the time of uploading of videos by content creators on YouTube is essential in several respects. Gaming content creators like Scape211 and Omniverse choose to create and upload gaming content involving special events during the point of release of such events in games. This helps them to target a specific target audience for whom such videos are relevant, thereby improving engagement. However, the results and analysis of Munger and Phillips (2022) have shown that viewership of far-right videos skyrocketed during 2017. The time of uploading videos, which influenced the perceptions of people, was found to be aligned with the incidence of political elections. This is because social media platforms like YouTube are the most popular destinations for people to spend their leisure time in. Therefore, the time of uploading videos is necessary to be aligned with associated situations and processes which are relevant to the viewers.

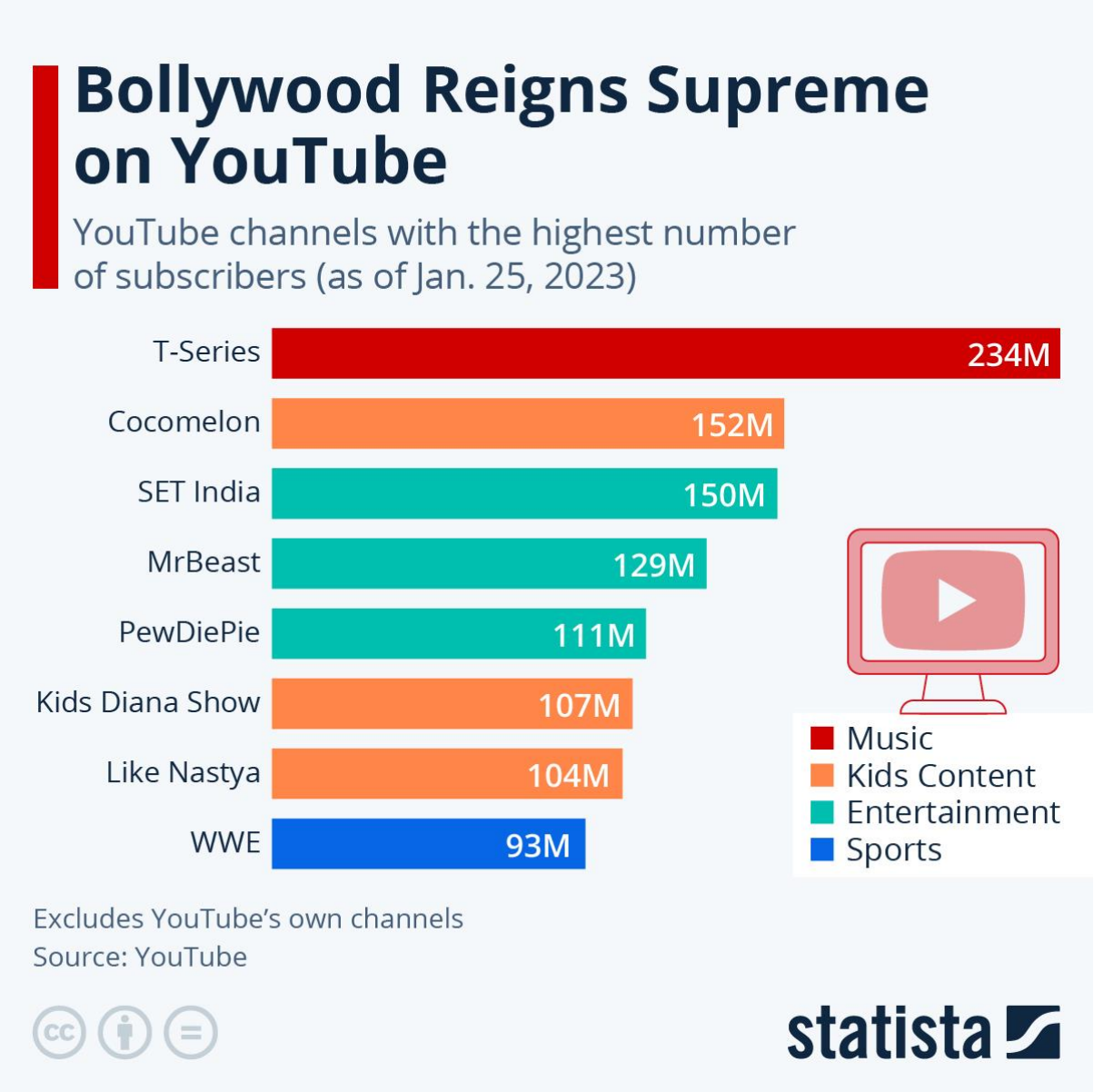


Figure 8: Dominance of Bollywood among YouTube channels Source: (Statista, 2023)

In the above figure, the YouTube channel T-series, which broadcasts mostly music videos of Bollywood music songs and movie shorts, is comparatively at the highest ranking among other channels. The managers, content creators and marketers of the T-series channel use time uploading videos as an effective tool to engage with its humongous subscriber base. The firm uses the release dates of new movies to release the songs of such movies on its YouTube channel. Therefore, the time of uploading videos in this regard helps fans of such movies to be engrossed and engaged with their channel. On the other hand, the producers and filmmakers of such movies also benefited because the T-series channel, with the release of their films, helps in marketing the film among large numbers of users worldwide, thereby promoting sales. Again, the songs of popular Bollywood content creators like Tulsi Kumar, Melody Badshah, Arijit Singh, Akasa, Arman Mallick, Astha Gill and Arjun Kanungo

are also released by the T-series channel exclusively in a manner which helps engage audiences. Therefore, the identification of the different occasions helps in identifying the opportunities for different types of Bollywood content based on the vast number of Indian festivals, which helps in engaging the Indian audience through T-series on YouTube.

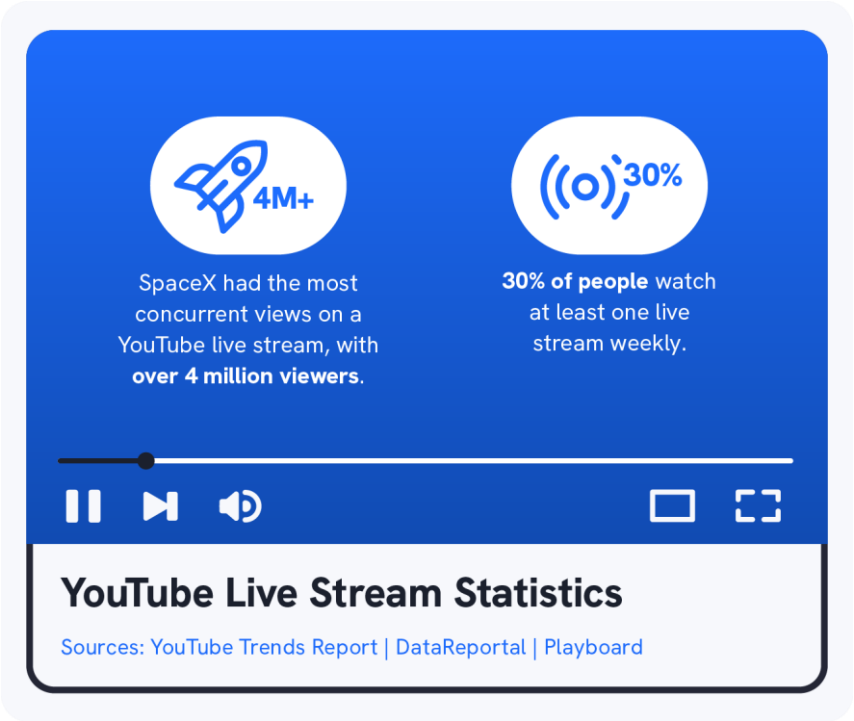


Figure 9: Live Stream popularity of YouTube Source: (Uscreen, 2022)

In the above figure, the popularity of YouTube in engaging users with the live stream feature is given. Here, SpaceX was found to have obtained the majority of concurrent views, attracting more than 4 million viewers as per the trends report of YouTube. Again, from a global perspective, 30% of people accessing YouTube were found to have watched at least one live stream per week. Therefore, the above infographic provides an outline of the preference of the audience towards the live streaming of online videos on YouTube. As a result, the engagement ratio of users on YouTube can be manipulated even further with the help of live streams and the time of uploading videos. This is because live streams on YouTube often include live posting of comments, which is more attractive than fully published videos for a considerable section of the global population. Such populations can thus be engaged effectively with the time of uploading videos through live streams.

3.2.6 The relationship between Category/Genre and Engagement ratio in YouTube videos

According to Chen *et al.* (2019), the measurement of user engagement on YouTube through interactive tasks facilitates innumerable applications for the optimisation of user experience, which ranges from e-learning to gaming. The authors in that paper have presented a non-intrusive estimator of engagement by leveraging facial recordings of users in naturalistic conditions during actual gameplay. The authors have also used a Face Engage dataset with more than 700 realistic gaming videos contributed by users. Secondly, the authors of Chen *et al.* (2019) have used a Face Engage system to capture relevant features of gamers from the front-facing recordings so that task engagement can be inferred. The authors have also implemented two Face Engage pipelines, one for user facial features and the other for id deep learning-enabled estimator. In this regard, the authors have found that user facial motions are engagement indicative. In spite of the challenging nature of videos on YouTube, Face Engage attained an accuracy of 83.8% along with leave-one-user-out precision of 79.9%. They were both considered superior to face motion-based value. Therefore, it is evident that the engagement ratio in YouTube videos as a dependent variable is positively correlated with the entertainment category and is higher for gaming YouTubers like Scape211 and Unstoppables than educational ones.

However, Bernad-Mechó and Valeiras-Jurado (2023) have addressed the multimodal engagement strategies, especially in the dissemination of scientific concepts and consider the case study of science videos on YouTube and TED talks. In this regard, the growing interest in the dissemination of science offered new opportunities for the communication of scientific concepts to different audiences. However, using YouTube as a platform for science dissemination has challenges of adaptation to audiences who do not share similar academic backgrounds. In this context, the authors have aimed to unveil the manner in which engagement strategies in science dissemination formats of online videos and face-to-face talks can be realised with the aid of complex multimodal ensembles and highlighting the differences among them. The authors have used the recordings that were annotated with the help of the Multimodal Video analysis software. The results and findings of that paper showed that both filmic and embodied modes were paramount in realising strategies to keep audiences on YouTube engaged. The engagement strategies showed that the emphasis of TEDx Averages 17.26% whereas that of PBS average is 58.76%. By contrast, attention-getting was 8.06% for the TEDx average and 52.30% for the PBS average. Dialogic involvement was 18.60% for the TEDx average and 3.33% for the PBS average. Consequently, the engagement strategies in the control of

responses were 27.7% on the TEDx average and 46.19% on the PBS average. Humour was 0.68% on the TEDx average and 11.07% on the PBS average, as per Bernad-Mechó and Valeiras-Jurado (2023). Therefore, category or genre influences the engagement ratio of YouTube videos depending on the popularity of such genres as well as that of the YouTubers themselves.

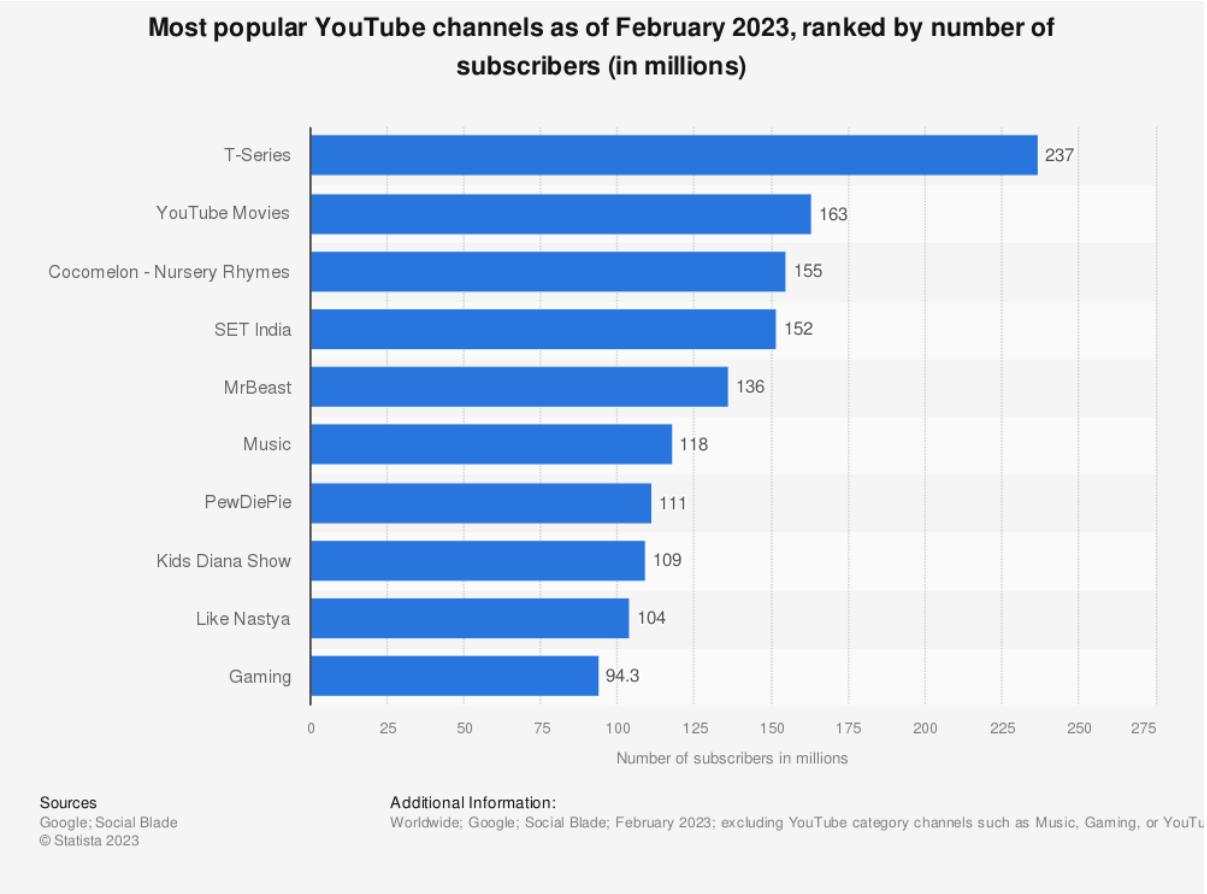


Figure 10: Popularity of different YouTube channels Source: (Statista, 2023)

In the above figure, the most popular channels on YouTube were ranked according to the number of subscribers. Here, T-series has topped the list with 237 million subscribers. This was followed by YouTube movies and Cocomelon Nursery Rhymes, with 163 million and 155 million subscribers, respectively. By contrast, SET India, MrBeast, Music and PewDiePie recorded 152, 136, 118 and 111 million subscribers, respectively. Therefore, the engaging videos created by such YouTube content creators help such channels grow their number of subscribers over time.

However, Munaro *et al.* (2021) have questioned whether to engage or not engage by identifying the genres and features of video content on YouTube which affect the engagement of digital consumers.

These authors presented and tested a model for the popularity of YouTube videos based on engagement actions such as views, comments, likes and dislikes. The language elements of subjectivity, emotional valence and linguistic style in videos of different genres have influenced the engagement of subscribers and online video popularity on YouTube. Therefore, medium and long-length videos posted during non-business weekdays and hours with subjective language styles and less active events received more engagement in the form of likes, comments and views.

Figure 11: Content choice of YouTube users Source: (Uscreen, 2022)

In the above figure, a brief outline of the choice of genre and content chosen by users on YouTube is given. Here, the data obtained from the Statista database shows that 694,000 video hours were streamed every minute on YouTube. Additionally, a majority share of 70% of viewers have revealed that their mood shapes their searches and, as such, the choice of content viewed by them. Therefore, in the regression between the dependent variable engagement ratio and independent variable genre/category, the mood of users acts as an instrumental variable which secretly influences the relationship of the former two variables from the shadows.

3.2.7 The relationship between Tags and Engagement ratio in YouTube videos

According to Iqbal Khan and Ahmad (2022), the use of hashtags is an incredible tool which is interrelated across different social media channels and aids digital content creators in gaining attention. The content and media posted by micro-celebrities on different social media platforms like YouTube and Twitter help in the online engagement of the target audience. The authors in that paper have found that the type, language, length, mentions, links, and hashtags have a significant relationship with the online engagement of social media platforms. In this regard, the influence of hours and day of the week in which videos are posted also have a significant influence. The usage of hashtags by popular YouTube channels like T-series is often designed to post links and associate eminent personalities like Akshay Kumar, Kriti Sanon and others regarding songs and Bollywood content featuring them. Therefore, the huge fan following of such Bollywood actors and personalities can be engaged by not only the posting of Bollywood content but also by creating links to their official YouTube channels on the individual videos themselves. By contrast, the use of hashtags can also be used for non-entertainment purposes as well.

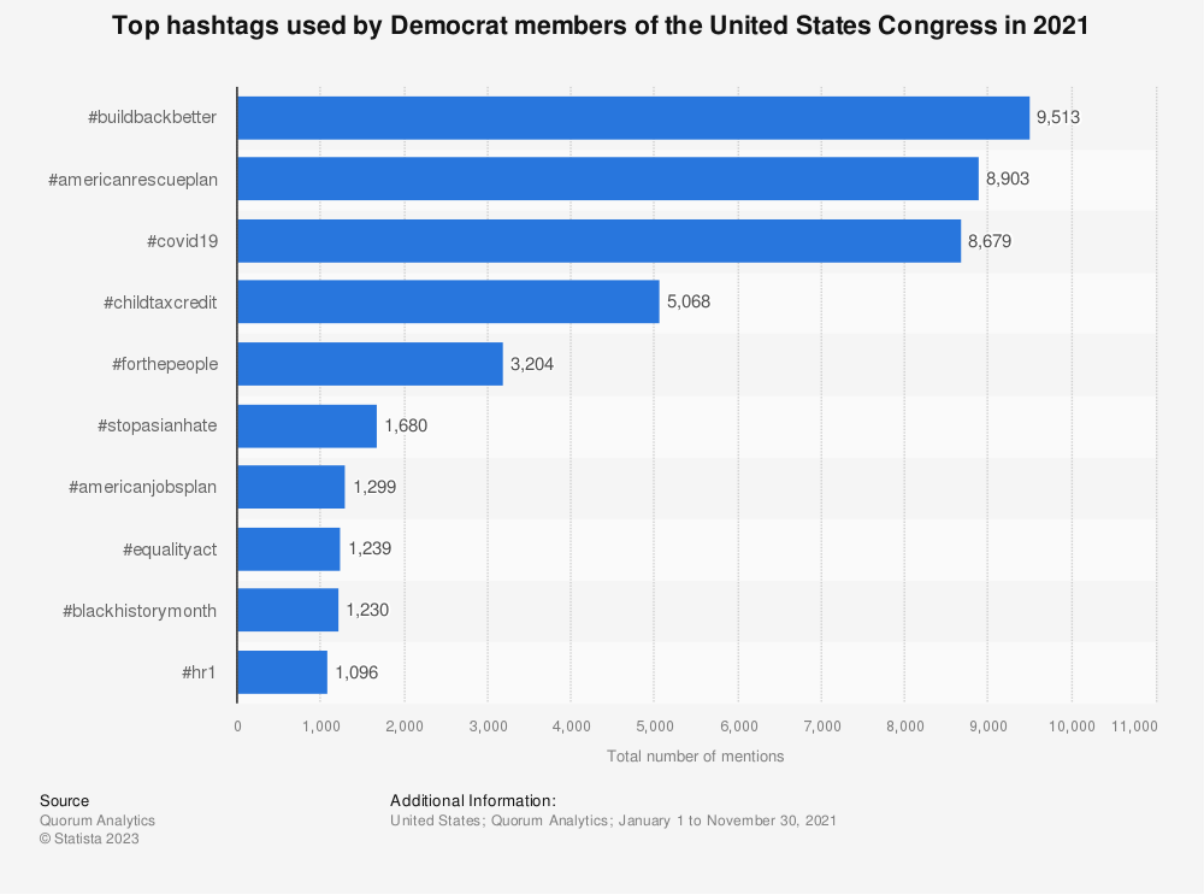


Figure 12: Popular hashtags used by democrats in US Congress Source: (Statista, 2022)

In the above figure, the most popular hashtags which were used by members of the Democratic Party in the United States Congress in 2021 are given. The most popular among them was #buildbackbetter, which was used 9513 times on social media websites like YouTube. This was followed by #americanrescueplan and #covid19 for 8903 and 8679 times, respectively. By contrast, #childtaxcredit and #stopasianhate were used 5068 and 1680 times, respectively. Therefore, the differences in the number of hashtags used in different social media platforms like YouTube are dependent on the gravity of issues and, thus, their relevance to larger numbers of people who participate in social media. As such, the use of hashtags also helps politically conscious people to engage large numbers of users on YouTube.

Furthermore, Manca (2021) has addressed the manner in which social media is used for new memory ecologies through the use of hashtags. The constant use of hashtags on different posts on YouTube and other social media websites creates links to previous incidents like the holocaust. Therefore, the usage of hashtags engages users on YouTube by acting as reminders for past incidents and issues which have repercussions in contemporary times.

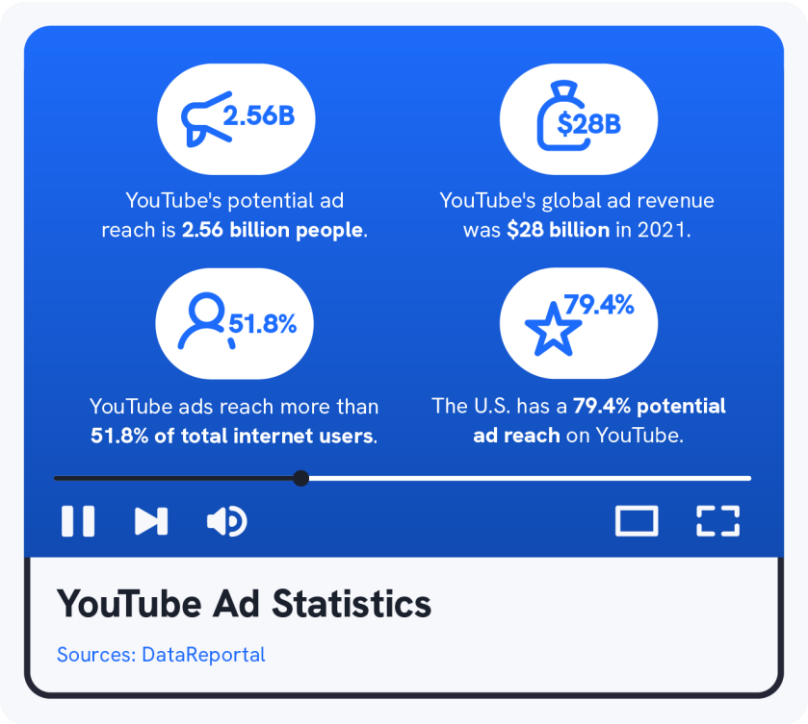


Figure 13: Viral nature of YouTube Source: (Uscreen, 2022)

In the above figure, the viral nature of YouTube videos is given as per the potential revenue generated by commercial advertisements. Here, the above infographic shows that the potential ad reach of YouTube is 2.56 billion people, whereas the global revenue from advertisements was $28 billion in 2021. Again, the reach of YouTube advertisements was more than 51.8% of the global internet users, out of which the US had 79.4% of potential advertisement reach on YouTube. This commercial reach of advertisements is possible with the help of hashtags and linking of different videos with channels of potentially creditable personalities and entities who endorse such commercial products. Therefore, hashtags also help in engaging large sections of operations accessing YouTube, thereby generating revenue for businesses.

3.2.8 Critical Comparison and contrast of the Hypothesis to be used for each iv to dv. Via heat map for assigning the weight

The utilisation of an overall heat map can specifically take place for the increase in the piece of evaluation regarding the existence of the different factors and the impact of the factors upon the considered research phenomena. The type of degrees that are presented through the utilisation of a heat map can easily depict the intensity of the relationship that exists between the overall research

phenomena and the types of factors, resulting in the provision of increased insight (Kant and Yadele, 2023). In the case of the analysis between the engagement score and the factors such as the duration, likes, comments and views, the time of uploading followed by the genre and tags has led to the development of an idea regarding the effect which the factors have on the YouTube video engagement process. The intensity of this relationship can be identified through the utilisation of a heat map.

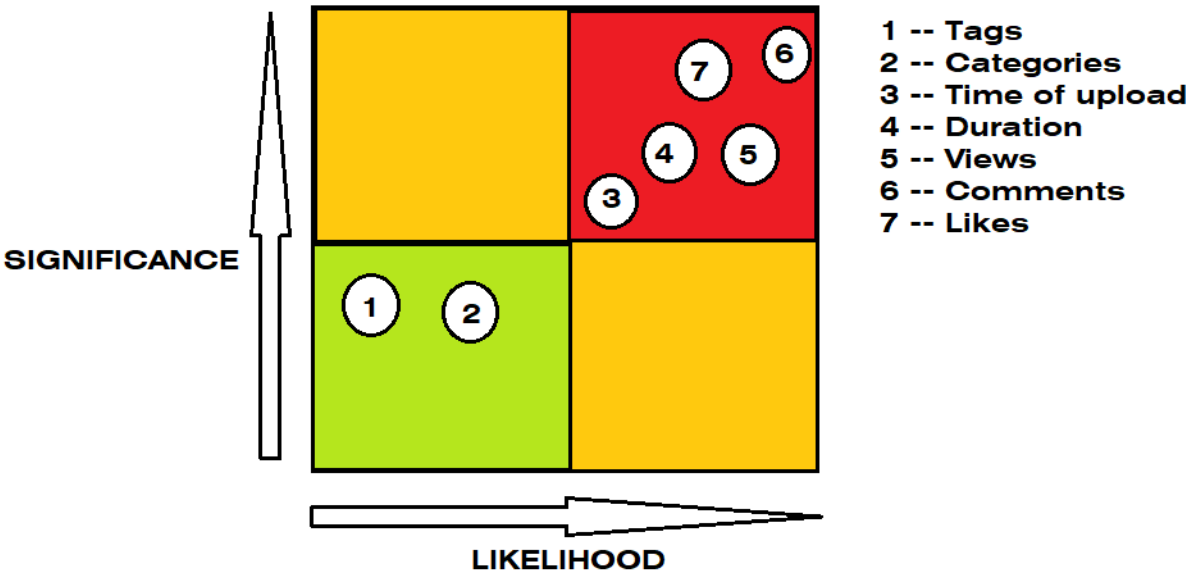


Figure 14: Heat Map Source: (self-designed)

In the case of the relationship between the likes obtained by the YouTube videos and the engagement rate that is connected to the same, it was first observed through the research of Levinson, Greenfield, and Signorelli (2020) that the number of likes achieved carries a significant relation with the type of engagement that is observed leading to the success of a Youtuber. It has been further presented by Salah *et al.* (2022) that the rise the downfall in engagement towards YouTube videos for frequent users of the application can be particularly calculated by the division of the parameters, such as the likes, comments and dislikes, from the number of views that are received. As per the explosions that have been made by Salah *et al.* (2022), based on the type of content that is posted by the professionals, different households or business processes operating in the adjacent areas in the locality can identify the number of benefits or advancements which can take place within the decision making or strategy development processes ultimately resulting in the betterment of the lifestyles or productivity.

However, according to Mulders and Wendt (2022), in comparison to the overall effect that is achieved through the analysis of the relationship between likes and the engagement ratio, the connection that is established between the types of comments posted and the engagement of the views towards the video carries greater importance and promises further results. The inner emotions or the feelings that have been developed regarding the type of understanding that the views achieved from the content or the greater expectations that they have significantly developed after watching the video can be easily evaluated using the comments that have been posted, according to Mulders and Wendt (2022). A higher influence of the comments upon the type of engagement has been detected from the research of Amarasekara and Grant (2019). The authors have been able to state that with the presence of greater YouTube comments, the current state of the sensitive areas focused upon within the content created can be known, ultimately allowing the benefit of further analysis for the content creators, resulting in the improvement of future content.

It has been earlier identified through the research of Minaro *et al.* (2021) that the number of views that have been collected and the overall engagement generated are related to each other as the effects on the user upon viewing content include the changes in the personal pronouns, word utilisation and analytical thinking. Kuitunen (2019) has further highlighted that identification of the type of improvement which is expected from a YouTube channel must consist of the analysis based on the combination of both the views and the comments. Utilising any one of the attributes alone cannot provide information regarding the engagement ratio, as mentioned by Kuitunen (2019). Further identification of the importance of the video duration has been analysed in the research of Gupta, Polla Ravi and Wang (2023). Similar to the research that has been presented by Saurabh and Gautam (2019), the authors have been able to identify that if the duration of the video is mainly between 0 to 40 seconds, the largest level of ineffectiveness is measured in terms of impact creation on the target audience. The minimum length of the video, therefore, has to be between 10 to 15 minutes for the clear conveying of the targeted messages to the different users. Hence, on this basis, the authors have ruled out that the current introduction of the video format, such as the YouTube shorts, provides the least effectiveness.

In the case of the relation between the time of uploading the video and the corresponding engagement ratio, the views provided by Munger and Phillips (2022) have focused on one hand towards the political or social environment that is experienced by the citizens within a specific nation or region governance the time at which a particular type of video gets uploaded. On the other hand, Le *et al.* (2022) have been able to combine the factor of the overall security of video uploading through the utilisation of the figures based on the time of uploading. The combination of the data sets, including the training and testing data sets and the division of those data sets with the time of

uploading the video, helps in the management of the uploading security through the extraction of the video features. Hence, equal importance has been witnessed in the case of the uploading time as a factor of engagement with the videos.

In the case of the other factors, such as the categories of the videos and the tags that have been attached when uploading, the research process initiated by Manca (2021) has been able to identify the change that is brought within the videos, especially in terms of visibility by the utilisation of hashtags. Further research carried out by Doyle, Su and Kunkel (2022) has, however, been able to identify the negative relationship between the hashtags used and the online engagement that is noticed. This occurrence mainly takes place as the type of quality which the video demonstrates provides greater engagement, as per the authors, in comparison to the hashtags. The category, hence, based on the quality of the videos, such as the standard definition videos, the high definition and the ultra-high definition videos, induce greater engagement.

3.2.9 The connection between the machine learning regression model and YouTube video analysis

The machine learning regression model can be critically connected with the YouTube video analysis in various ways, such as the worship prediction engagement analysis or video length optimisation. According to the study of Zhang *et al.* (2020), the regression model can be critically used to predict the number of weavers a YouTube video can likely receive by analysing the various features of the video description, the tags, historical data, and the video title. The viewership prediction by using the regression model can be helpful for the content creators to understand the future performance of video. This understanding can also help them to plan their creation strategy accordingly. Similarly, this can also consider engagement analysis to predict user engagement metrics, including comments, likes, or shares. This analysis collects data on video attitude and chooses the feature or input variable that can impact viewer engagement (Lim *et al.,* 2020). The regression technique aims to find relationships between the selected feature and engagement metrics like advanced models like random fosters or gradients for boosting to build up predictive models. On the other hand, as per the view of Wang *et al.* (2020), regression models are also beneficial for video-like optimisation as it collects data from videos regarding the view retention rate and their lengths to measure view retention by analysing the drop-off point in the video. Similarly, this identifies the ideal with viewer length that emphasises maximum view retention and what time to analyse the relationship between

video length and viewer retention. Furthermore, segmentation can be considered by the regression model to identify different views, habits, and preferences. For example, different views may have different tolerance for video length, and by understanding the preferences of audiences, the content creators can create content accordingly to get maximum reach of audiences.

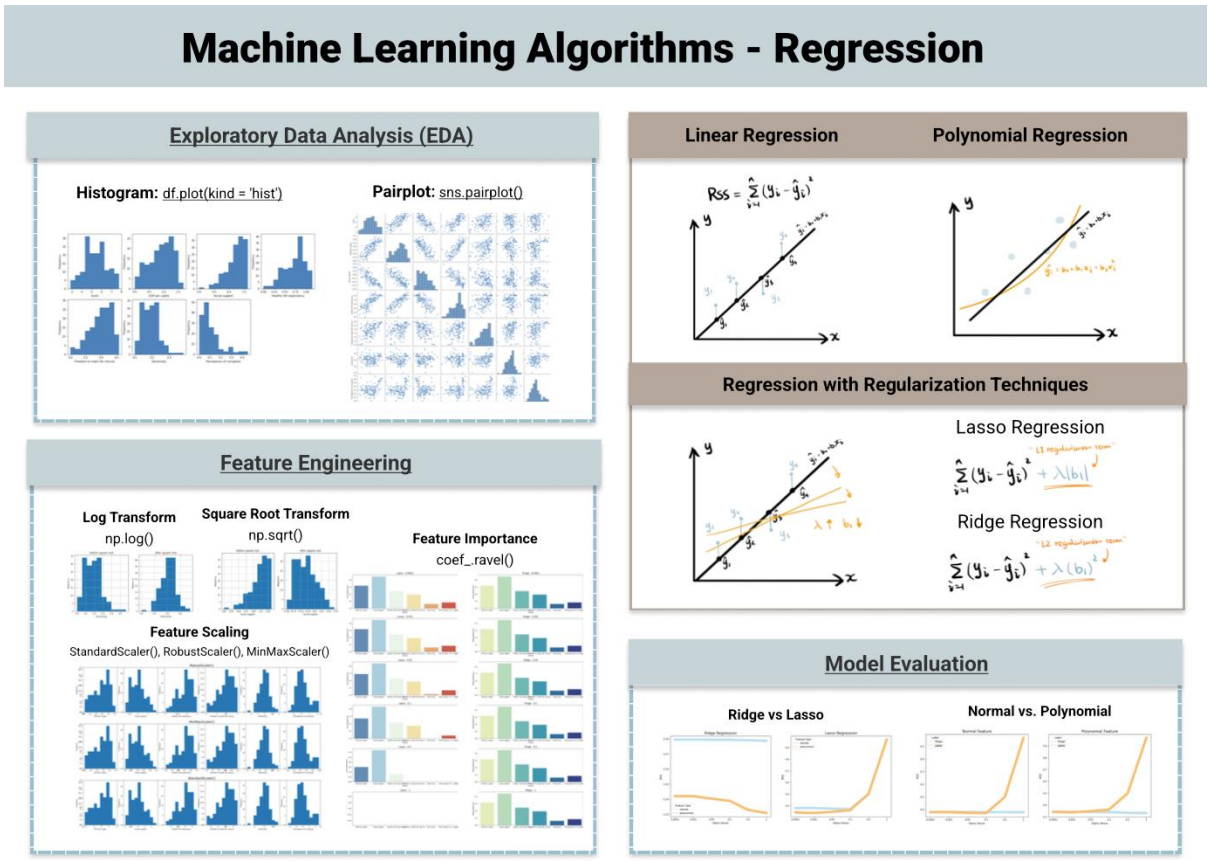


Figure 15: Different types of Machine learning regression model) (Source: Gong, 2022)

The above-mentioned figure shows the four types of machine learning algorithms or Regression for classification. As per Figure 15, there are four types of regression models, including Linear regression, Lasso regression, Ridge regression, and Polynomial regression. The linear regression identifies optimal linear relationships between independent variables and dependent variables. According to Gong (2022), linear regression uses MSE as a cost function to minimise the sum of squared differences between actual values and predicted, does not include regulations, and typically has a low model complexity in case the relationship between variables is nonlinear. Similarly, the Lasso Regression aims to address a Linear relationship between variables while emphasising sparsity in the model between penalising sum coefficients to become exactly zero, which provides the advantage of

automatic feature selection and helps to prevent overfitting by reducing the complexity of the model. Moreover, the Ridge regression reduces multicollinearity and helps prevent overfitting while operating smoothly when all features are critically applicable. Lastly, the relationship between the independent and dependent variables is non-linear. Polynomial regression is used in cases of curved or oscillatory relationships to capture non-linear relationships in the data.

3.3 Literature Gap

The overall literature gap that can be considered in the case of the contribution of the factors such as the views, likes, comments, video durations and time of uploads, etc., is specifically related to the further analysis that has not been carried out within the studies that have been completed by the authors leading to the rise in the number of limitations of exploration. Some of the research works have provided information regarding the engagement of customers through YouTube videos, specifically in terms of education and knowledge extraction. However, the primary question that still remains is the amount of effect on the perception of the people within the entire population, including those who have educational requirements, with respect to parameters such as likes, comments, video durations, time of uploads, etc. The lack of research regarding the improvement of YouTube channels based on the types of videos which are posted by the concerned influencers has resulted in severe limitations in the literature. Other research works have also been able to focus towards the importance of the engagement ratio for the improvement of the social media presence of different bodies, such as the influencers or the companies having a heavier presence online. Further focus has also been provided towards the measurement of the type of factors which can identify user engagement and modify the same through the greater concentration and involvement of the stakeholders (Saura, Palacios-Marqués and Iturricha-Fernández, 2021). However, critical focus on the perspective of the YouTubers for the measurement of the metrics has not been taken into account leaving behind the greater question of the different strategies that could have been developed only through the perspective of a single person who wants to build a brand that will be highly appreciated by viewers. The lack of research regarding the same has resulted in greater limitations that require further attention.

Other research works have further focused towards the importance of the increase in the views, likes and comments within YouTube videos for the influencers operating within the competitive environment. However, such research processes have mainly focused on the further impacts which can take place on the functioning of the companies and the types of agendas that they maintain in terms of marketing the products (Agarwal and Damle, 2020). Company perspective is highly taken into account within the investigations, which has resulted in the negligence towards the type of activities that are undertaken by the influencers themselves for increasing the weightage of their channels in the competition. Hence, this research takes into account such limitations and attempts to fulfil them respectively through the usage of suitable methodologies.

3.4 Chapter Summary

This chapter takes into account the relationships that are present between the overall factor of engagement of the audience towards YouTube videos and the other parameters such as the durations of the videos followed by the types of categories, the time of uploading, the likes, comments and views registered. A critical comparison of the intensity of the relationships present between the variables, followed by an exploration of the current gaps in the literature, has allowed this chapter to increase the amount of information regarding the views and opinions that have been significantly presented by the previous authors. Through the inclusion of such aspects in the form of themes, the literature review chapter has been able to provide a greater boost to the process of the fulfilment of research aims and objectives.

4.0 Significance of the Study

The importance of this research project extends beyond the walls of academia, having an impact on numerous stakeholders throughout the digital content ecosystem. This study promises to give content creators a thorough understanding of the underlying factors that influence engagement, providing them with useful information to help them optimise their content strategies and build deeper relationships with their audiences.

The results of this study can be used by advertisers and brands to guide their selection of partners, venues, and ad targeting tactics. In order to create successful advertising campaigns on YouTube, it is essential to understand the elements that draw in and engage audiences.

From a scholarly standpoint, this study adds to the body of knowledge about online video consumption habits that is constantly growing. It provides researchers with a complex lens through which to analyse the changing dynamics of online media and sheds light on the intricate interactions between creators, content, and viewers in the digital age.

This research paper aims to shed light on the various elements that contribute to the allure and enduring appeal of Casey Neistat's YouTube channel. We aim to unravel the complexities of what constitutes engagement in the digital age and, in doing so, further enhance our understanding of the wider landscape of online content creation and consumption through empirical analysis, predictive modelling, and in-depth exploration.

5.0 Methodology

This study aims to investigate the engagement patterns observed in the YouTube videos of Casey Neistat, a prominent content creator. By employing a rigorous methodological approach, we seek to uncover the factors that contribute to the high levels of engagement shown in Neistat's videos. The methodology employed in this study integrates various techniques like data collecting, feature engineering, regression modelling, and data visualization to analyse and understand the complex patterns of engagement observed within Casey Neistat's YouTube channel. Our project seeks to make a valuable contribution to the wider discussion on YouTube content consumption behaviour, content creation strategies, and the evolving dynamics between creators and their audiences in the digital era. This will be achieved through the development of an engagement score, the prediction of engagement factors, and the visualization of significant relationships.

5.1 The Process of Data Collection in Analysing YouTube Videos

The phase of data collecting assumes a crucial significance as it serves as the primary means of acquiring the necessary information for our inquiry.

5.1.1 The YouTube Video API: An Access Point to Data

In order to utilise the extensive range of knowledge included inside Casey Neistat's YouTube channel, we employed the YouTube Video API. The provided application programming interface served as a direct access point to a comprehensive dataset that included video metrics, metadata, and engagement information. The utilisation of the API facilitated the retrieval of data in a highly efficient manner, enabling us to analyse a wide range of films from the channel and guaranteeing that our research is grounded on a thorough sample.

5.1.2 Points of Interest in the Data: Collected Information

A comprehensive range of data points crucial for our investigation was collected from the YouTube Video API. The aforementioned items encompassed:

- Video metrics encompass essential data points, including view counts, positive and negative reactions in the form of likes and dislikes, as well as comments, which collectively offer valuable insights into viewer engagement and audience responses.
- The video metadata encompasses several elements, including the video title, description, and tags. These components provide valuable contextual and thematic information, enabling us to effectively categorise the material.
- Video Duration: The duration of each video was measured, enabling us to incorporate this element into the calculations of engagement scores.
- The inclusion of the published date and time of each video enabled us to do a temporal analysis, allowing us to identify and examine trends and patterns.
- The categorization of videos into distinct content genres or topics facilitates the study of material and enables category-specific analysis.
- The engagement score was calculated by using the acquired data, resulting in a measure that indicates audience involvement for each video.

5.1.3 Ethical Considerations

Recognising the ethical implications associated with data acquisition from a publicly accessible site such as YouTube is of utmost importance. We have complied with the Terms of Service and API usage standards set out by YouTube, demonstrating adherence to privacy and copyright regulations, and only using publicly accessible data.

In conclusion, **t**he YouTube Video API was utilized to gather an extensive dataset comprising various metrics including views, likes, comments, video duration, video genre, publication date, tags, and other relevant information from Casey Neistat's channel.

5.2 Data Cleaning:

The dataset's integrity was upheld through meticulous data cleaning procedures, which involved addressing missing values, eliminating duplicates, and validating data accuracy.

5.3 Feature Engineering

The process of feature engineering holds significant importance inside our research project, which centers around the analysis of YouTube videos pertaining to Casey Neistat's channel. This procedure entails the creation of novel characteristics or the alteration of preexisting ones in order to derive beneficial insights from our dataset. In this discussion, we explore the diverse feature engineering strategies employed to enhance the optimization of our analysis.

I. The duration of the video in seconds.

The process of data transformation involved converting the duration of videos from the YouTube format, denoted as 'PT M S' (e.g., 'PT10M45S' representing a duration of 10 minutes and 45 seconds), into a conventional format of seconds. This modification ensured consistency and facilitated the process of comparing videos.

II. Management of Tags:

The creation of tag columns was implemented in order to effectively utilize the significance of video tags. As part of this approach, a novel binary feature was created. Videos lacking tags were assigned a numerical value of zero, whilst videos possessing tags were assigned a numerical value of one. The utilization of binary categorization facilitated the assessment of the influence of tags on engagement.

III. Standardized scores for key metrics:

The engagement score was derived by computing z-scores for four key engagement measures, namely views, likes, comments, and time. Z-scores are utilized to normalize measures, so placing them onto a uniform scale, and subsequently highlighting their relative significance by means of weight assignment. The engagement score comprehensively captures viewer involvement in an equitable manner, facilitating the evaluation of the overall engagement level of each video.

IV. The Classification of Publication Timing:

The concept of "time-of-day" refers to the division of a 24-hour period Categorization: In order to investigate the impact of publishing time on engagement, we classified video publication times into discrete segments: early morning, morning, afternoon, evening, and night. The utilization of this categorisation facilitated the identification of prospective temporal patterns and viewer preferences associated with distinct periods throughout the day.

Early morning	5 am to 7am
Morning	7.01 am - 12.00pm
Afternoon	12.01 pm to 5.00pm
Evening	5.01pm- 8.00pm
Night	8.01pm-4.59am

Table 1: The Classification of Publication Timing

V. The formulation of an Engagement Score

The components of engagement were determined by evaluating significant engagement data, including likes, comments, views, and video time, in order to calculate our engagement score. The standardization process involved the utilization of z-scores for each statistic.

- Weight Assignment: In order to underscore the significance of each indicator related to user involvement, weights were allocated in the following manner: likes (0.2), comments (0.2), views (0.5), and duration (0.1).

Variables	Assigned Weights
Likes	0.2
Comments	0.2
Views	0.5
Duration	0.1

Table 2: Weight Assignment

Understanding Weight Assignment

The weight assignment in the calculation of the engagement score is grounded on the underlying assumption that specific engagement metrics possess a greater level of significance in determining a video's popularity and influence on YouTube's algorithm, in comparison to other metrics. The assigned weights for each metric, specifically likes, comments, views, and duration, are indicative of their respective significance within the engagement ecosystem of YouTube.

The metric of "likes" on a video serves as an indicator of favourable reception and constructive feedback from the audience. The algorithm employed by YouTube places significant weight on the number of likes a video receives, as it is perceived as a robust indicator of user satisfaction. Videos that exhibit a high ratio of likes to dislikes are more prone to being recommended to a wider audience. Therefore, by assigning a weight of 0.2 to likes, one recognises and acknowledges the substantial influence they have on engagement.

The metric of "Comments (0.2)" indicates the level of user engagement through various forms such as discussions, feedback, and interactions. Comments on YouTube are highly regarded due to their ability to enhance user engagement and increase the overall duration of users' interactions with the platform. Hence, the allocation of a weight of 0.2 to comments acknowledges their significance in promoting engagement.

The measurement of views serves as a fundamental metric for assessing the extent of a video's audience reach. The algorithm employed by YouTube exhibits a preference for videos that possess the ability to captivate a larger viewership, as these videos are instrumental in augmenting both the duration of viewership and the revenue generated through advertisements. Assigning a weight of 0.5 to views recognises the significant influence they have in determining the success of a video.

The concept of duration refers to the temporal extent of viewer engagement with a video. Extended durations of watching videos on YouTube are indicative of increased viewer engagement and retention, which holds significant importance as a pivotal element within the algorithm employed by the platform. There is a higher probability of videos being recommended if they have longer durations. Although duration may not hold as much significance as other metrics, it still plays a role in determining engagement and therefore justifies a weight of 0.1.

The assignment of weights to these metrics is based on an informed estimation of their relative importance, considering that YouTube's algorithm is proprietary and is not fully disclosed. Adopting a pragmatic methodology, this study aims to simulate the potential prioritisation of engagement metrics in YouTube's recommendation system. By allocating weights in this manner, we are effectively optimising our calculation of engagement score to align with the objectives of the platform, even when specific algorithmic details are not provided.

The computation of the engagement score for each video involved multiplying the z-scores of the metrics by their corresponding weights and then adding the obtained values. The utilization of this formula facilitated the attainment of a well-proportioned portrayal of viewer engagements.

Normalization was conducted in order to normalize the engagement scores across videos. This was achieved by the implementation of min-max normalization, which involved rescaling the values to ensure they fell within the range of 0 to 1.

5.4 Visualization Techniques and Exploratory Data Analysis

Data visualization was employed in this study to examine the associations between engagement ratio and different video features. A range of data visualization tools and methodologies were deployed for this purpose. In order to gain a comprehensive understanding of the linkages, various visual representations were utilized, including scatter plots, time series plots, heatmaps, and category-wise

engagement graphs. These visualizations were chosen for their ability to effectively convey clear insights into the aforementioned relationships.

The utilization of visualization techniques facilitated the identification of patterns, trends, and correlations within the dataset, enabling the generation of significant insights into the dynamics of viewer engagement.

5.5 Outlier Treatment

When we were cleaning up and preparing our data set for analysis, one of the most important steps was figuring out what to do with the numbers that were way off from the rest. Outliers are data points that are very different from the rest of the data in a set. They can have a big effect on statistical analyses and machine learning models. We used the Interquartile Range (IQR) method, which is a reliable way to find and deal with outliers, to solve this problem. We found the first quartile (Q1) and the third quartile (Q3) for each number attribute. Together, these two numbers made up the IQR. We found the upper and lower whiskers for outlier detection by using a multiplier factor set by the user. The upper whisker value was used to replace data points that were higher than the upper whisker. The lower whisker value was used to replace data points that were lower than the lower whisker. This treatment did a good job of reducing the effect of outliers on our dataset, making it a more solid and reliable starting point for further analysis. Visualising boxplots before and after outlier treatment made it clear how much the data was changed. This showed that outlier treatment had a big effect on how the data was distributed. This step makes sure that our analyses and models are built on data that better shows the underlying patterns and relationships in the dataset. This makes our findings more reliable.

5.6 Data Preparation and Preprocessing for Engagement Score Predictive Modelling

5.6.1 Feature Selection:

We start by splitting our dataset into features (X) and the target variable (y). Tag (Yes=1, No=0), Category Name, Time of Day, Likes, Views, Comments, and Duration (Seconds) are some of the features (X). We will make our predictions based on these things.

5.6.2 One-Hot Encoding:

"Category Name" and "Time of Day" are examples of categorical variables. Most algorithms for machine learning need numbers as input, so we use one-hot encoding on these categorical variables. This process turns them into numbers that the algorithm can use, making binary columns for each category with either a 0 or a 1.

5.6.3 Data Splitting:

My dataset contains 1100 rows. To test how well our machine learning model works, we divide the data into sets for training and testing. The training set (X_train and y_train) is used to teach the model, while the testing set (X_test and y_test) is used to see how well the model works with new, untried data. In this case, test_size=0.2 sets aside 20% of the data for testing, and random_state=42 uses a random seed to make the results repeatable.

5.6.4 Standardisation:

Numbers need to be standardised or normalised so that they all have the same scale. The data are changed so that the mean is 0 and the standard deviation is 1. This step is important for many machine learning algorithms because it keeps larger scale features from taking over smaller scale features. To do this standardisation, we use the StandardScaler from scikit-learn. It is fit to the training data and then used on both the training data and the testing data to make sure the scaling is the same.

5.7 Regression Modelling

The predictor variables encompassed in this study consisted of video category, published time, published date, tags, views, likes, comments, and duration. The aforementioned variables were utilized for the purpose of forecasting the engagement score, which served as the dependent variable.

Predictive Models: In order to ascertain the variables that have an impact on video engagement, regression modeling approaches were utilized. In this study, a range of regression models were employed, encompassing Multiple Linear Regression, Decision tree regression, Random Forest Regression, K-Nearest Neighbours (KNN) Regression, Gradient Boosting Regression, Neural Network Regression, and Extreme Gradient Boosting Regressor (XGBRegressor).

5.7.1 Multiple Linear Regression

Multiple Linear Regression is a widely used statistical and machine learning technique that seeks to establish a linear association between a dependent variable and multiple independent variables. This methodology involves the prediction or explanation of the dependent variable, commonly known as the target or outcome variable, through the consideration of multiple independent variables, also referred to as features or predictors. The model computes coefficients for each independent variable, which indicate their relative strengths and directions of impact on the dependent variable. Being the independent variables x and the dependent variable y, the basic model for each MLR is:

$$y = \beta 0 + \beta 1x1 + \cdots \beta mxm + \varepsilon$$

The effectiveness of this approach is contingent upon various assumptions, such as the presence of linearity, the independence of errors, homoscedasticity, multivariate normality, and minimal multicollinearity among the independent variables. The conventional procedure for Multiple Linear Regression entails several key stages, including the collection of data, preprocessing of the data, construction of the regression model, evaluation of the model's performance, interpretation of the coefficients, and prediction based on the model. The utilisation of this tool extends to diverse

domains including economics, finance, healthcare, and social sciences, facilitating researchers and analysts in acquiring valuable insights, making well-informed decisions, and forecasting outcomes through the utilisation of evidence derived from data analysis.

5.7.2 Decision Tree Regression

Decision Tree Regression is a highly adaptable and extensively utilised machine learning methodology employed for the purpose of regression tasks. In contrast to conventional linear regression, Decision Tree Regression aims to partition the feature space into separate regions and make predictions for the target variable by calculating the average of data points within each region. This methodology demonstrates a high level of efficacy in capturing intricate, non-linear associations. Decision trees possess a comprehensible hierarchical structure, wherein nodes symbolise distinct features, branches symbolise decision paths, and leaves encapsulate anticipated outcomes. The algorithm employs a recursive approach to expand the tree structure by carefully selecting the most optimal features for dividing the data. Additionally, the utilisation of pruning techniques can effectively mitigate the risk of overfitting. Decision Tree Regression is utilised in various fields such as finance, healthcare, and environmental science, facilitating the development of interpretable models for analysing relationships and making predictions.

5.7.3 K-Nearest Neighbours (KNN) Regression

The K-Nearest Neighbours (KNN) Regression algorithm is a flexible and intuitive machine learning technique that is employed for both regression and classification purposes. The algorithm functions based on the principle of proximity, wherein predictions are made by calculating the average of the values of the K-nearest neighbours to the target point. The value of K is determined by the user. In contrast to parametric approaches, KNN Regression does not impose any assumptions regarding the underlying data distribution, rendering it applicable to a wide range of scenarios. The method demonstrates exceptional proficiency in capturing non-linear relationships and proves to be particularly advantageous in situations where the characteristics of the data are not thoroughly comprehended. The simplicity and flexibility of the K-Nearest Neighbours (KNN) algorithm render it an appealing option for beginners. However, it is crucial to take into account its sensitivity to the selection of the parameter K and the computational burden it incurs when dealing with extensive datasets. However, KNN Regression is extensively utilised across various domains, ranging from the prediction of real estate prices to medical prognosis and recommendation systems. In these

applications, the ability to comprehend local patterns and establish relationships based on neighbouring instances is imperative for achieving precise predictions.

5.7.4 Random Forest Regression

The Random Forest Regression method is a robust and flexible machine learning approach that effectively combines the capabilities of numerous decision trees to produce precise predictions. The operational mechanism involves the construction of a collection of decision trees during the training phase, followed by the aggregation of their outputs to generate predictions. The utilisation of this ensemble approach serves to improve the accuracy of predictions and enhance generalisation, while also mitigating the issue of overfitting. In the Random Forest algorithm, every decision tree is trained on a randomly selected subset of the dataset and a randomly selected subset of features. This approach introduces variability and diversity among the individual trees. The presence of diverse elements, in conjunction with the amalgamation of forecasts, yields a regression model that is notably efficient. The Random Forest Regression technique is extensively employed across diverse domains, including finance, ecology, and healthcare, owing to its capacity to effectively manage intricate, non-linear associations, as well as its robustness against outliers and noise present in the data. Moreover, it offers valuable insights regarding the importance of features, thereby proving its significance in the context of feature selection and interpretation within predictive modelling tasks.

5.7.5 Gradient Boosting Regression

Gradient Boosting Regression is widely recognised as a highly effective machine learning methodology due to its remarkable predictive precision and resilience. The algorithm under consideration falls within the ensemble learning framework, wherein it operates by iteratively training a series of decision trees. Each subsequent tree is designed to rectify the mistakes made by its predecessor. Gradient Boosting is a machine learning technique that aims to optimise a loss function and minimise the errors in the model's predictions. This approach has proven to be highly effective for various tasks, including regression and classification. One notable application of this technique is the Gradient Boosting Machine (GBM). Gradient Boosting Machines (GBM) construct decision trees by leveraging the gradient of the loss function, thereby creating a model that iteratively enhances its predictive capabilities. Gradient Boosting Regression is extensively utilised in various domains such as finance, healthcare, and online advertising owing to its capacity to effectively handle intricate associations within data, resilience against atypical observations, and

aptitude to capture intricate patterns. In addition, this technique enables the refinement of hyperparameters, rendering it a valuable instrument for attaining cutting-edge outcomes in diverse predictive modelling scenarios.

5.7.6 Neural Network Regression

Neural Network Regression is an advanced machine learning methodology that utilises artificial neural networks to make predictions of numerical values. It is a specific type of deep learning model that utilise interconnected layers of neurons to process and transform data in order to generate predictions. The adaptability of neural network regression renders it highly proficient in capturing intricate, non-linear relationships present in complex datasets. By employing a multi-layered design and diverse activation functions, the neural network has the capacity to acquire knowledge and depict complex patterns, rendering it well-suited for applications such as predicting stock prices, modelling climate patterns, and forecasting demand. Nevertheless, the process of training neural networks typically necessitates significant computational resources and a sufficient amount of labelled data. The flexibility, scalability, and potential for high accuracy of neural network regression render it an indispensable tool in modern predictive modelling and data analysis.

5.7.7 Extreme Gradient Boosting Regressor (XGBRegressor)

The XGBRegressor, a vital component of the Extreme Gradient Boosting (XGBoost) machine learning framework, is a robust regression algorithm specifically developed for the purpose of forecasting continuous numerical values. The XGBRegressor algorithm is based on the principles of gradient boosting and utilises ensemble learning techniques to construct a reliable predictive model. This is achieved by iteratively improving the predictions of numerous weak models, typically decision trees. The distinguishing characteristics of XGBRegressor lie in its versatility and robustness. The algorithm integrates regularisation techniques, efficient tree pruning, and exhibits robust handling of missing data, thereby reducing susceptibility to overfitting. The proficiency of the system extends to multi-core central processing units (CPUs), wherein the utilisation of parallel processing techniques facilitates the acceleration of model training. Due to its capacity to identify and assess the significance of features, as well as its wide-ranging applications in various fields including finance, healthcare, and e-commerce, XGBRegressor has emerged as a highly preferred option for regression tasks. It consistently demonstrates high levels of accuracy and dependability in the field of predictive modelling.

The process of model validation was conducted rigorously in order to assure the robustness and generalizability of our regression models. This was achieved through the implementation of techniques such as cross-validation.

5.8 Model performance evaluation techniques

1) Key Evaluation Metrics for YouTube Video Analysis Regression Models

Within the context of YouTube video analysis, the assessment of regression models frequently relies on a set of crucial metrics that provide insights into the predictive capabilities of the model. Three commonly employed metrics in various fields are Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). These metrics offer significant insights into the predictive capabilities of a regression model in relation to engagement scores for videos on platforms such as YouTube.

a) Mean Squared Error (MSE)

The Mean Squared Error (MSE) is a statistical metric used to measure the average squared difference between the predicted and actual values in a dataset.

The Mean Squared Error (MSE) measures the average of the squared differences between the predicted and actual engagement scores across a given set of videos. The computation is done using subsequent formula:

$$MSE = \frac{1}{n}\sum(\hat{y}_i - y_i)^2$$

In this formula, *n* represents the number of videos in your dataset, *y^i* denotes the predicted engagement score for the *i*-th video, and *yi* represents the actual engagement score for the *i*-th video. MSE quantifies how much the model's predictions deviate from the actual scores, with higher values indicating larger prediction errors.

b) The Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) is a statistical metric commonly used to evaluate the accuracy of a predictive model.

The root mean square error (RMSE) is considered a more interpretable metric due to its alignment with the unit of measurement used for the engagement scores. The square root of the mean squared error (MSE) is computed as follows:

$$RMSE = \sqrt{MSE}$$

The Root Mean Square Error (RMSE) is a valuable metric for knowing the magnitude of prediction inaccuracies within a practical setting. Lower root mean square error (RMSE) values indicate that the model's predictions exhibit greater proximity to the actual engagement scores.

c) Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) is a metric used to quantify the mean absolute error between predicted and actual engagement scores. The calculation is performed using the prescribed formula:

$$MAE = \frac{1}{n} \sum |\hat{y}_i - y_i|$$

The Mean Absolute Error (MAE) offers understanding of the average prediction error exhibited by the model. The metric under consideration exhibits a lower sensitivity to outliers in comparison to both mean squared error (MSE) and root mean squared error (RMSE).

It's important to note that the Mean Absolute Percentage Error (MAPE) could not be calculated because there were times when the actual engagement scores were 0. MAPE involves dividing by the actual value, which means dividing by zero when the actual score is zero.

2) Scatter Plots

Scatter plots allow us to visually compare actual and predicted values to assess our predictions' accuracy. Scatter plots quickly and easily reveal systematic deviations in our predictions.

3) Residual plots

Residual plots help us analyse prediction error distribution. These plots show patterns or trends in our model's errors, ensuring we're not making systematic predictions errors.

4) Learning Curve

Lastly, we use learning curves to see how our model's performance changes with training dataset size. Learning curves help us optimise our model by revealing overfitting and underfitting.

6.0 Data Analysis

6.1 Descriptive Statistics

6.1.1 Numerical Data

	Views	Likes	Comments	Duration (Seconds)	Engagement Score
count	1.100000e+03	1.100000e+03	1100.000000	1100.000000	1100.00000
mean	2.866041e+06	7.674883e+04	6071.305455	493.528182	0.062654
std	4.052141e+06	7.903336e+04	16112.121927	174.026355	0.055962
min	1.470720e+05	2.025000e+03	117.000000	28.000000	0.000000
25%	1.381321e+06	3.781050e+04	2334.250000	407.750000	0.038820
50%	2.031741e+06	6.233700e+04	4117.500000	496.000000	0.052134
75%	3.099148e+06	9.025375e+04	6285.250000	590.000000	0.070384
max	7.882453e+07	1.106300e+06	400184.000000	2122.000000	1.000000

Table 3 Descriptive Statisitcs of Numerical Variables

Views: The dataset contains videos with a wide range of view counts, from a minimum of approximately 147,072 to a maximum of around 78,824,530. On average, videos have about 2,866,041 views, with a standard deviation of approximately 4,052,141.

Likes: The number of likes on videos varies, with a minimum of 2,025 likes and a maximum of 1,106,300 likes. The average number of likes is approximately 76,749, with a standard deviation of about 79,033.

Comments: Video comments also show significant variability, ranging from a minimum of 117 comments to a maximum of 400,184 comments. The average number of comments is approximately 6,071, with a standard deviation of around 16,112.

Duration (Seconds): Video durations span from a minimum of 28 seconds to a maximum of 2,122 seconds (approximately 35 minutes). The average video duration is approximately 493.53 seconds (about 8 minutes and 13 seconds), with a standard deviation of roughly 174.03.

Engagement Score: The engagement score for videos varies between 0 and 1. On average, videos have an engagement score of approximately 0.0627, with a standard deviation of about 0.056.

6.1.2 Categorical Data

(People & Blogs	862
Travel & Events	176
Howto & Style	19
Nonprofits & Activism	11
Science & Technology	7
Autos & Vehicles	7
Sports	6
Pets & Animals	5
Education	4
Film & Animation	2
Music	1
Name: Category Name, dtype: int64,	
Afternoon	689
Evening	203
Night	169
Morning	37
Early Morning	2
Name: Time of the Day, dtype: int64,	
0	843
1	257
Name: Tag (Yes=1, No=0), dtype: int64)	

Table 4 Descriptive Statistics of Categorical Variables

Category Name: The majority of videos in the dataset belong to the "People & Blogs" category, with 862 videos. Other categories have fewer videos, such as "Travel & Events," "Howto & Style," and "Nonprofits & Activism."

Time of the Day: Most videos are published in the afternoon, followed by the evening and night. "Morning" and "Early Morning" have fewer video publications.

Tag (Yes=1, No=0): Out of the videos, 843 do not have tags (labeled as 0), while 257 videos have tags (labeled as 1).

6.2 Exploratory Data Analysis

6.2.1 Univariate Graphs

Distribution of variables

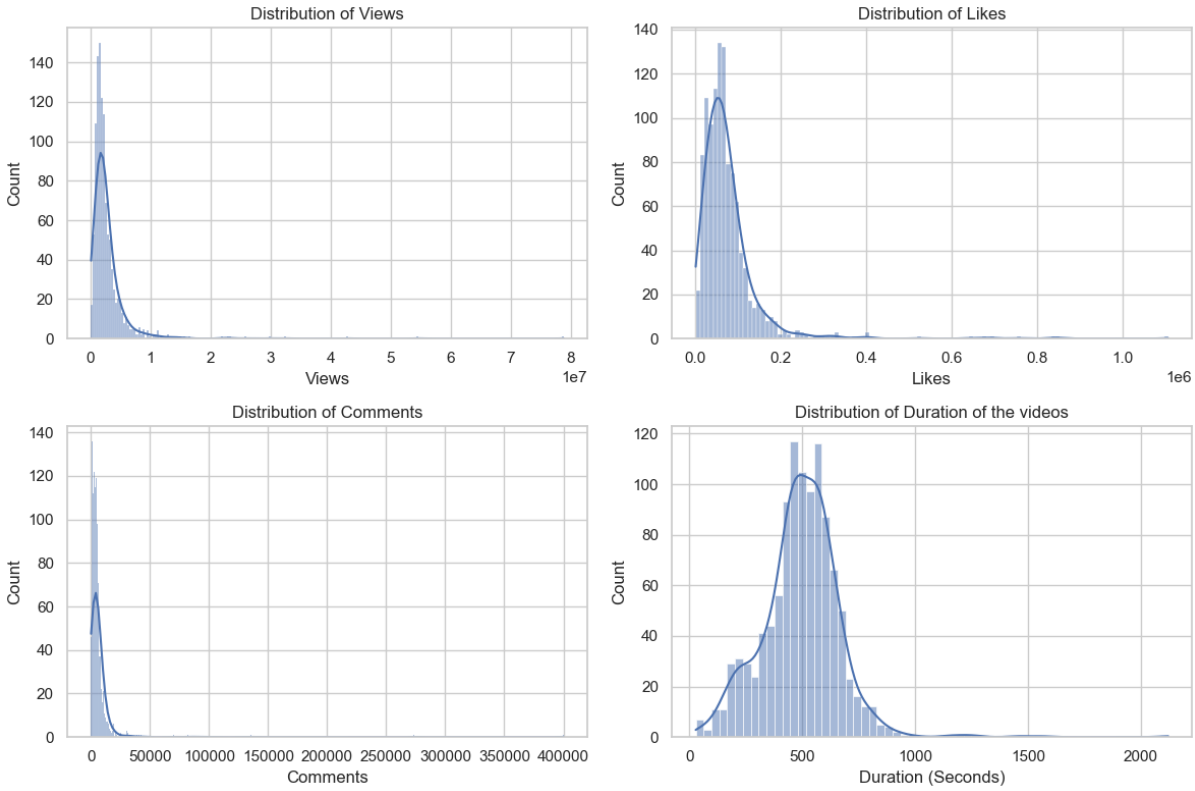


Figure 16: Distribution of Variables

View Distribution:

The longer right tail of the histogram indicates a slightly right-skewed distribution of "Views".

The majority of videos have under 10 million views.

This suggests that most videos in your dataset have typical view counts, with a few outliers having very high views.

Likes Distribution:

The right tail of "Likes" is longer and slightly right-skewed.

Most videos have fewer than 200,000 likes.

Similar to views, most videos receive a typical number of likes, with a few receiving much more.

Comment Distribution:

The distribution of "Comments" is slightly right-skewed.

Comment counts are below 25,000 for most videos.

Like views and likes, this suggests that most videos get usual comments, with a few getting a lot.

Video Duration Distribution:

The distribution of "Duration (Seconds)" is symmetrical and approximately normally distributed.

About 16 minutes and 40 seconds, most videos last 50 to 1000 seconds.

This suggests your dataset has a wide range of video lengths.

6.2.2 Multivariate graphs

Distribution of variables over time

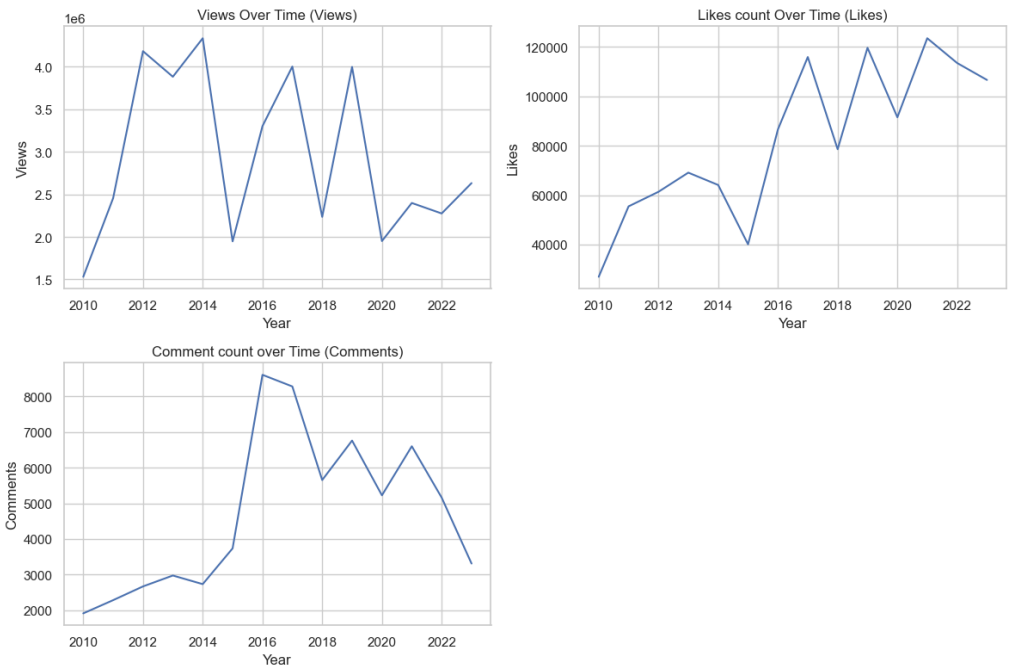


Figure 17: Distribution of variables over time

Changes in Views (Views):

The line plot shows how "Views" has changed over time.

It shows that the number of times videos have been watched has changed over time.

Surprisingly, there was a big jump in the number of views around the year 2014. This suggests that videos uploaded around that time got a lot of views.

After that, there is a general upward trend, which shows that the number of views has slowly grown over time. The recent drop in the number of views for the most recent videos is normal, since views tend to build up slowly over time.

The changes in Likes Over Time (Likes):

This line plot shows how the number of "Likes" has changed over time.

The graph shows that there have also been changes in the number of likes.

The number of likes seems to reach its highest point around the year 2021, which means that videos posted during that time got a lot of likes.

Like the number of views, the number of likes for the most recent videos has gone down recently. This is a common pattern because likes build up slowly.

Changes in the number of comments (Comments):

The line graph shows how the number of "Comments" has changed over time.

It shows that the number of comments has changed, just like the number of views and likes.

The number of comments reached its highest point around the year 2016, which suggests that videos from that time got a lot of comments.

As with views and likes, the number of comments on the most recent videos has been going down, which is similar to how engagement usually builds up slowly over time.

Views Vs. Category:

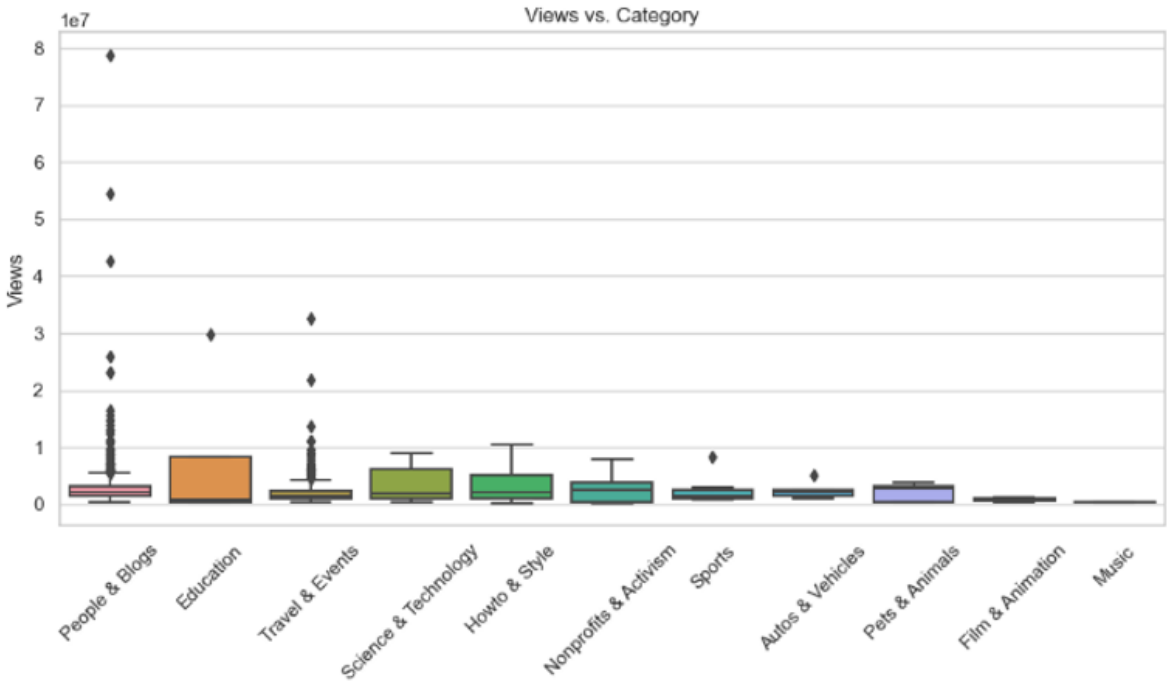


Figure 18: Views Vs Category

The presented boxplot illustrates the distribution of video view counts across various categories.

The horizontal axis denotes the various categories of videos, while the vertical axis represents the numerical count of views.

The depicted boxes illustrate the interquartile range (IQR), wherein the lower and upper boundaries of the boxes correspond to the first quartile (Q1) and third quartile (Q3), respectively.

Analysis: It can be observed that videos belonging to the "Education" category exhibit a relatively higher median count of views when compared to videos in other categories. Nevertheless, it is worth noting that there exist outliers in various categories, with particular emphasis on the People and Blogs category, which encompasses videos exhibiting significantly high or low view counts.

Likes Vs. Category

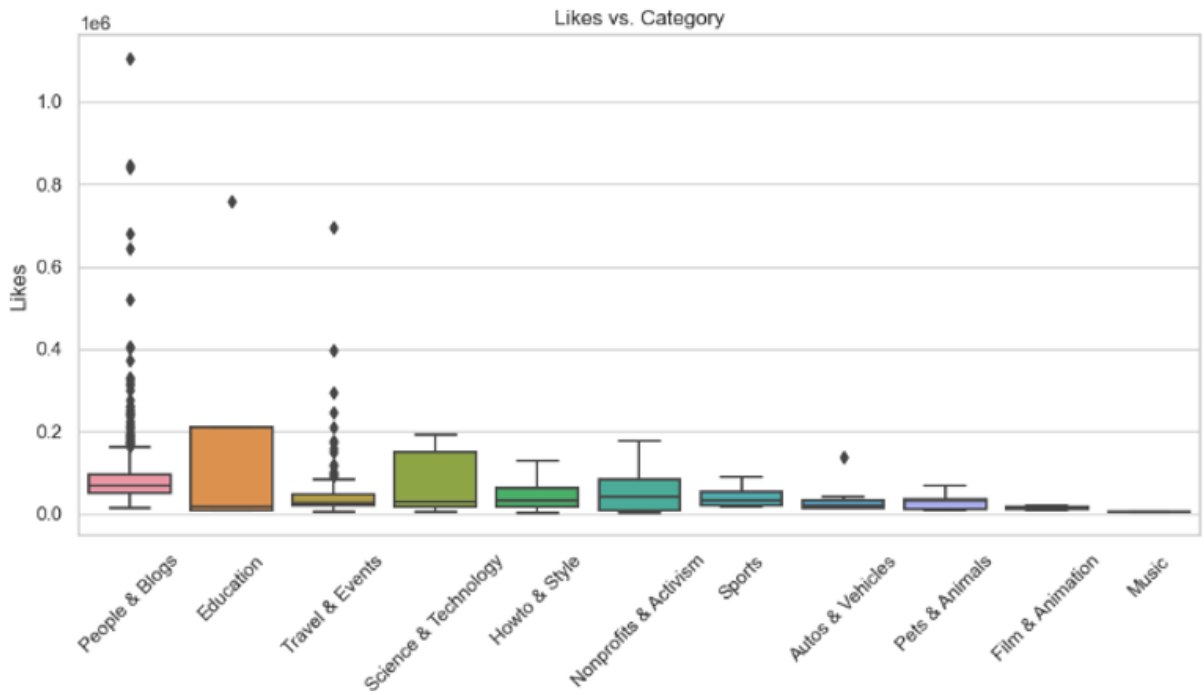


Figure 19: Likes Vs Category

The presented boxplot illustrates the distribution of the number of likes across various categories of videos.

In a manner similar to the initial boxplot, the horizontal axis denotes the various categories of videos, while the vertical axis represents the count of likes received.

The boxes in the diagram represent the interquartile range (IQR), while outliers are visually depicted.

The analysis reveals that "Education" videos generally exhibit a higher median number of likes. However, it is important to note that there are certain exceptions within different categories, particularly in the People and Blogs category.

Comments Vs Category

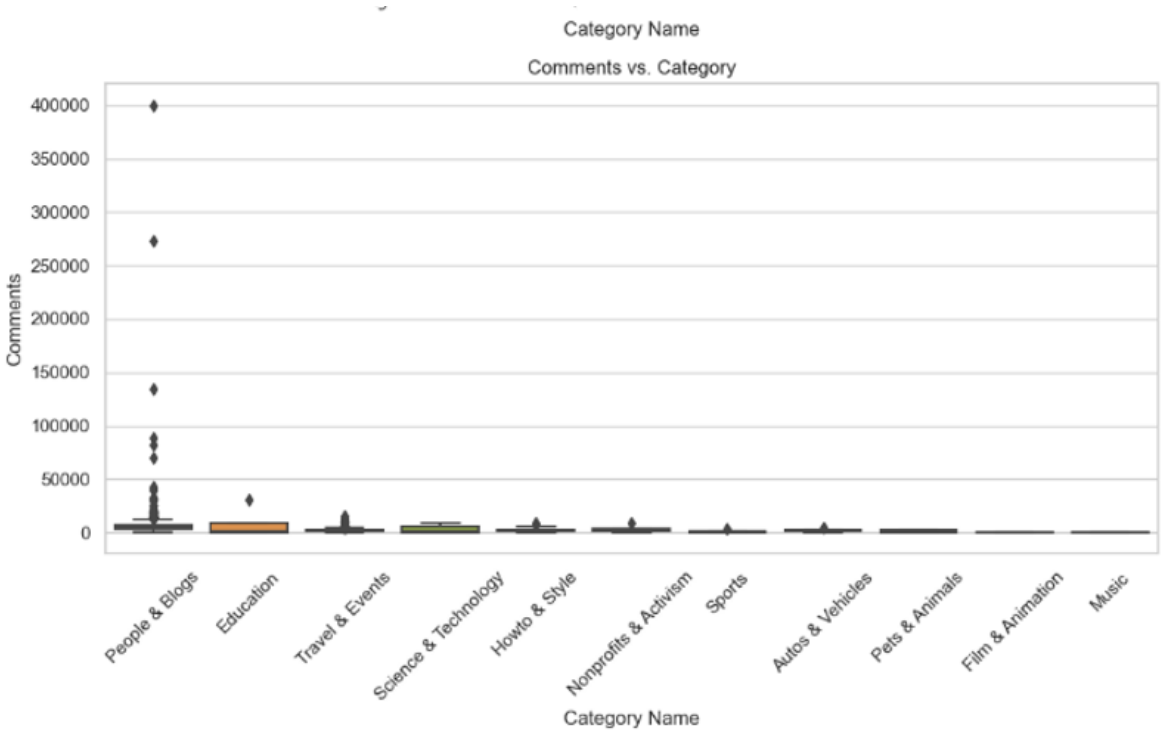


Figure 20: Comments Vs Category

The presented boxplot illustrates the distribution of comment counts across various video categories.

Once more, it should be noted that the x-axis in this context denotes the various categories of videos, while the y-axis corresponds to the quantity of comments received.

The boxes in the visual representation symbolise the interquartile range (IQR), while any data points that fall outside the boxes are considered outliers.

Analysis: It has been observed that videos belonging to the "Education" category tend to attract a higher number of comments compared to other categories. However, it is worth noting that there may be certain exceptional cases within the "People and Blogs" category that deviate from this trend.

Engagement Scores Vs Category

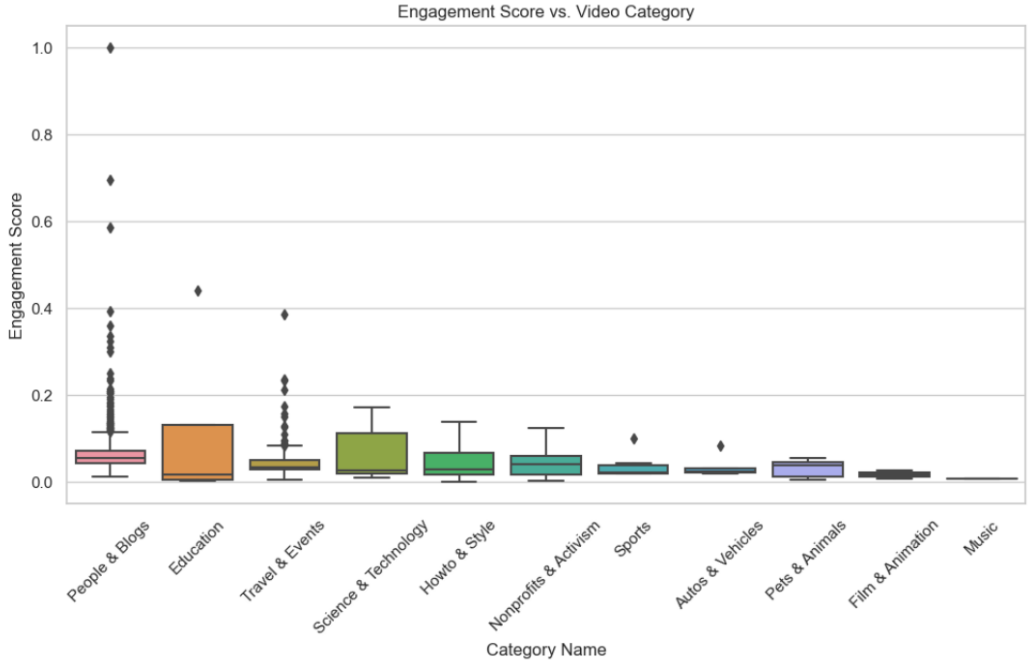


Figure 21: Engagement Scores Vs Category

The horizontal axis of the graph represents the various video categories, while the vertical axis represents the measure of engagement, known as the engagement score.

Analysis: It has been observed that educational videos exhibit a relatively elevated median engagement score in comparison to other categories. Nevertheless, it is important to note that certain categories, such as 'People and Blogs' and 'Travel and Events', exhibit potential outliers, suggesting the presence of videos with remarkably high engagement scores.

Average Views by the Time of the Day

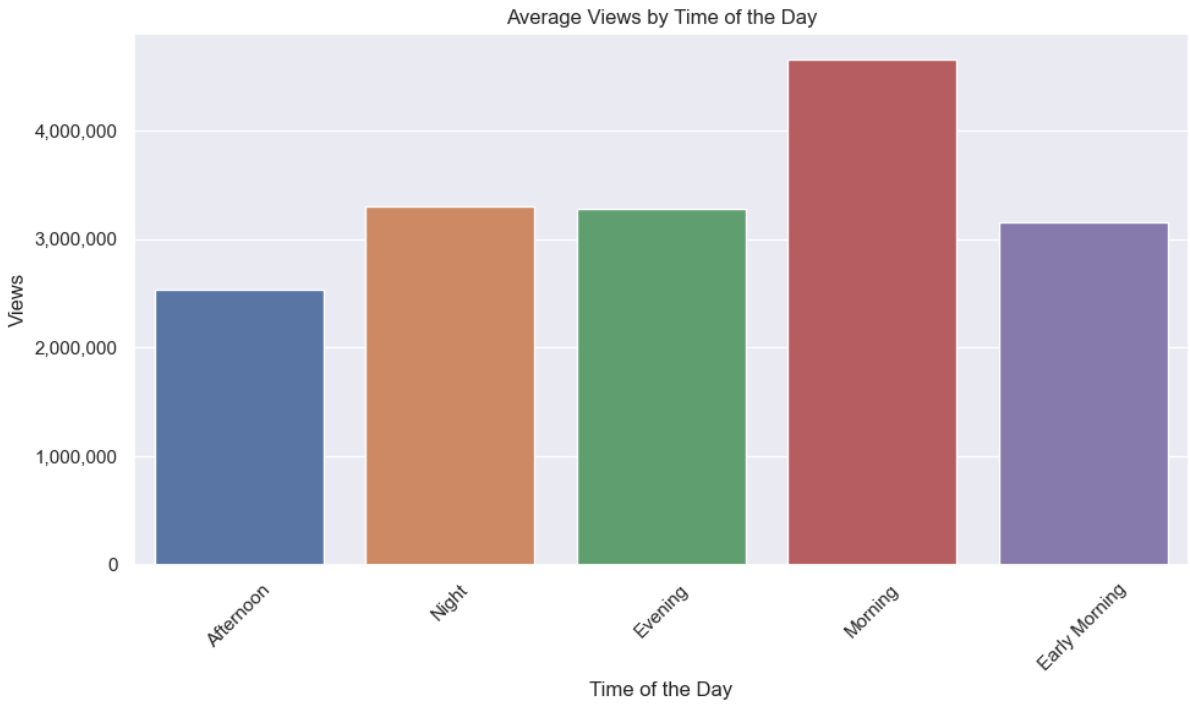


Figure 22: Average Views by the Time of the Day

The bar plot illustrates the average number of views for videos that have been uploaded during various time periods throughout the day.

The vertical bars in the graph correspond to distinct temporal intervals, with their respective heights denoting the mean number of views for videos that were uploaded within those intervals.

Analysis: The data reveals that videos uploaded during the morning hours (7 am and 12 pm) attract the highest average views. Subsequently, videos uploaded during the night, evening, early morning, and afternoon receive progressively lower average views. This finding indicates that videos that are published in the morning exhibit a higher average viewership in comparison to videos published during other times of the day.

Average Views by Having and Not Having Tags

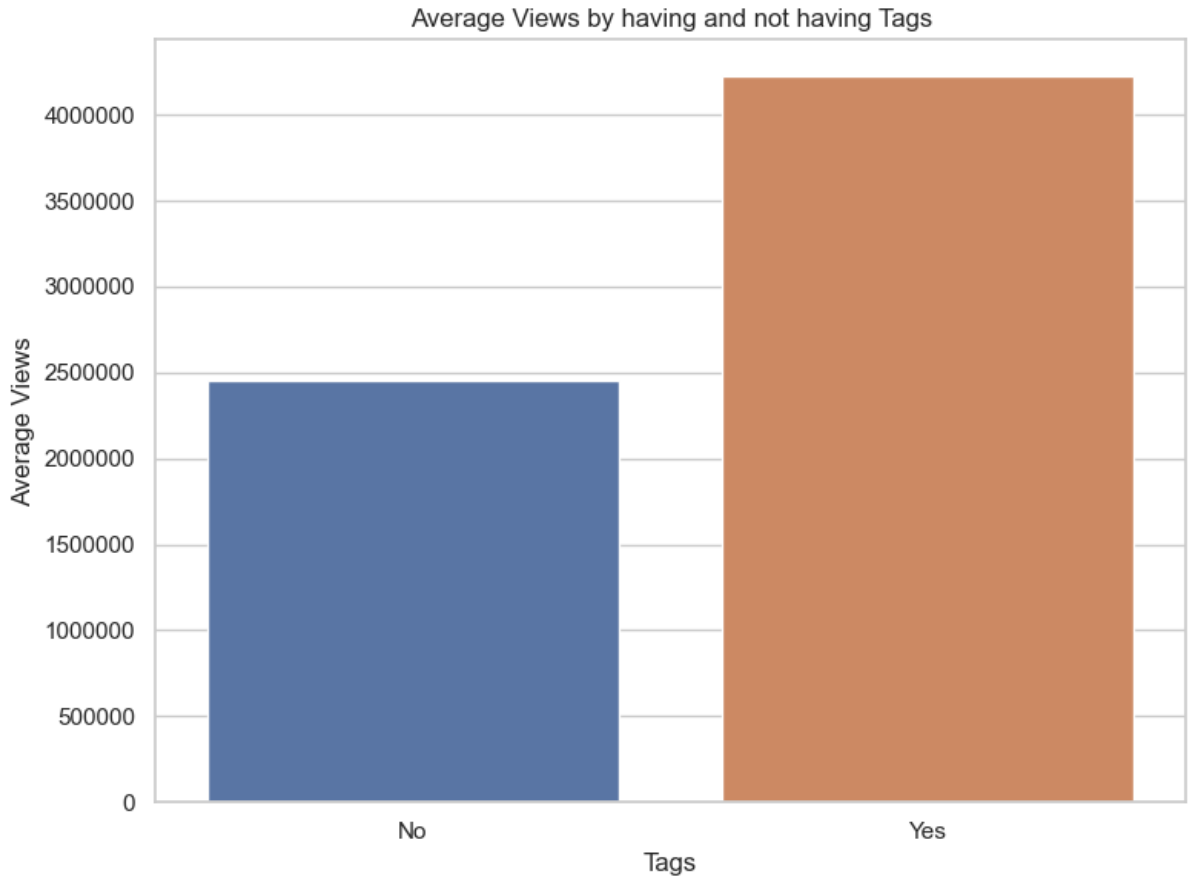


Figure 23: Average Views by Having and Not Having Tags

The average number of views for both tagged and untagged videos is plotted as a bar chart. It demonstrates how tags affect a video's popularity.

Videos with Tags (Tag=Yes): These videos typically receive 40 million views on average.

There appears to be a correlation between using tags and increased video views.

On the other hand, videos with no tags (Tag=No) only get about 25 million views on average.

This suggests that, on average, fewer people watch videos that haven't been tagged.

The data demonstrates that tagging videos is a common and efficient method of making them more discoverable and drawing in a larger audience.

Engagement Score by Having and Not Having Tags

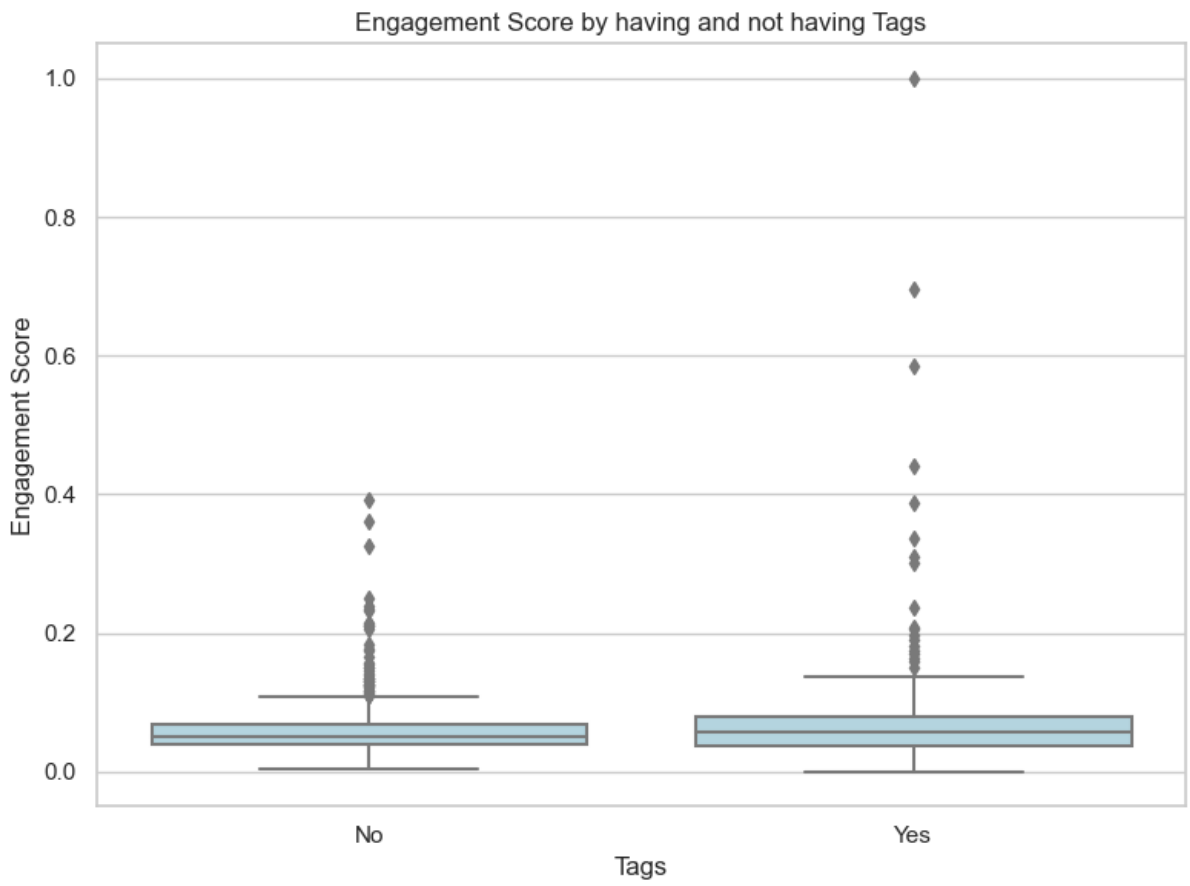


Figure 24: Engagement Score by Having and Not Having Tags

In line with the previous bar graph, it is observed that videos with tags generally exhibit higher levels of Engagement Score in comparison to videos without tags.

6.3 Outlier Treatment

Before Outlier Treatment

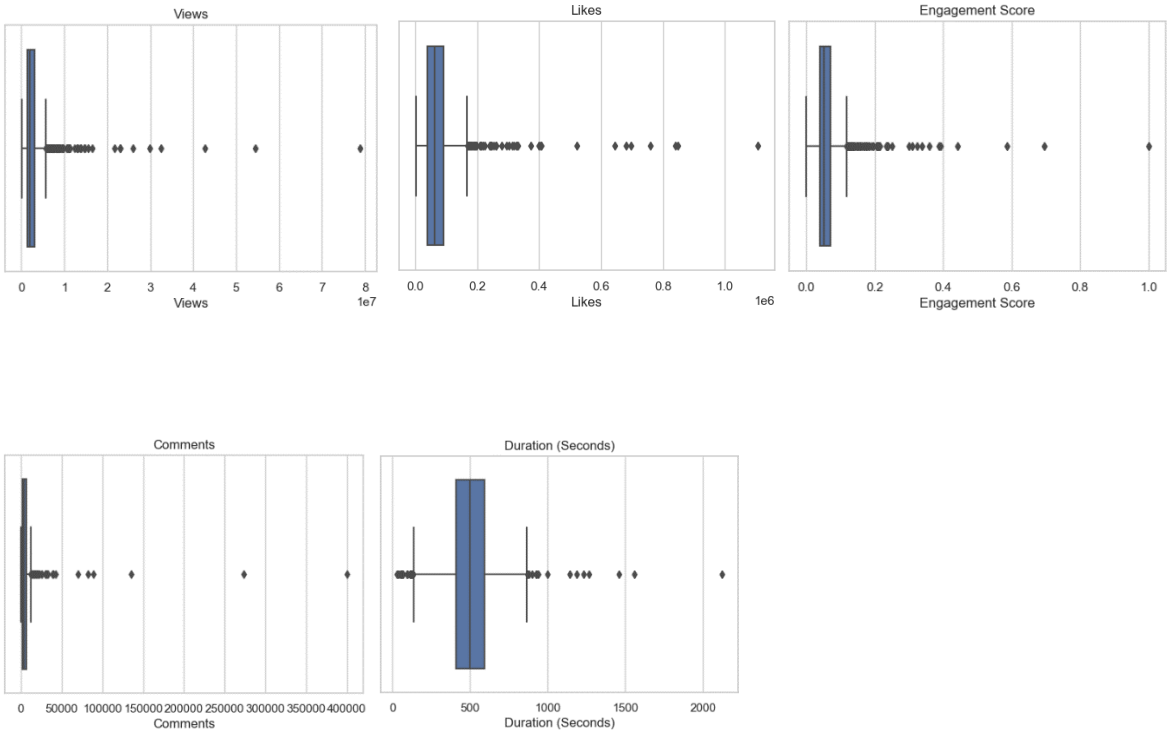


Figure 25: Before Outlier Treatment

After Outlier Treatment

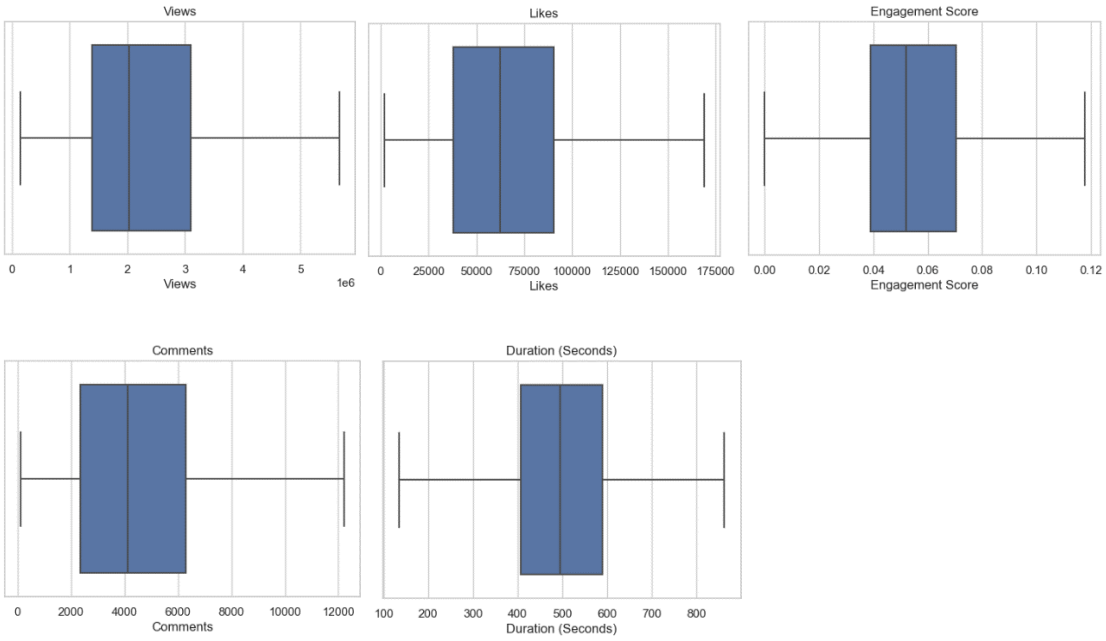


Figure 26: After Outlier Treatment

6.4 Baseline Model Evaluation for Engagement Score Prediction

In this analysis, we used different metrics to get a starting point for judging how well engagement score predictions work. The predictions of the baseline model were made by giving each observation the mean engagement score of the whole dataset.

The Mean Squared Error (MSE) at the start is found to be about 0.00067. This is a measure of the average squared difference between the actual engagement scores and the scores that were predicted. Mean Squared Error (MSE) is the square root of the Root Mean Squared Error (RMSE), which is about 0.0259. This number shows the average error made when trying to predict engagement scores.

The Mean Absolute Error (MAE) of the baseline model is about 0.0204, which is the average absolute difference between the actual engagement scores and the scores that were predicted.

6.5 Regression Models - Results

6.5.1 Multiple Linear Regression

The performance metrics of the model are presented below:

```
Mean Squared Error: 1.2621520483328495e-05
Root Mean Squared Error: 0.0035526779312693817
Mean Absolute Error: 0.002072416445382901
Mean Squared Error (Train): 1.9344721384726094e-05
Root Mean Squared Error (Train): 0.004398263451036794
Mean Absolute Error (Train): 0.001971080728755711
```

The Mean Squared Error (MSE) is a quantitative metric used to assess the degree of alignment between the predictions made by the model and the observed engagement scores. In this particular

scenario, the mean squared error (MSE) is calculated to be approximately 1.26e-05, indicating a significantly small magnitude. This suggests that, on average, the model's predictions exhibit a high degree of proximity to the actual values. A lower mean squared error (MSE) is typically indicative of superior performance.

The Root Mean Squared Error (RMSE) is a metric used to assess the predictive accuracy of a model. The root mean square error (RMSE) is a mathematical measure derived from the mean squared error (MSE) and is commonly employed to assess the magnitude of error in predictive models. The model's predictions exhibit a high level of precision, as evidenced by the root mean square error (RMSE) of approximately 0.00355. The margin of error in these predictions is consistently minimal.

The Mean Absolute Error (MAE) is a metric used to evaluate the performance of a model by quantifying the average absolute deviation between its predicted values and the true engagement scores. The mean absolute error (MAE) of approximately 0.00207 indicates that, on average, the model's predictions deviate by approximately 0.00207 units from the actual engagement scores.

The Mean Squared Error (Train) refers to the calculation of the MSE on the training data. The second mean squared error (MSE) is similar to the initial MSE, but it is calculated based on the dataset used for training the model. In this instance, the value is approximately 1.93e-05, suggesting that the model demonstrates a strong fit to the training data.

The Root Mean Squared Error (Train) refers to the calculation of the RMSE on the training data, which is analogous to the RMSE computed for the testing data. The value is approximately 0.00440, indicating that the model's predictions on the training data exhibit a high level of accuracy.

The Mean Absolute Error (Train) refers to the calculation of the MAE on the training data. The calculated value is approximately 0.00197, suggesting that the predictions made by your model using the training data are, on average, deviating by approximately 0.00197 units in terms of engagement score.

In general, the Multiple Linear Regression model exhibits strong performance. The model exhibits minimal errors on both the training and testing datasets, suggesting its ability to generate precise forecasts for engagement scores of YouTube videos.

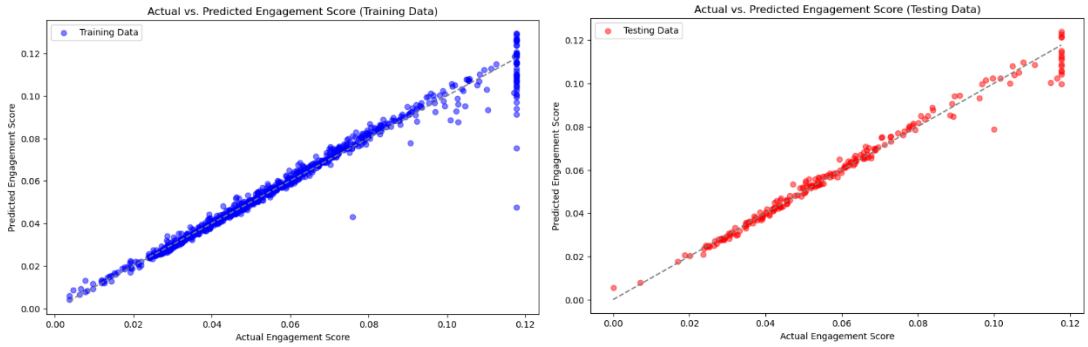


Figure 27: Multiple Linear Regression model Actual and Predicted Values

The predictions made by the model exhibit a high level of accuracy and demonstrate consistency when compared to the actual outcomes. This outcome is considered favourable and indicates that the model has successfully acquired a comprehensive understanding of the underlying patterns and relationships present within the data.

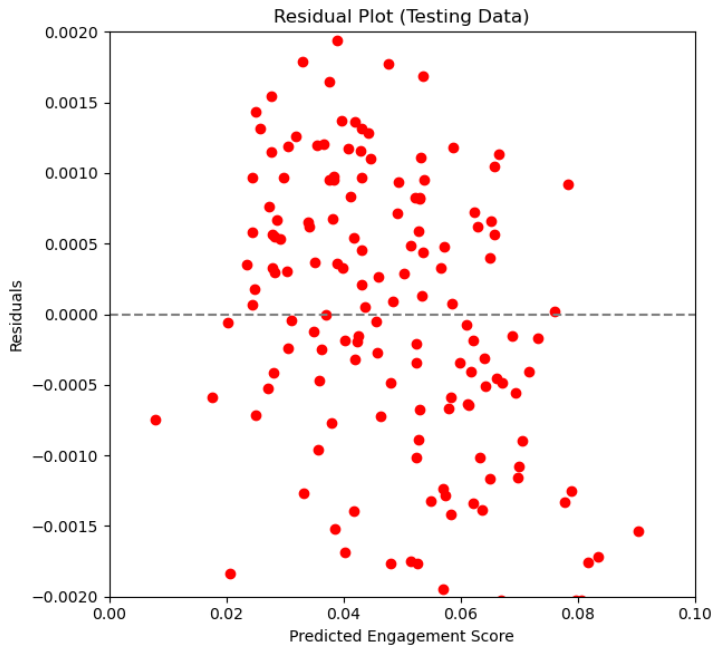


Figure 28: Multiple Linear Regression Residual Plot

Upon examination of the residual plot, it is evident that most of the residual values exhibit a random distribution in relation to the horizontal line, which represents the line of zero error. This observation suggests that our regression model provides a reasonably accurate fit. Nevertheless, certain residual

values exhibit a significant deviation from the previously mentioned horizontal line. This phenomenon may be given to the small magnitude of the y-axis on the graph. To clarify, the existence of outliers in the residual plot, particularly those that appear significantly distant from the horizontal line, may be influenced by the scaling of the y-axis.

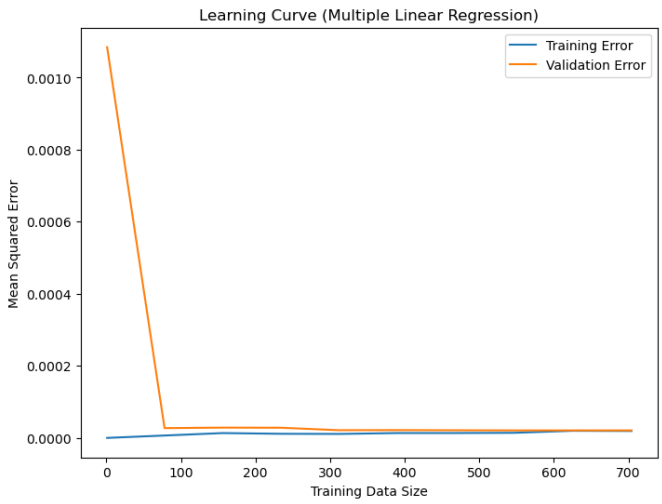


Figure 29: Multiple Linear Regression Learning Curve

With a small training dataset, the training error drops nearly to zero, but the testing error remains high. This is usually caused by overfitting, where the model fits the training data well but struggles to generalise to new data, increasing testing error. As the training dataset grows, usually around 100 samples, training and testing errors converge. This convergence indicates that the model is balancing fitting training data and generalising to unseen data, reducing overfitting. As the training dataset grows, especially beyond 600 samples, training and testing errors become almost touching. This indicates that model is robust and well-generalized, performing well on both the training and testing datasets.

6.4.2 Decision Tree Regression

The performance metrics of the model are presented below:

```
Mean Squared Error (Training): 3.0639794569796934e-35
Mean Squared Error (Testing): 2.9435282649827546e-05
Root Mean Squared Error (Training): 5.535322444970747e-18
Root Mean Squared Error (Testing): 0.005425429259498971
Mean Absolute Error (Training): 1.1039149392579681e-18
Mean Absolute Error (Testing): 0.003461923447383735
```

The Mean Squared Error (MSE) of the training data is exceptionally small, estimated to be approximately 3.06e-35. This observation suggests that the Decision Tree model has achieved a high level of accuracy in fitting the training data. The predictions of the model on the training data exhibit a high degree of proximity to the actual values, indicating the possibility of overfitting.

The Mean Squared Error (MSE) of the testing data is observed to be relatively low, with an approximate value of 2.94e-05. This observation indicates that the model exhibits satisfactory performance when applied to data that it has not been trained on, though with a slightly higher mean squared error (MSE) compared to the training data. The low mean squared error (MSE) suggests that the model's predictions on the testing dataset exhibit a high level of accuracy.

The Root Mean Squared Error (RMSE) of the training data is estimated to be approximately 5.54e-18. Similar to the mean squared error (MSE) of the training set, the remarkably low value indicates that the model exhibits a near-perfect fit to the training data. Nevertheless, the presence of a significantly low root mean square error (RMSE) could potentially suggest the occurrence of overfitting. This phenomenon occurs when the model excessively focuses on memorising the training data rather than effectively generalising from it.

The Root Mean Squared Error (RMSE) of the testing data is estimated to be approximately 0.00543. Although the testing RMSE is marginally higher than the training RMSE, it remains a negligible value, suggesting that the model's predictions on the testing data exhibit a high level of precision.

The Mean Absolute Error (MAE) value obtained for the training data is approximately 1.10e-18. Similar to the mean squared error (MSE) and root mean squared error (RMSE) used in training, this observation indicates that the model exhibits a high degree of accuracy in fitting the training data, but there is a possibility of overfitting.

The Mean Absolute Error (MAE) for the testing data is estimated to be approximately 0.00346. This metric demonstrates that, on average, the model's predictions on the testing data exhibit a deviation of approximately 0.00346 units in engagement score.

In conclusion, the Decision Tree Regression model exhibits minimal errors on both the training and testing datasets. Although it may appear to be an optimal scenario, the presence of such minimal errors, particularly in the training data, may indicate the occurrence of overfitting. The model may have learned to fit the training data too closely, potentially resulting in reduced generalization to unseen data.

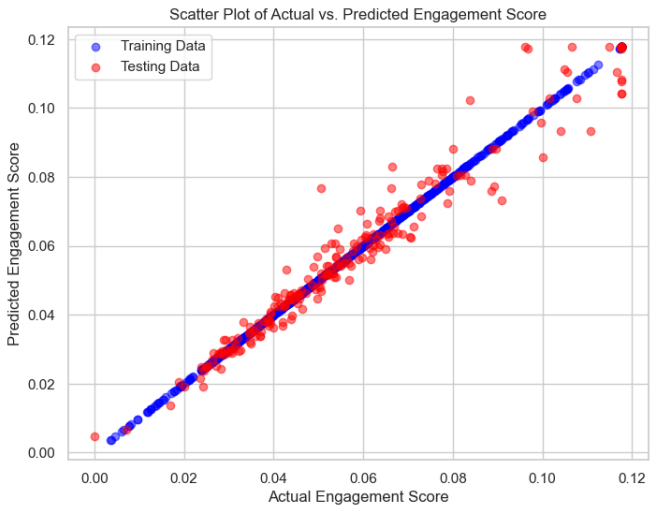


Figure 30: Decision Tree Regression Actual and Predicted Scatter Plot

The model has preserved the training data, resulting in a strong alignment along the diagonal. The testing data comprises new instances that were not encountered by the model during its training phase. Despite the model having acquired knowledge of the underlying patterns from the training data, its performance on unseen data may not be flawless. The dispersion of the testing data points along the diagonal indicates that the model's predictions exhibit a high degree of proximity to the actual values, although with a slightly lower level of accuracy compared to the utilisation of the training data.

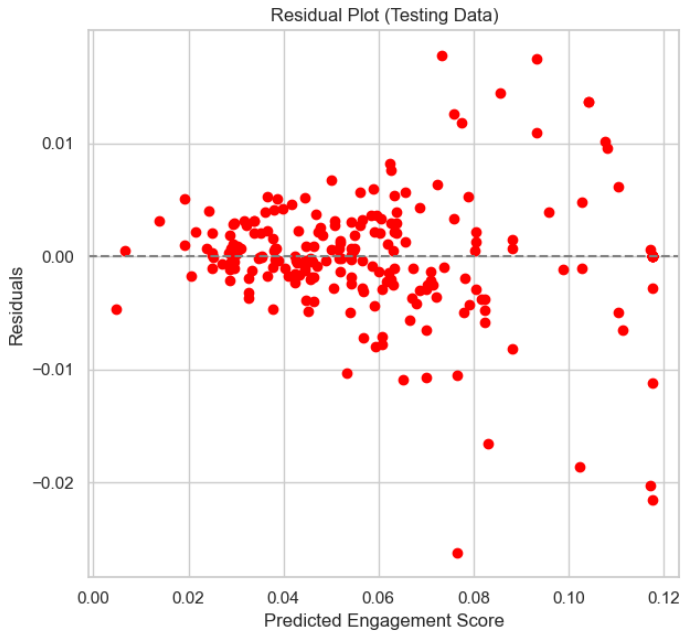


Figure 31: Decision Tree Regression Residual Plot

On the residual plot, we can see that most of the residual values are randomly spread around the horizontal line, which is the zero-error line. This means that our regression model fits pretty well. But it looks like some residual values are a long way from this horizontal line. The very small scale of the y-axis in the plot could be the reason for this effect. In other words, the scale of the y-axis may affect the number of outliers in the residual plot, especially those that look far from the horizontal line.

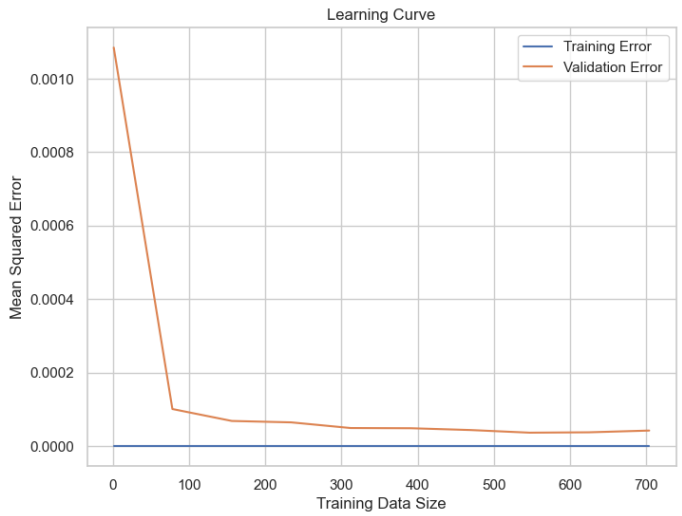


Figure 32: Decision Tree Regression Residual Plot

When using a small training dataset, the model overfits, resulting in almost zero training error. But this doesn't improve model performance. Because it memorises the training data rather than

learning the patterns, it performs poorly on unseen data (testing error). The initial gap between training and testing errors is large.

As you increase the training dataset, roughly 100 samples, the model encounters more diverse examples. This helps it learn data patterns rather than specific instances. Thus, testing error decreases and approaches training error. This convergence indicates that model is balancing training data and generalisation to new data.

6.5.3 K-Nearest Neighbours (KNN) Regression

The performance metrics of the model are presented below:

```
Mean Squared Error (Train): 4.555465315073462e-05
Mean Squared Error (Test): 0.00015618994188221086
Root Mean Squared Error (Train): 0.006749418726878236
Root Mean Squared Error (Test): 0.012497597444397497
Mean Absolute Error (Train): 0.0036399523172057956
Mean Absolute Error (Test): 0.005292555489943322
```

The Mean Squared Error (MSE) of the training data is estimated to be approximately 4.56e-05. The mentioned value denotes the mean squared deviation between the observed and predicted engagement scores within the training dataset. A low mean squared error (MSE) indicates a strong fit of the K-nearest neighbours (KNN) model to the training data, implying that the predictions made on the training data are generally in close range to the actual values.

The Mean Squared Error (MSE) of the testing data is estimated to be approximately 0.000156. The observed mean squared error (MSE) exhibits a slight elevation compared to the MSE obtained during the training phase, as anticipated. Nevertheless, the current value remains comparatively low, suggesting that the K-nearest neighbours (KNN) model exhibits strong performance when applied to previously unseen data. The predictions made by the model on the testing data exhibit a high level of accuracy.

The Root Mean Squared Error (RMSE) of the training data is estimated to be around 0.00675. The root mean square error (RMSE) is derived from the mean square error (MSE) and serves as a metric for quantifying the average discrepancy between predicted and actual values. A low root mean

square error (RMSE) indicates that the model's predictions on the training data exhibit a high level of proximity to the actual values, on average.

The Root Mean Squared Error (RMSE) of the testing data is estimated to be around 0.0125. Comparable to the mean squared error (MSE) observed during testing, the root mean squared error (RMSE) observed during training exhibits a slightly higher value, albeit remaining relatively low. This suggests that the predictions made by the KNN model on the testing data exhibit a satisfactory level of precision.

The Mean Absolute Error (MAE) for the training data is estimated to be around 0.00364. MAE denotes the mean absolute deviation between the observed and predicted values of engagement scores. This metric serves as an additional indicator of the precision of predictions. A low mean absolute error (MAE) indicates that the K-nearest neighbours (KNN) model's predictions on the training data exhibit a relatively small average deviation from the true values.

The Mean Absolute Error (MAE) of the testing dataset is estimated to be approximately 0.00529. Similar to the evaluation of mean squared error (MSE) and root mean squared error (RMSE), the assessment of mean absolute error (MAE) reveals a slightly elevated value for the testing MAE in comparison to the training MAE. However, this suggests that the predictions made by KNN model on the testing data exhibit a reasonable level of accuracy.

In conclusion, the KNN Regression model exhibits strong predictive capabilities in estimating engagement scores for YouTube videos. The model demonstrates minimal errors on both the training and testing datasets, indicating its ability to effectively generalise to new, unseen data.

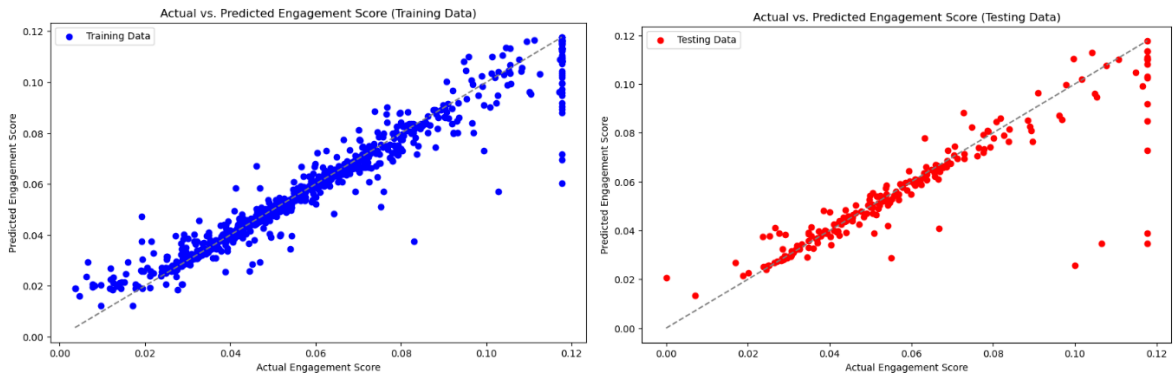


Figure 33: K-Nearest Neighbours (KNN) Regression Actual and Predicted Values Scatter Plot

The slight scattering of training data points along the diagonal suggests that the model is not overfitting the training data, which means it is not fitting the data points too closely and picking up noise. Instead, it generalises well enough to allow for some variation within the training set. The test data are new examples. The fact that the testing data points are scattered in the same way as the training data along the diagonal shows that the model is making fairly accurate predictions for both sets of data.

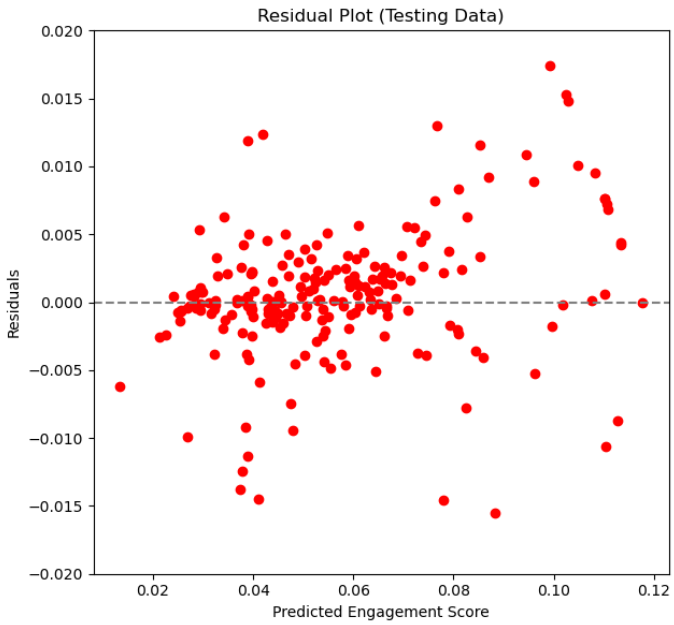


Figure 34: K-Nearest Neighbours (KNN) Regression Residual Plot

Similar to the other models, the residual plot shows that the majority of the residual values are dispersed randomly around the horizontal line, which stands in for the zero-error line, indicating that

our regression model fits the data reasonably well. Some residual values, however, seem to be significantly off from this horizontal line. The y-axis in the plot's extremely small scale may be to blame for this phenomenon. In other words, the size of the y-axis can have an impact on the presence of outliers in the residual plot, especially those that appear to be far from the horizontal line.

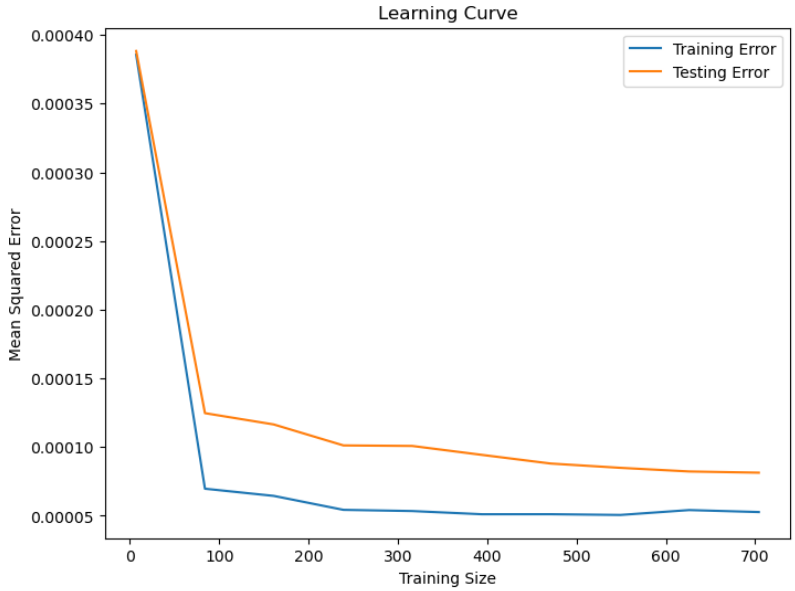


Figure 35: K-Nearest Neighbours (KNN) Regression Learning Curve

The learning curve follows a pattern seen with limited data. Training and testing errors are high initially due to the small training dataset. This is expected because the model struggles to understand the patterns and overfits the dataset noise. However, both errors decrease as the training dataset grows. The training error always remains lower than the testing error, indicating a slight overfitting.

A significant change occurs around 100 training datasets. The training and testing errors converge as the model finds a better balance between fitting training data and generalising to unseen data. Although the training error remains slightly lower, the reduced gap suggests that the model is becoming less overfitting and more generalizable.

6.5.4 Random Forest Regression

The performance metrics of the model are presented below:

```
Mean Squared Error (Train): 0.00000860
Mean Squared Error (Test): 0.00002001
Root Mean Squared Error (Train): 0.00293228
Root Mean Squared Error (Test): 0.00447333
Mean Absolute Error (Train): 0.00182658
Mean Absolute Error (Test): 0.00284967
```

The Mean Squared Error (MSE) of the training data is estimated to be approximately 8.60e-06. The mentioned value denotes the mean squared deviation between the observed and predicted engagement scores within the training dataset. A low mean squared error (MSE) indicates a high level of fitness between the Random Forest model and the training data, implying that the predictions made on the training data are generally in close proximity to the actual values.

The Mean Squared Error (MSE) of the testing data is estimated to be approximately 2.00e-05. The observed mean squared error (MSE) exhibits a slight increase compared to the MSE obtained during the training phase, as anticipated. Nevertheless, the current value remains comparatively low, suggesting that the Random Forest model exhibits strong performance when applied to data that it has not encountered before. The predictions made by the model on the testing data exhibit a high level of accuracy.

The Root Mean Squared Error (RMSE) of the training data is estimated to be around 0.00293. The root mean square error (RMSE) is mathematically derived as the square root of the mean squared error (MSE), serving as a metric to quantify the average discrepancy between predicted and actual values. A low root mean square error (RMSE) indicates that the model's predictions on the training data exhibit a high level of proximity to the true values, on average.

The Root Mean Squared Error (RMSE) of the testing data is estimated to be approximately 0.00447. Comparable to the mean squared error (MSE) observed during testing, the root mean squared error (RMSE) observed during training exhibits a slightly higher value, though remaining relatively low. This suggests that the Random Forest model's predictions on the testing data exhibit a satisfactory level of precision.

The Mean Absolute Error (MAE) of the training data is estimated to be around 0.00183. The MAE denotes the mean absolute error, which quantifies the average absolute discrepancy between the

observed and predicted values of engagement scores. It constitutes an additional metric for evaluating the precision of predictions. A low mean absolute error (MAE) indicates that the Random Forest model's predictions on the training data exhibit a high level of proximity to the corresponding actual values, on average.

The Mean Absolute Error (MAE) for the testing data is estimated to be approximately 0.00285. Similar to the evaluation of mean squared error (MSE) and root mean squared error (RMSE), the assessment of mean absolute error (MAE) reveals a slightly elevated value for the testing MAE in comparison to the training MAE. However, this suggests that the predictions made by Random Forest model on the testing data exhibit a reasonable level of accuracy.

In conclusion, the Random Forest Regression model exhibits exceptional efficacy in forecasting engagement scores for YouTube videos. The model demonstrates a notable capacity to generalise to new, unseen data, as indicated by its minimal errors observed in both the training and testing datasets.

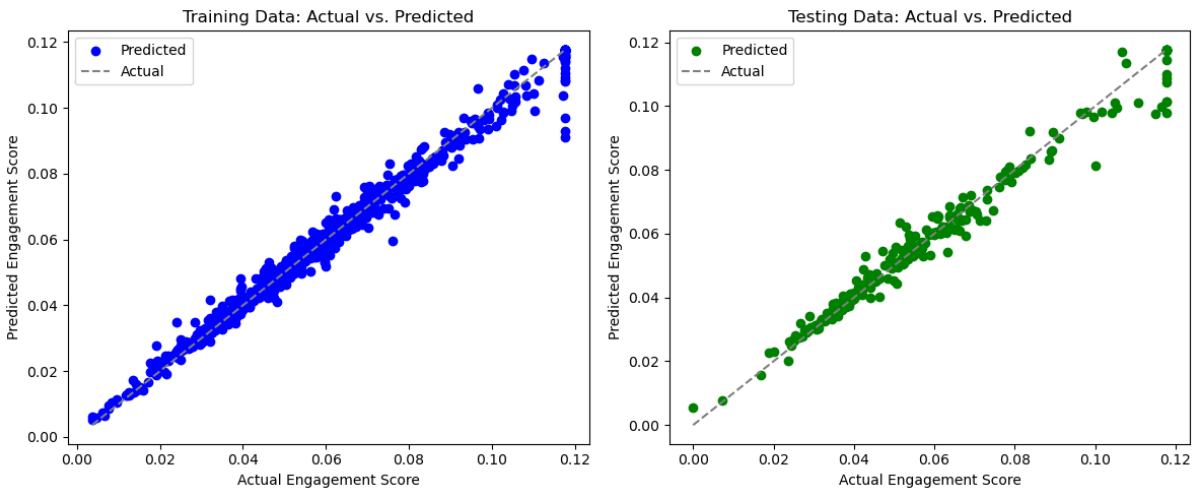


Figure 36: Random Forest Regression Actual and Predicted Values Scatter Plot

The slight scattering of training data points along the diagonal suggests that your model is not overfitting the training data. Instead, it generalises well enough to allow some variation within the training set. The fact that the testing data points have a similar scattering pattern along the diagonal as the training data shows that your model is making fairly accurate predictions for both sets of data.

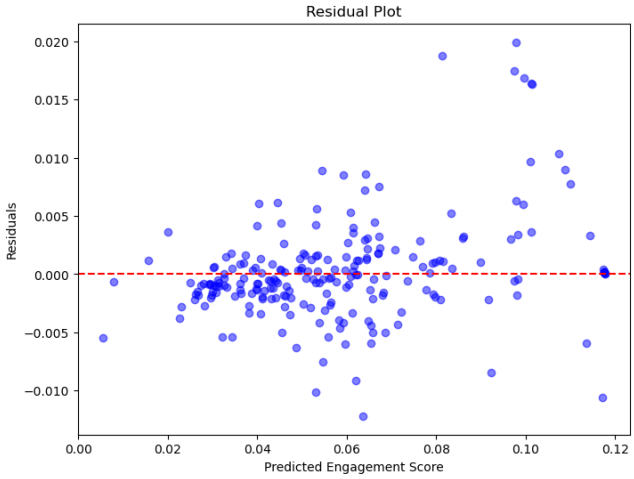


Figure 37: Random Forest Regression Residual Plot

The majority of the residual values in the residual plot are randomly distributed around the horizontal line, which stands in for the zero-error line, indicating that our regression model fits the data reasonably well. However, some residual values appear to be significantly off from this horizontal line, similar to the other models. The y-axis in the plot's extremely small scale may be the cause for this phenomenon.

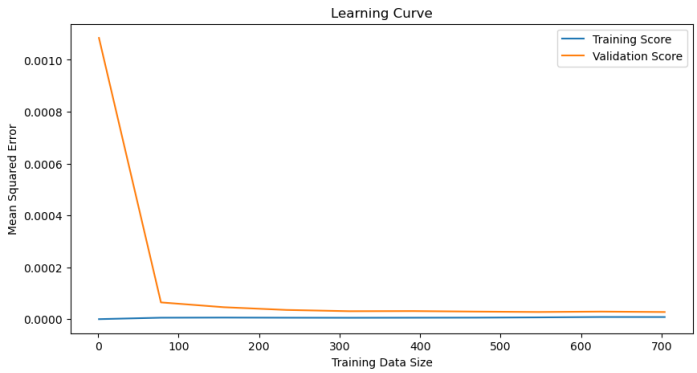


Figure 38: Random Forest Regression Learning Curve

The model overfits with a small training dataset, resulting in almost zero training error.The model encounters more diverse examples as the training dataset increases and roughly after 100 samples, it focuses on learning data patterns rather than memorising specific instances. Consequently, testing error decreases and approaches training error. The convergence means it is better at fitting training data and generalising to new data.

6.5.5 Gradient Boosting Regression

The performance metrics of the model are presented below:

```
Mean Squared Error (MSE) for Testing Data: 1.4939608932921407e-05
Mean Squared Error (MSE) for Training Data: 3.172018492536667e-06
Root Mean Squared Error (RMSE) for Testing Data: 0.003865179029866716
Root Mean Squared Error (RMSE) for Training Data: 0.001781016140448106
Mean Absolute Error (MAE) for Testing Data: 0.0021862617948932963
Mean Absolute Error (MAE) for Training Data: 0.0011605056883281454
```

The mean squared error (MSE) for the testing data is approximately 1.49e-05. A low mean squared error (MSE) signifies that the Gradient Boosting model yields precise predictions when applied to previously unseen data.

The mean squared error (MSE) for the training data is approximately 3.17e-06. The mentioned value denotes the mean squared deviation between the observed and predicted engagement scores within the training dataset. A low mean squared error (MSE) in training indicates a strong fit of the model to the training data, effectively capturing the inherent patterns within.

The root mean square error (RMSE) for the testing data is estimated to be approximately 0.00387. The root mean square error (RMSE) is calculated as the square root of the mean squared error (MSE), providing a measure of the average error in predictions, expressed in the same units as the target variable. A low root mean square error (RMSE) suggests that the model's predictions on the testing dataset are, on average, in close proximity to the true values.

The root mean square error (RMSE) for the training data is estimated to be approximately 0.00178. The training root mean squared error (RMSE) is also low, suggesting that the Gradient Boosting model yields accurate predictions on the training dataset, similar to the testing RMSE.

The Mean Absolute Error (MAE) for the testing data is estimated to be approximately 0.00219. The MAE denotes the mean absolute error, which quantifies the average absolute discrepancy between the observed and predicted values of engagement scores. This metric serves as an additional indicator of the precision of predictions. A low mean absolute error (MAE) indicates that

the predictions made by the Gradient Boosting model on the testing dataset exhibit a high level of proximity to the true values, on average.

The mean absolute error (MAE) for the training data is estimated to be approximately 0.00116. Similar to the mean absolute error (MAE) observed in the testing phase, the MAE in the training phase is also relatively low. This suggests that the model's predictions on the training data exhibit a high level of proximity to the true values, on average.

In conclusion, the Gradient Boosting Regression model exhibits exceptional efficacy in forecasting engagement scores for YouTube videos. The model demonstrates a notable capacity to generalise to new, unseen data, as evidenced by its minimal errors observed in both the training and testing datasets.

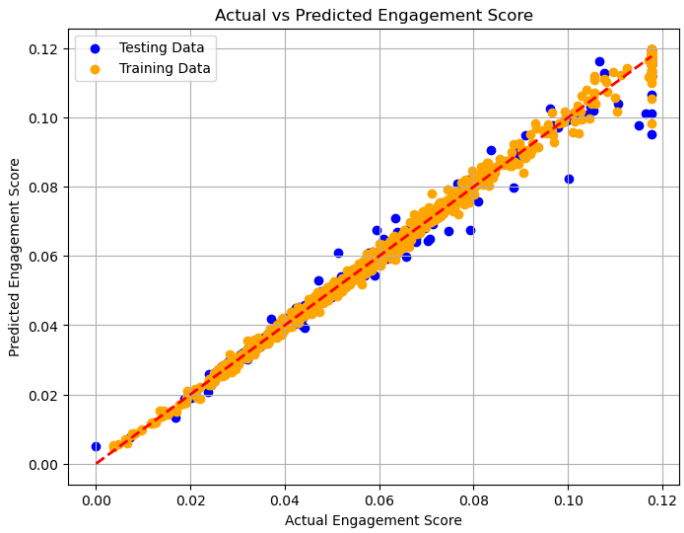


Figure 39: Gradient Boosting Regression Actual and Predicted Values Scatter Plot

The slight scattering of training data points along the diagonal suggests that your model is not overfitting the training data. The testing data points being scattered in a similar way along the diagonal as the training data shows that your model is making fairly accurate predictions for both sets.

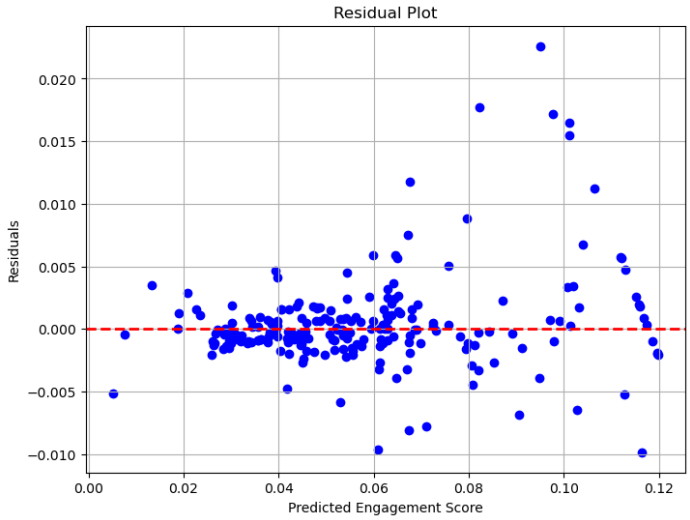


Figure 40: Gradient Boosting Regression Residual Plot

The residual plot shows that most residual values are randomly scattered around the horizontal line, which represents the zero-error line, indicating a good regression model fit. However, like other models, some residual values appear far from this horizontal line. This may be due to the plot's extremely small y-axis.

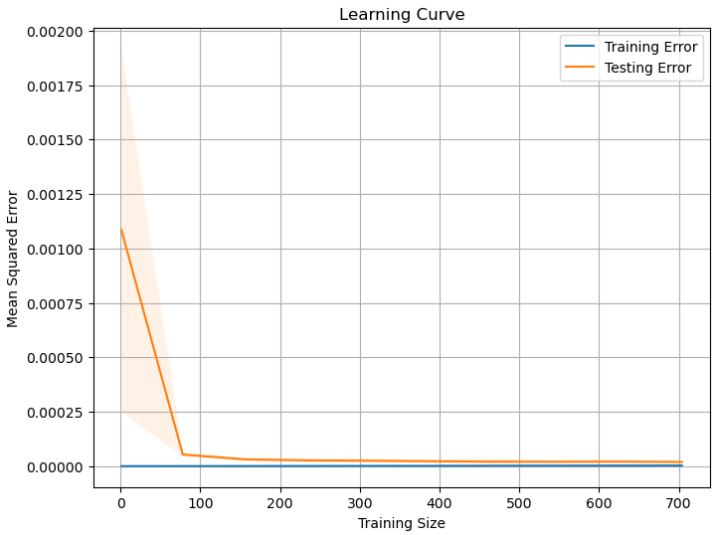


Figure 41: Gradient Boosting Regression Learning Curve

With a small training dataset, the model has nearly zero training error. This suggests the model memorises the training data rather than learning the patterns, resulting in poor generalisation. Since the model struggles to apply its knowledge to unseen data, the initial testing error is much higher. As the training dataset increases to 100 samples, the model encounters more diverse examples. This helps it focus on fundamental data patterns, reducing testing error. At approximately 150 in the training size, the testing and training errors become closely parallel, indicating that model has found

a balance between fitting training data and generalising to new data. This suggests that model has captured the essential data patterns and may not benefit from additional training data.

6.5.6 Neural Network Regression

The performance metrics of the model are presented below:

```
Mean Squared Error (Train): 0.00097440
Mean Squared Error (Test): 0.00145979
Root Mean Squared Error (Train): 0.03121536
Root Mean Squared Error (Test): 0.03820719
Mean Absolute Error (Train): 0.02204670
Mean Absolute Error (Test): 0.02352057
```

The mean squared error (MSE) for the training data is approximately 0.00097440. The metric presented herein serves to quantify the mean squared deviation between the observed and predicted engagement scores within the training dataset. A low mean squared error (MSE) value signifies that the Neural Network model yields precise predictions on the training data.

The mean squared error (MSE) for the testing data is estimated to be approximately 0.00145979. Similar to the mean squared error (MSE) in training, this metric denotes the mean of the squared discrepancies between the observed and predicted engagement scores on the testing dataset. A comparatively low mean squared error (MSE) in testing indicates that the Neural Network model exhibits effective generalisation capabilities when presented with new, unobserved data.

The Root Mean Squared Error (RMSE) value is calculated for the training data. The root mean square error (RMSE) for the training data is approximately 0.03121536. It quantifies the average discrepancy between predicted values and actual values, using the same units as the target variable. Although the aforementioned value exhibits a relatively higher magnitude in comparison to alternative models, it is crucial to acknowledge that the interpretation of Root Mean Square Error (RMSE) is depending upon the scale of the target variable.

The root mean square error (RMSE) value obtained for the testing data is approximately 0.03820719. Similar to the root mean squared error (RMSE) of the training data, this particular value exhibits a

higher magnitude in comparison to certain alternative models. Nevertheless, the current value remains comparatively low, indicating that the Neural Network model offers reasonably precise forecasts for the testing dataset.

The mean absolute error (MAE) for the training data is estimated to be approximately 0.02204670. The MAE denotes the mean absolute error, which quantifies the average absolute discrepancy between the observed and predicted values of engagement scores. It represents an additional metric for evaluating the precision of predictions. A relatively low mean absolute error (MAE) suggests that the predictions made by your Neural Network model on the training data exhibit a tendency to closely approximate the true values, on average.

The Mean Absolute Error (MAE) was calculated for the testing data. The Mean Absolute Error (MAE) for the testing data is estimated to be approximately 0.02352057. The training mean absolute error (MAE) exhibits a relatively low magnitude, suggesting that the predictions made by Neural Network model on the testing data are, on average, in close proximity to the true values.

In conclusion, the Neural Network Regression model demonstrates strong performance in its ability to accurately predict engagement scores for YouTube videos. Although the root mean square error (RMSE) values may exhibit relatively higher magnitudes in comparison to alternative models, it is crucial to take into account the magnitude of the target variable.

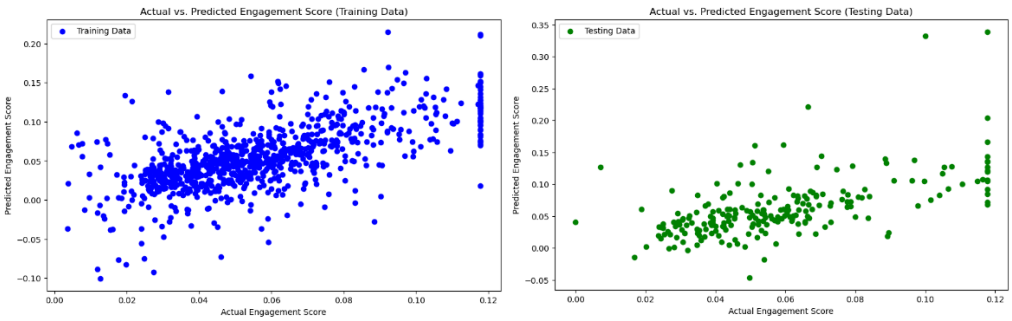


Figure 42: Neural Network Regression Actual and Predicted Values Scatter Plot

The fact that the data points are spread out in both the training and testing datasets shows that the model's predictions aren't perfect. This could also be because the plot is on a very small scale. The

fact that the scattering pattern of the testing data is similar to that of the training data is a good sign. It means that the performance of the model is the same in both the training and testing datasets. This means that the relationships learned from the training data are being applied fairly well to the testing data.

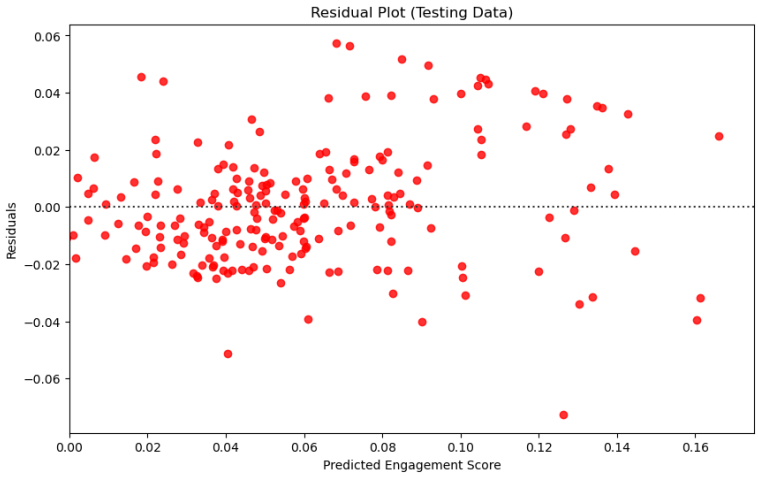


Figure 43: Neural Network Regression Residual Plot

Based on the residual plot, our regression model fits well because most residual values are randomly distributed around the horizontal line, which represents the zero-error line. Some residual values appear far from this horizontal line, like the other models. Plotting's extremely small y-axis may explain this. Outliers in the residual plot, especially those far from the horizontal line, can be affected by y-axis scale.

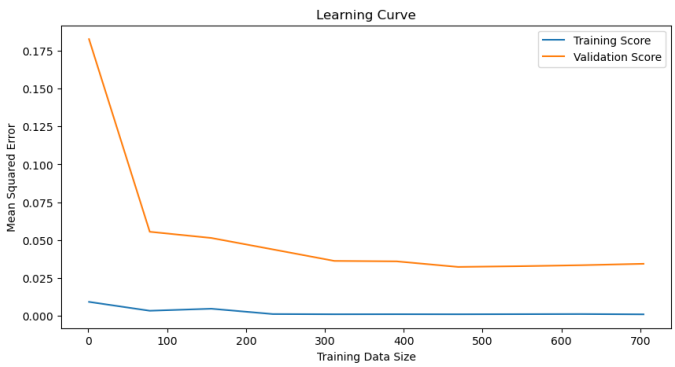


Figure 44: Neural Network Regression Learning Curve

With a small training dataset, your model has almost zero training error. This suggests that the model can perfectly fit the training data, but overfitting may prevent it from generalising to new data. Thus,

the testing error is much higher at the start, indicating that the model's excellent performance on training data does not apply to unseen examples.

As the training dataset grows to 100 samples, model encounters more diverse examples. This reduces testing error by forcing it to learn data patterns rather than memorise training data. Notably, the gap between training and testing errors remains wider than in other models.

6.5.7 Extreme Gradient Boosting Regressor (XGBRegressor)

The performance metrics of the model are presented below:

```
Mean Squared Error: 1.1912632134964598e-05
Root Mean Squared Error: 0.0034514681129867906
Mean Absolute Error: 0.002008383642961417
```

The Mean Squared Error (MSE) is a metric used to quantify the average squared difference between the predicted and actual engagement scores. The mean squared error (MSE) obtained is approximately 1.191e-05.

The Root Mean Squared Error (RMSE) is a statistical metric that is calculated as the square root of the Mean Squared Error (MSE). It is commonly used to assess the average prediction error in the same units as the target variable. The root mean square error (RMSE) is estimated to be approximately 0.00345.

The Mean Absolute Error (MAE) is a metric used to measure the average absolute difference between predicted and actual engagement scores. The Mean Absolute Error (MAE) achieved is approximately 0.00201.

Based on the performance metrics, it can be inferred that the XGBRegressor model demonstrates a notable degree of precision when forecasting engagement scores for Casey Neistat's videos.

Evaluation of the model on the training data:

```
Mean Squared Error (Train): 4.919803260557441e-07
Root Mean Squared Error (Train): 0.0007014130923042028
Mean Absolute Error (Train): 0.0005175764128670042
```

In order to conduct a more comprehensive evaluation of the model's performance, we proceeded to assess its performance on the training data.

The Mean Squared Error (MSE) for the training data is estimated to be approximately 4.92e-07.

The Root Mean Squared Error (RMSE) for the training data is estimated to be approximately 0.000701.

The Mean Absolute Error (MAE) for the training data is estimated to be approximately 0.000518.

The performance of the model on the training data suggests a solid fit, exhibiting minimal difference between the predicted and actual engagement scores.

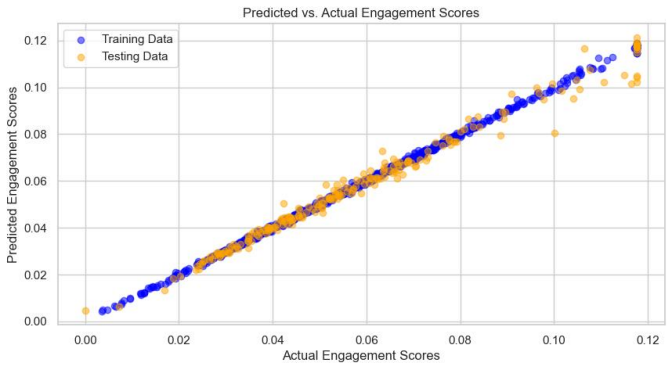


Figure 45: Extreme Gradient Boosting Regressor (XGBRegressor) Actual and Predicted Values Scatter Plot

The training data have been retained by the model, which is why there is a tight alignment along the diagonal. The testing data are new examples that the model did not see while it was being trained. Even though the model has learned the underlying patterns from the training data, it may not work perfectly on this new data that it has never seen before. The fact that the testing data points are a

little bit spread out along the diagonal shows that the model's predictions are very close to the real values, but not as accurate as when using the training data.

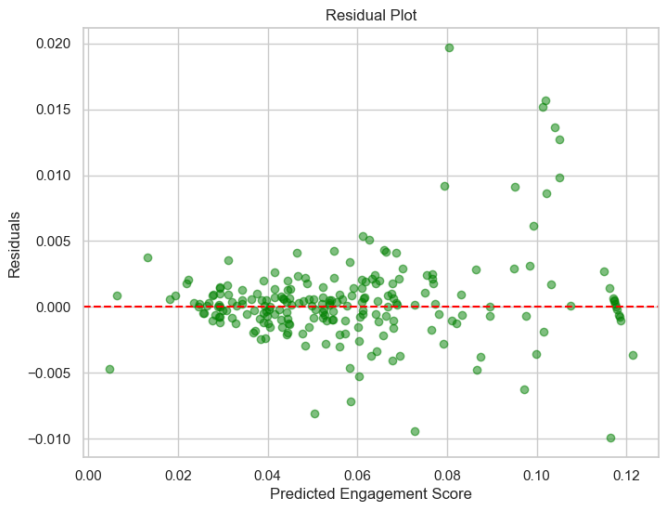


Figure 46: Extreme Gradient Boosting Regressor (XGBRegressor) Residual Plot

In the residual plot, most residual values are randomly scattered around the horizontal line, which represents the zero-error line, indicating a good regression model fit. Like the other models, some residual values are far from this horizontal line. The plot's extremely small y-axis may explain this.

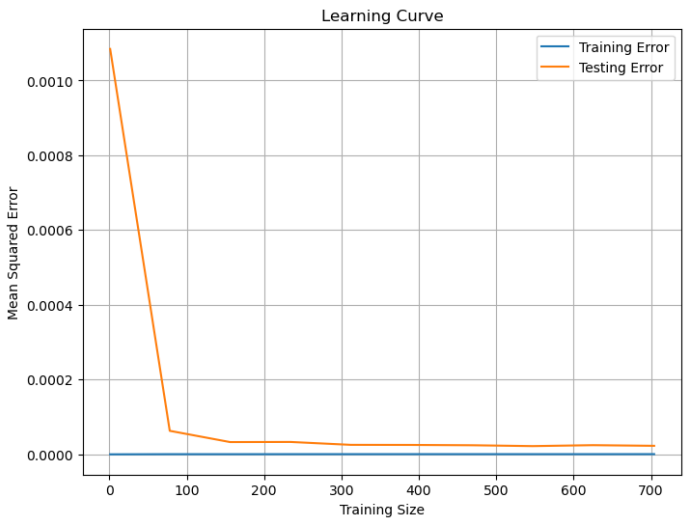


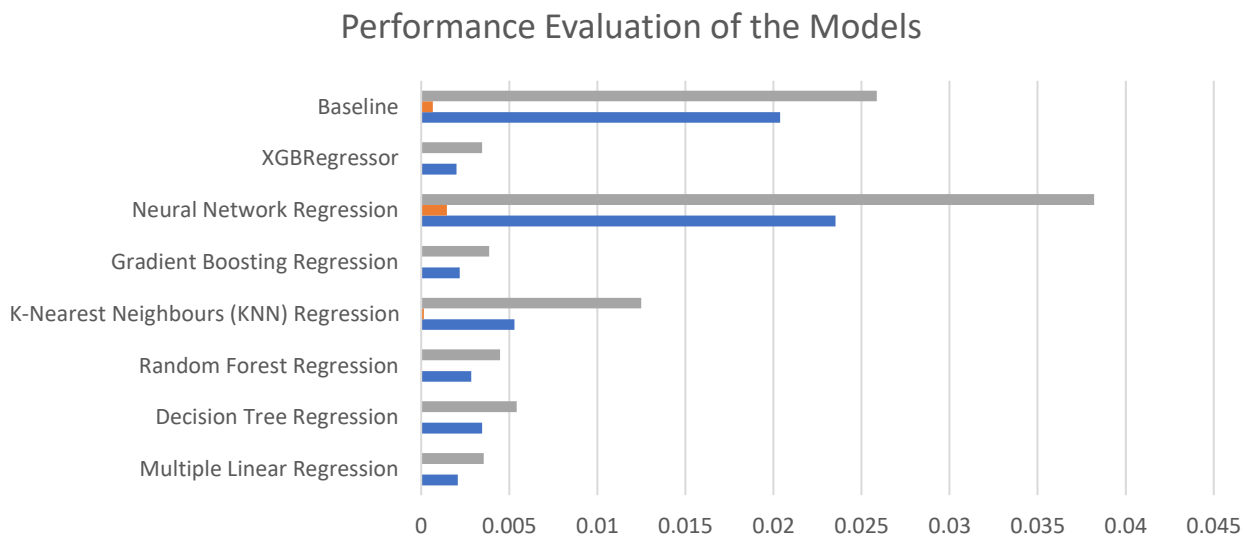
Figure 47: Extreme Gradient Boosting Regressor (XGBRegressor) Learning Curve

The model has almost no training error with a small dataset. This suggests the model memorises training data rather than learning patterns, resulting in poor generalisation. The initial testing error is higher because the model struggles to apply its knowledge to unseen data. As the training dataset grows to 100 samples, the model encounters more variety. This reduces testing error by focusing on

fundamental data patterns. At roughly 300 training size, testing and training errors are nearly identical, indicating that the model has found a balance between fitting training data and generalising to new data. This suggests that model has captured key data patterns and may not need more training data.

6.6 Comparison of Models

Performance Evaluation of the models



	Multiple Linear Regression	Decision Tree Regression	Random Forest Regression	K-Nearest Neighbours (KNN) Regression	Gradient Boosting Regression	Neural Network Regression	XGBRegressor	Baseline
RMSE	0.003552678	0.005425429	0.00447333	0.012497597	0.003865179	0.03820719	0.003451468	0.025870292
MSE	1.26E-05	2.94E-05	0.00002001	0.00015619	1.49E-05	0.00145979	1.19E-05	0.000669272
MAE	0.002072416	0.003461923	0.00284967	0.005292555	0.002186262	0.02352057	0.002008384	0.020378709

Figure 48: Performance Evaluation of the models

The Multiple Linear Regression model exhibits the lowest Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values, suggesting precise predictions and superior overall performance compared to the other models.

Both the Decision Tree Regression and Random Forest Regression models demonstrate satisfactory performance, as indicated by their relatively low Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values. This suggests that both models are appropriate for the given task.

The K-Nearest Neighbours (KNN) algorithm The regression model, although relatively simple to implement, demonstrates elevated mean absolute error (MAE) and root mean squared error (RMSE) values, indicating less precise predictions in comparison to alternative models.

The Gradient Boosting Regression model demonstrates strong performance in terms of low Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values. As a result, it is considered a highly viable option for accurately predicting engagement scores.

The neural network regression model exhibits the highest mean absolute error (MAE) and root mean squared error (RMSE) values, suggesting comparatively less precise predictions in comparison to alternative models.

The XGBRegressor model exhibits comparable performance to Multiple Linear Regression, as evidenced by its low Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values. This indicates that XGBRegressor is a highly effective model.

The Baseline Model, although serving as a point of reference, demonstrated higher error metrics in comparison to the majority of regression models, thereby confirming the significance of utilising more sophisticated regression techniques.

In conclusion,

Based on the assessment criteria, Multiple Linear Regression and XGBRegressor demonstrate superior performance as models for forecasting engagement scores within the domain of YouTube

video analysis. These models exhibit the lowest levels of errors and showcase strong predictive capabilities.

Both Decision Tree Regression and Random Forest Regression yield competitive outcomes and can be deemed suitable options in scenarios where interpretability and ensemble methods are desired.

The K-Nearest Neighbours (KNN) algorithm is a non-parametric classification and regression method used in machine learning. Although regression is a commonly used method, it may not be the optimal choice due to its tendency to yield higher errors.

Gradient Boosting Regression provides a robust alternative to Multiple Linear Regression, exhibiting reduced Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values.

The Neural Network Regression model exhibited the highest Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values, indicating a comparatively lower level of accuracy in its predictive performance. The baseline model is utilised as a reference point to assess the enhancement attained through the selected regression methodology.

7.0 Conclusion

This YouTube video analysis of Casey Neistat's channel has provided valuable insights and met its goals. We began with a thorough analysis to understand viewer engagement factors. Visualising the complex relationships between views, likes, comments, video duration, publishing time, and tags was crucial. A suite of regression models that accurately predict engagement scores emerged from predictive modelling. These models showed the potential for data-driven content creation and marketing decision-making using MAE, MSE, and RMSE. Actionable insights from our research helped content creators and marketers. Strategic timing is key, as morning uploads (7.01 am–11:59 am) have the highest engagement. Tags boost video reach. Notably, Neistat's "Education" videos had the most likes, views, and comments, indicating audience preference. Regression models were evaluated to determine their efficacy. The models varied in accuracy, but Multiple Linear Regression and XGBRegressor were ideal for engagement score prediction.

7.1 Limitations

This Casey Neistat channel-specific YouTube video analysis project must consider several constraints. The study only used data from Casey Neistat's channel, which may not accurately represent YouTube's rich ecosystem. Therefore, extrapolating results to other channels requires caution. YouTube's API may limit data accessibility, limiting the analysis's scope. YouTube's recommendation algorithms, which affect viewer engagement, change over time, making long-term conclusions difficult. Casey Neistat's channel has a variety of content, but it focuses on a particular style, which may limit its applicability to other YouTube categories. Finally, trends, news stories, and competition can affect video engagement, but the dataset may not fully capture them, resulting in incomplete analyses. These limitations and considerations emphasise the need for cautious and nuanced interpretation of the study's findings.

7.2 Recommendations

Video Optimization**:** Utilize regression analysis to optimize video attributes like duration, keywords, and thumbnails. Implement SEO best practices to enhance video discoverability.

Audience Engagement: Leverage insights from regression models to tailor content that resonates with the audience. Engaging with viewers through comments, polls, or Q&A sessions can foster a loyal community.

Invest in Learning: Encourage content creators to stay updated on industry trends, algorithms, and analytics tools. Continuous learning can lead to more effective content strategies.

Continuous Data Analysis: Regularly update regression models with fresh data to adapt to changing viewer preferences. Stay informed about emerging tools and methodologies in data analysis.

7.3 Future Research Opportunities

Collaborations and Research: Work with online content creators and researchers to validate and improve the models. Sharing knowledge can benefit both parties.

User Segmentation: Consider clustering algorithms to segment viewers with similar engagement patterns and tailor content.

Data from outside sources: Include social media trends, events, and holidays that may affect viewership. Your models' predictive power may improve.

NLP: Use NLP to analyse user comments and sentiment to understand how feedback affects engagement.

Incorporate Time Series: Consider adding time series analysis to the models. This can track engagement trends and seasons.

Algorithmic Changes: Develop strategies to adapt to YouTube's recommendation algorithm changes and how they affect video engagement. YouTube's algorithm changes frequently, so tracking is crucial.

Multimodal Analysis: Consider adding audio sentiment, speech content, and closed caption text sentiment analysis to better understand viewer engagement.

Content Creation Trends: Monitor content creation trends like livestreaming, short-form videos, and interactive storytelling to assess engagement.

Research user experience (UX) : To determine how video quality, subtitles, and user interface design affect viewer engagement.

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9.0 Appendices

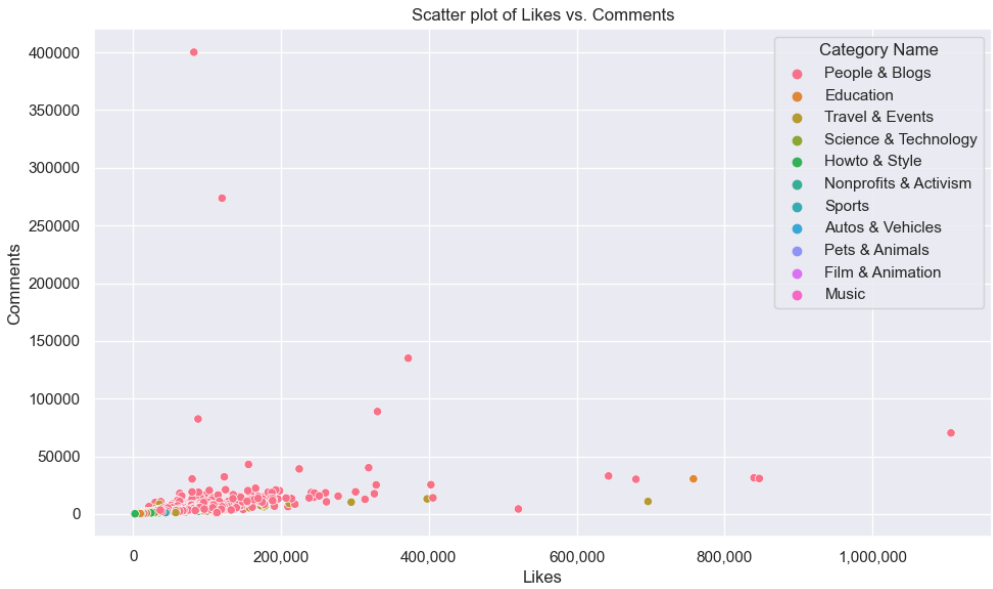

Figure 49: Appendix 1

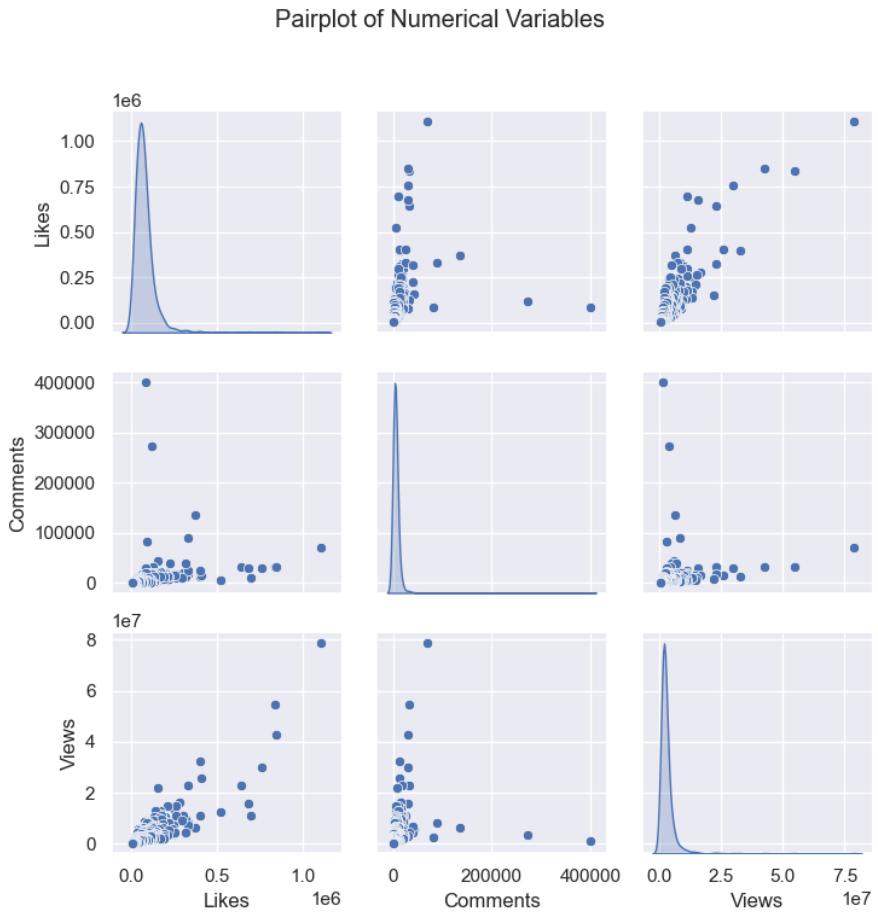

Figure 50:Appendix 2

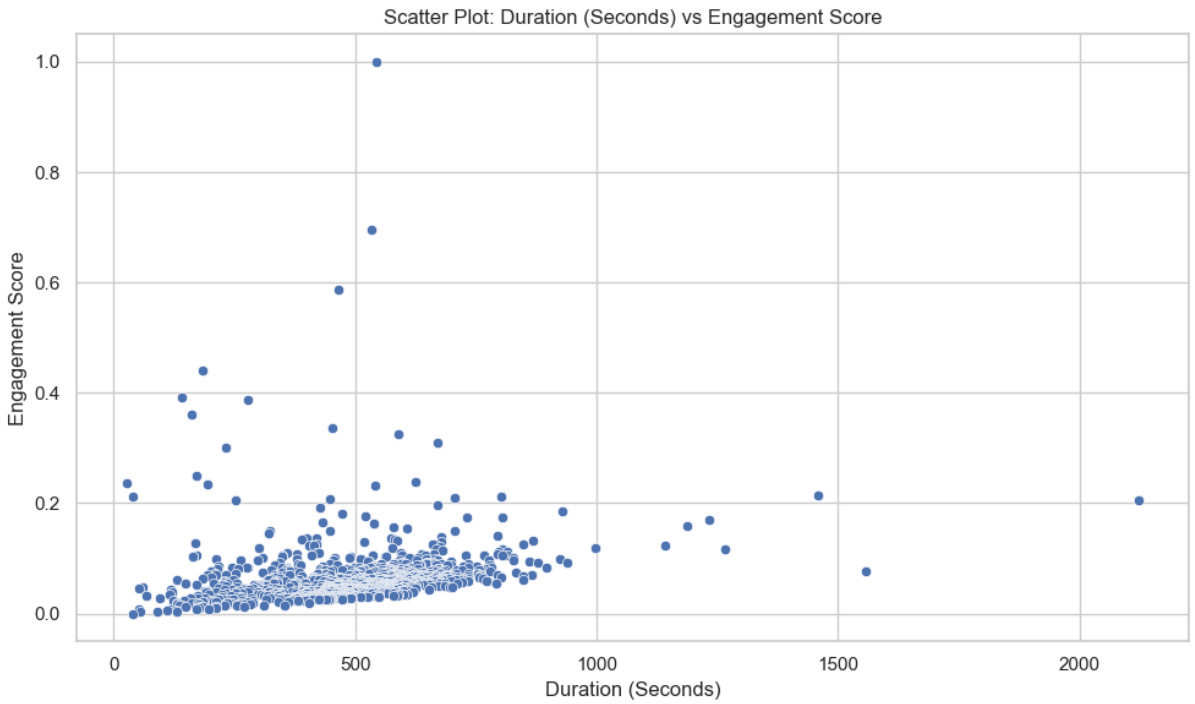

Figure 51: Appendix 3

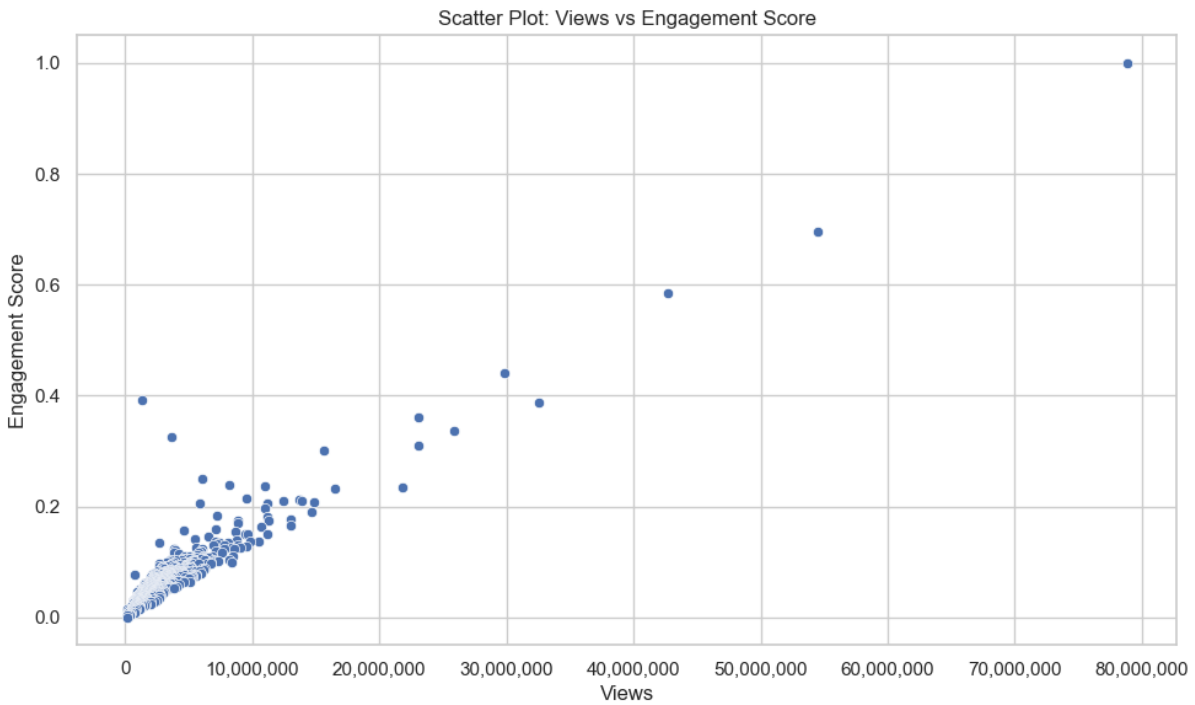

Figure 52: Appendix 4

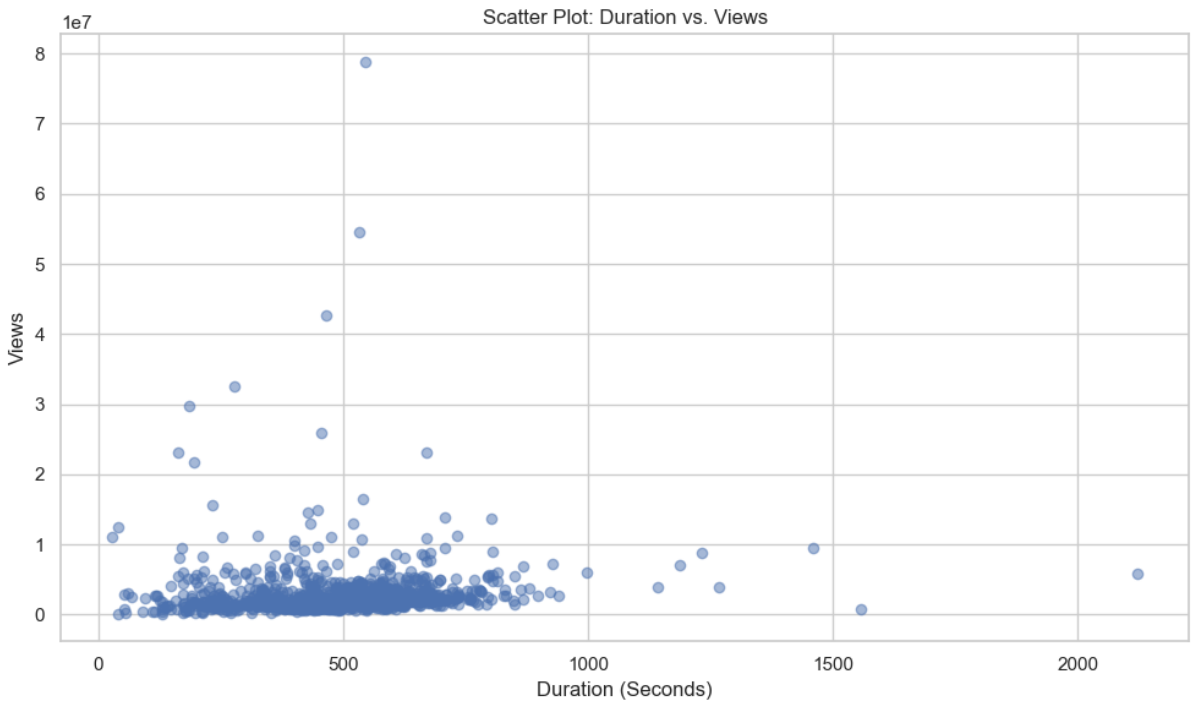

Figure 53: Appendix 5

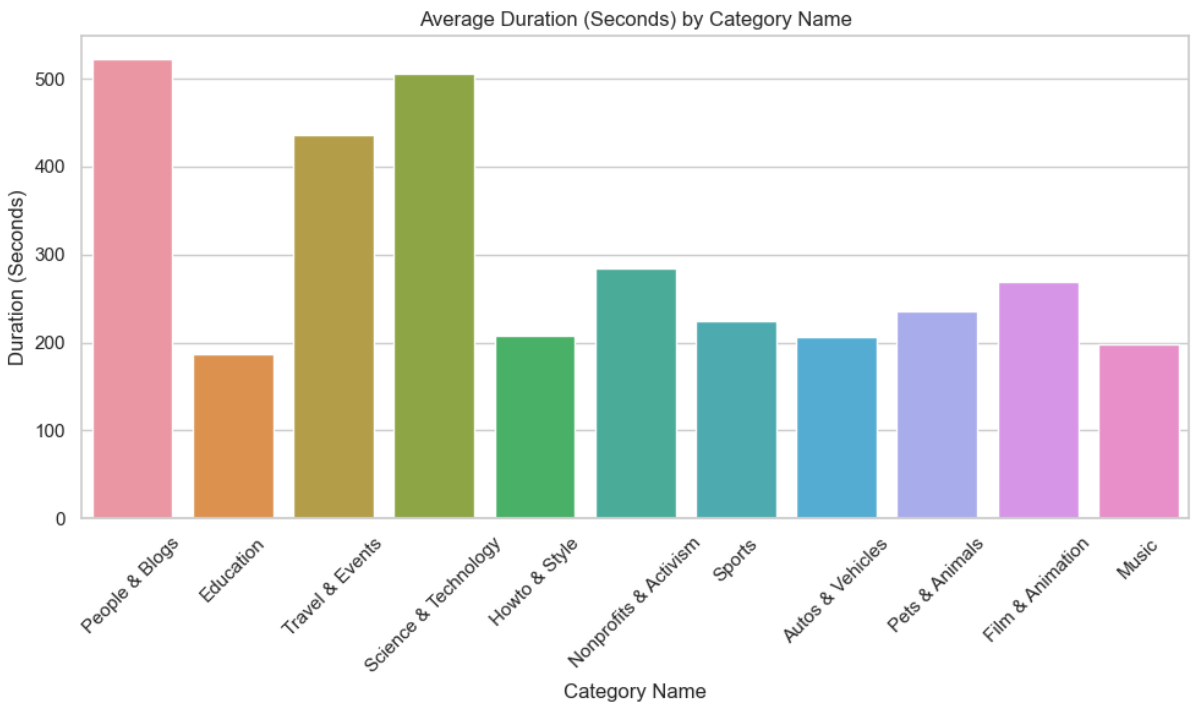

Figure 54: Appendix 6